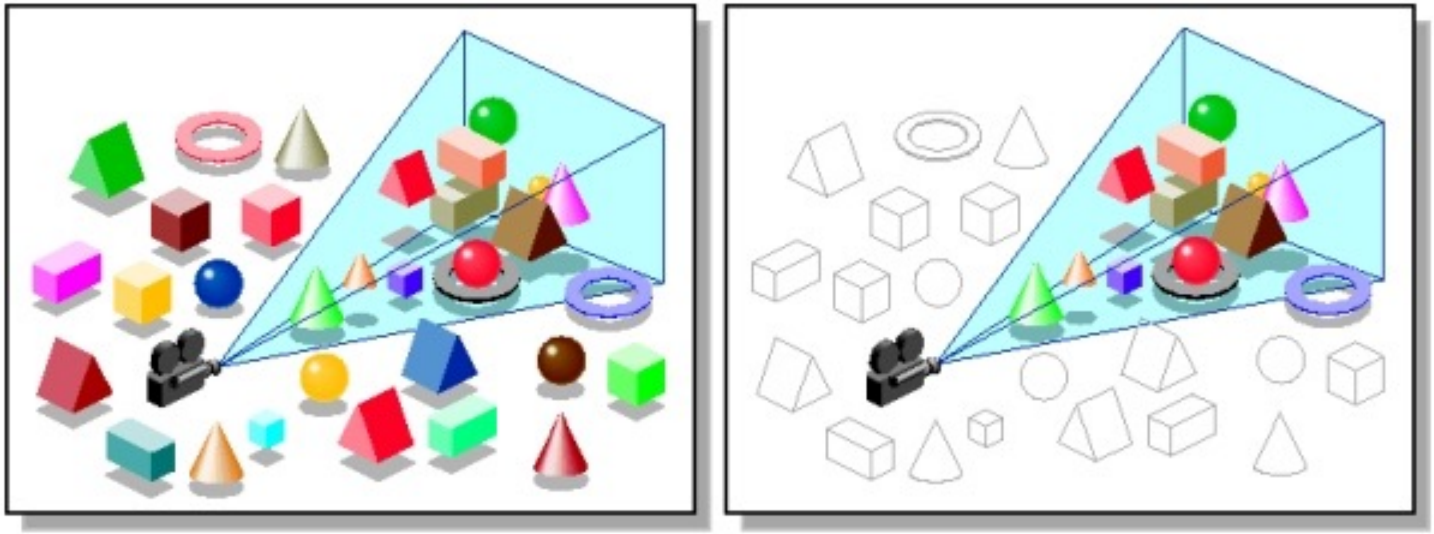
Clipping di Poligoni

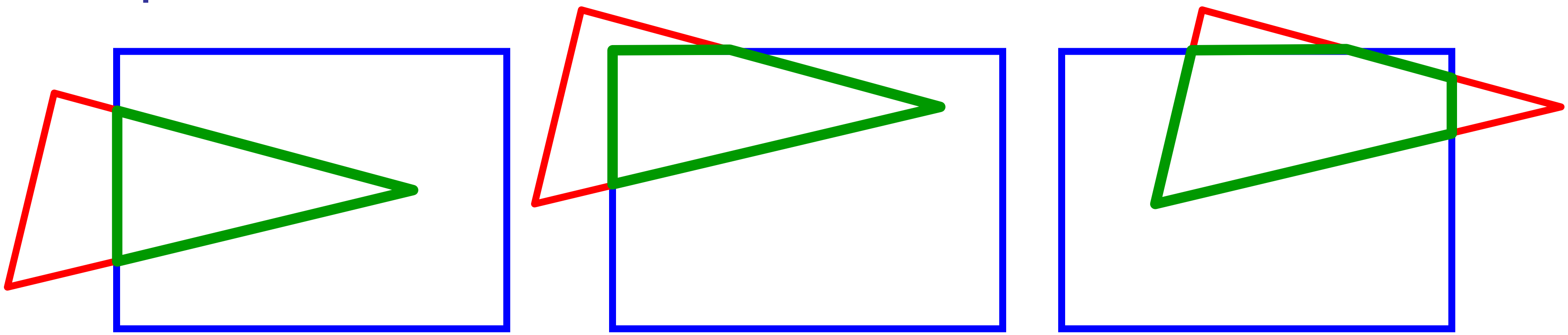


Clipping di Poligoni in 2D

Il Clipping di Poligoni è complicato!

Cosa accade ad un triangolo durante il clipping?

- possibili situazioni:



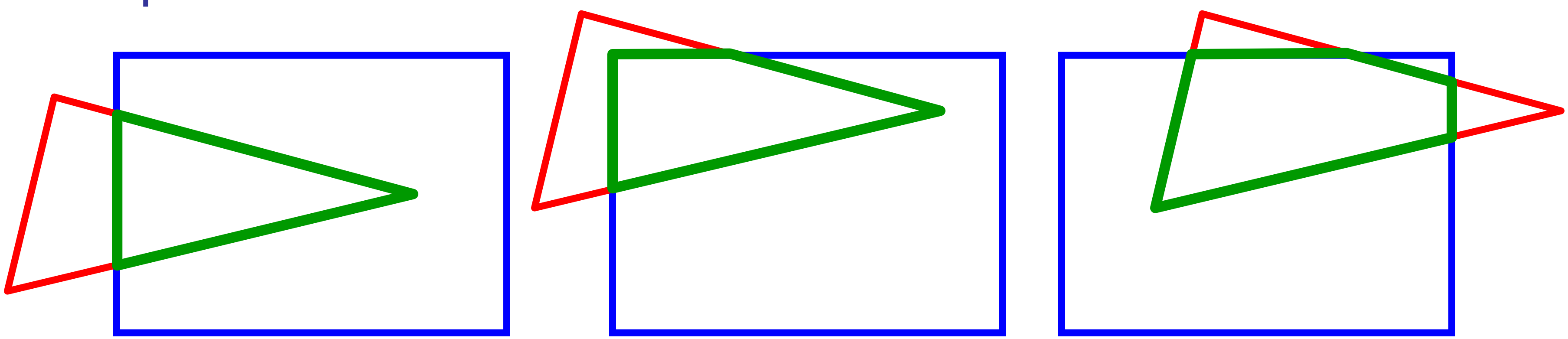
triangolo → triangolo   triangolo → quadrilatero   triangolo → pentagono

Clipping di Poligoni in 2D

Il Clipping di Poligoni è complicato!

Cosa accade ad un triangolo durante il clipping?

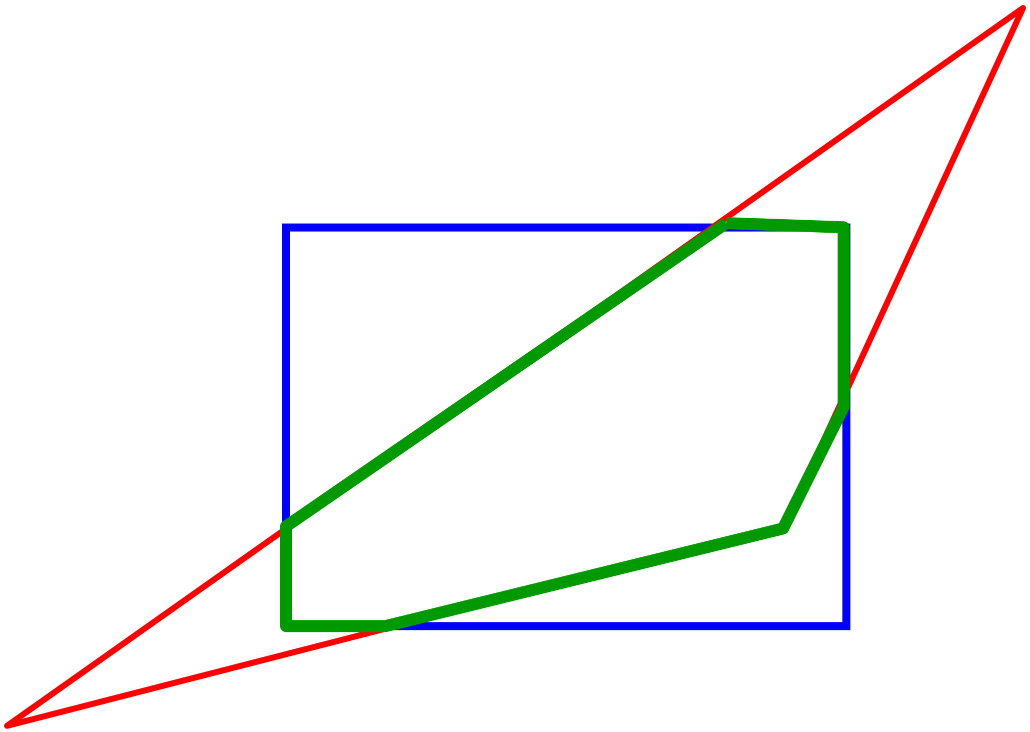
- possibili situazioni:



triangolo → triangolo   triangolo → quadrilatero   triangolo → pentagono

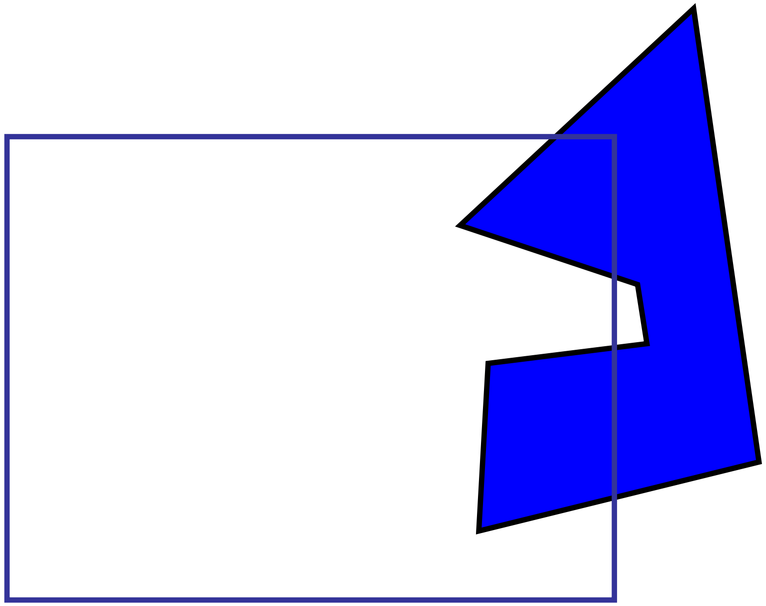
Quanti lati può avere un triangolo dopo il Clipping?

Quanti lati?



fino a 7 lati

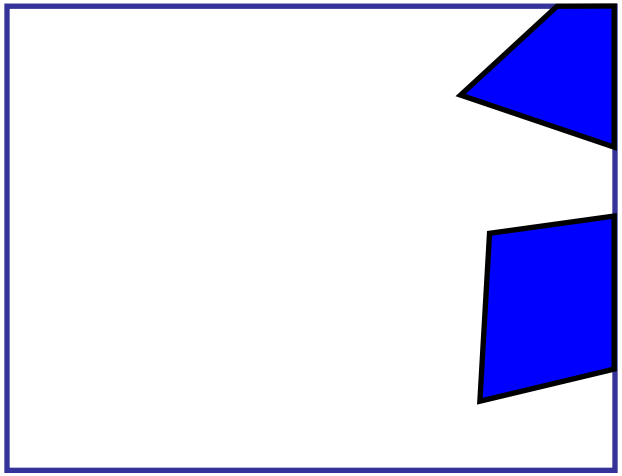
Il Clipping di Poligoni è complicato!



Poligono concavo → più poligoni

Il Clipping di Poligoni è complicato!

Un caso veramente difficile!

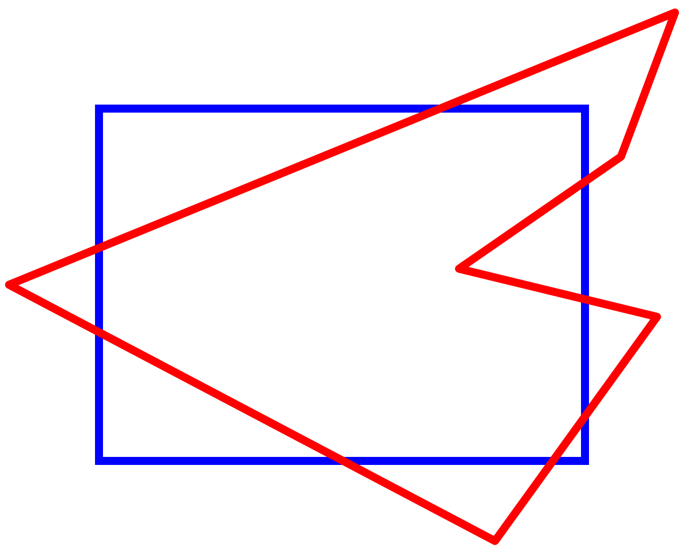


Sutherland-Hodgman Clipping

Idea base:

- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window

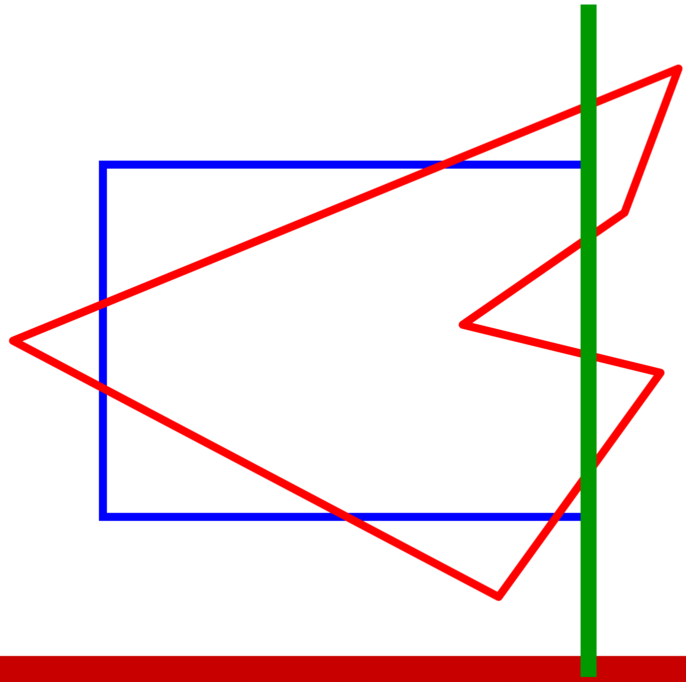
….vediamo



Sutherland-Hodgman Clipping

Idea base:

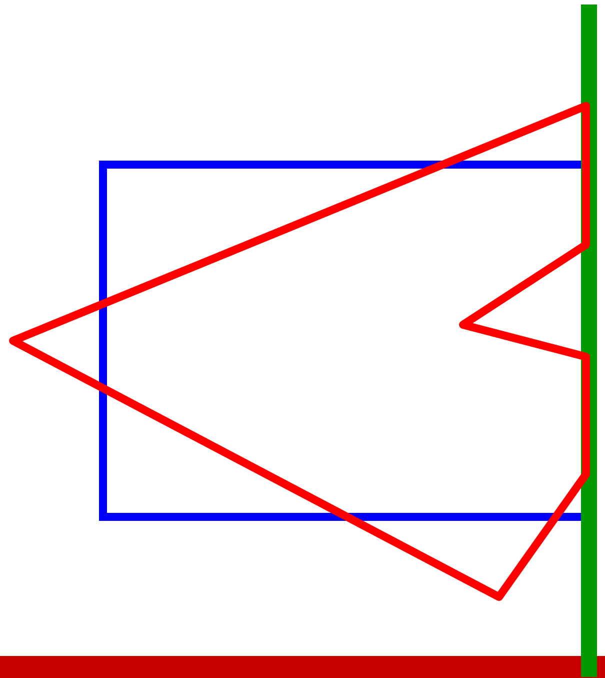
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

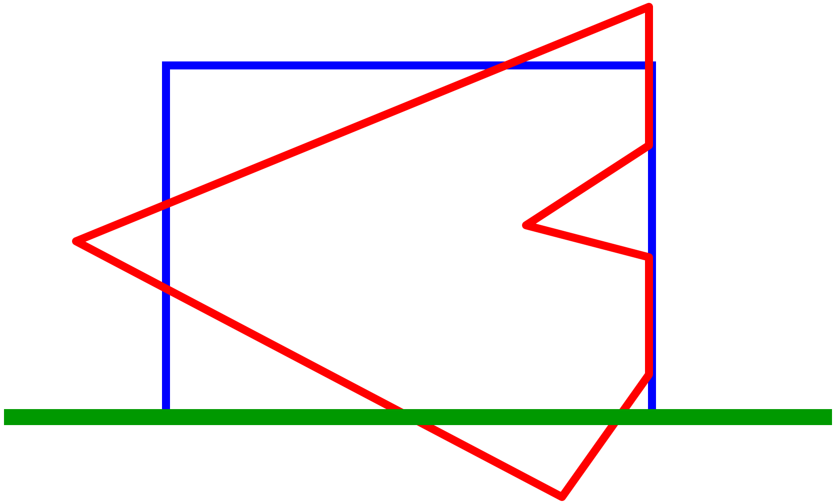
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

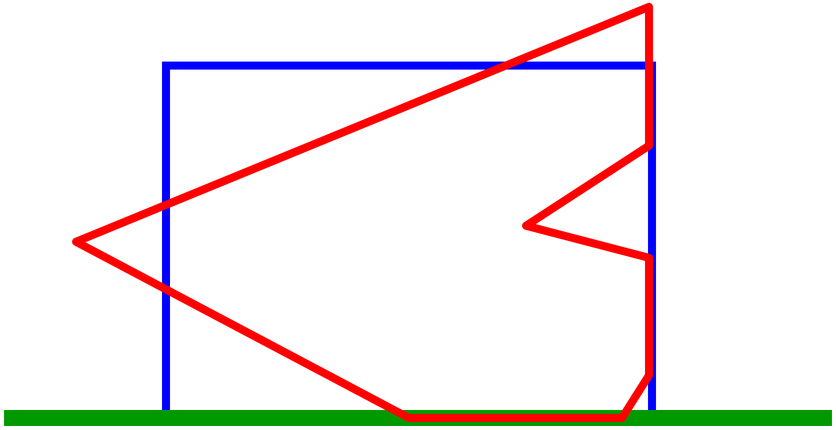
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

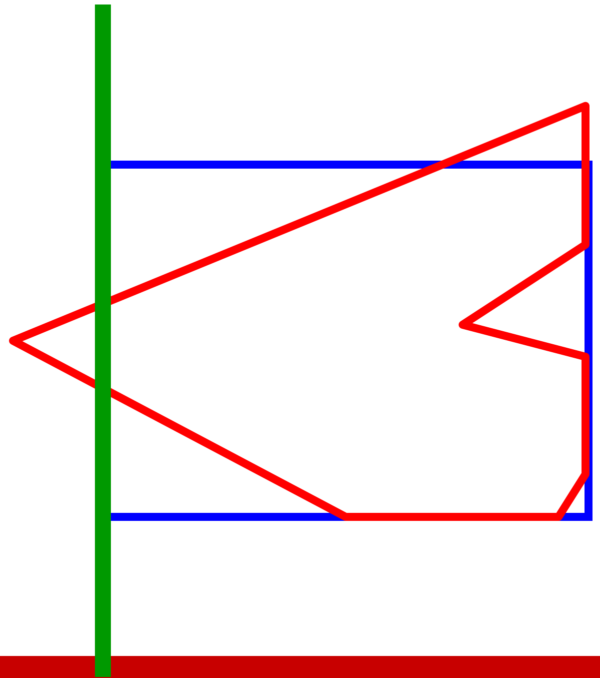
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

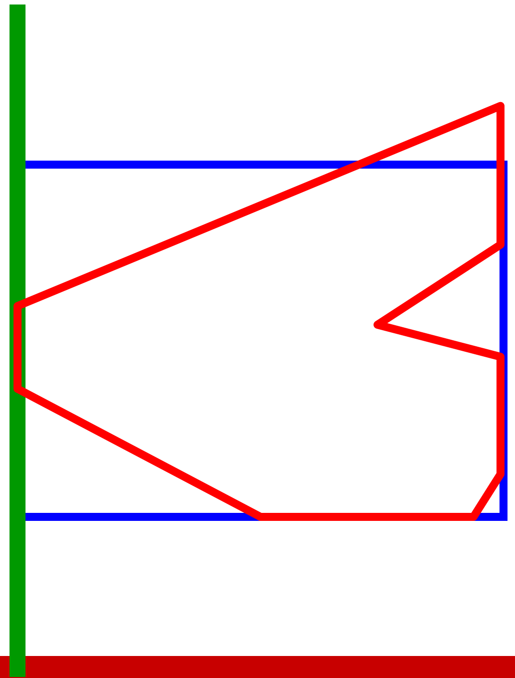
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

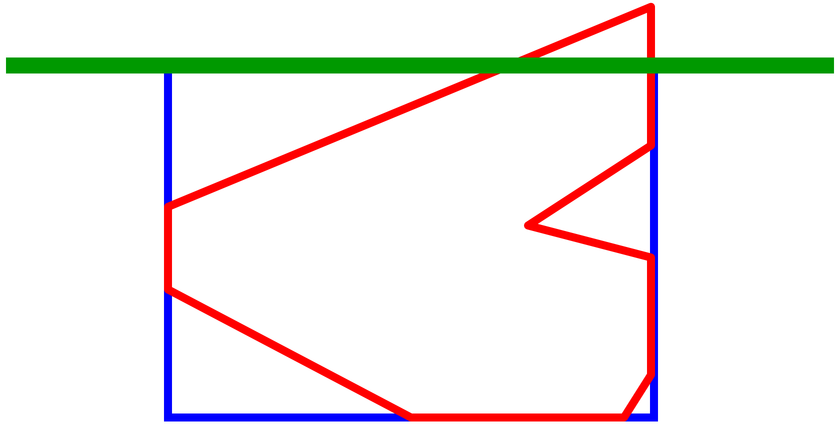
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

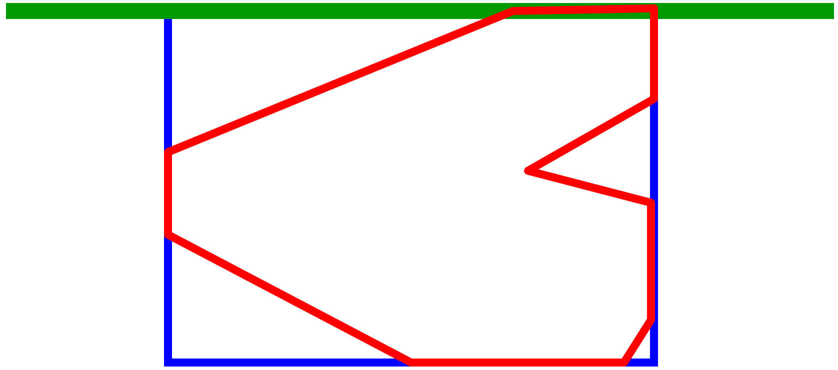
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

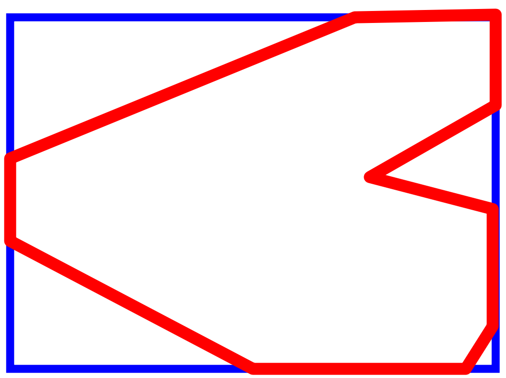
- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window



Sutherland-Hodgman Clipping

Idea base:

- si considera ogni lato della window individualmente
- si fa il clip del poligono rispetto ai lati della window
- dopo aver processato ogni lato, il clipping del poligono sarà completo

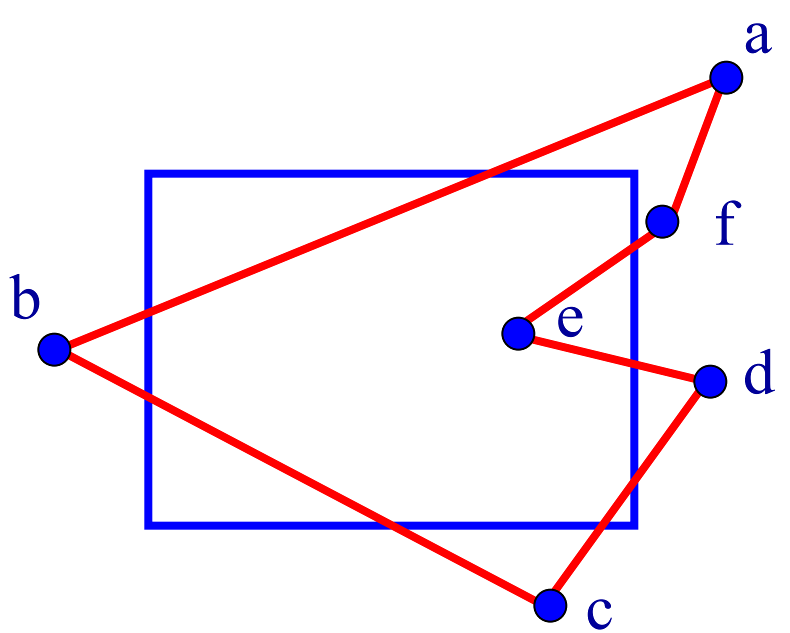


Sutherland-Hodgman Clipping

Input/output dell'algoritmo:

- input: lista dei vertici del poligono nell'ordine

  {a, b, c, d, e, f}



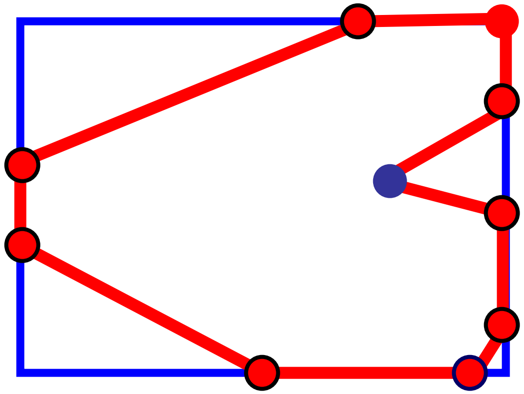
Sutherland-Hodgman Clipping

Input/output dell’algoritmo:

- input: lista dei vertici del poligono nell’ordine

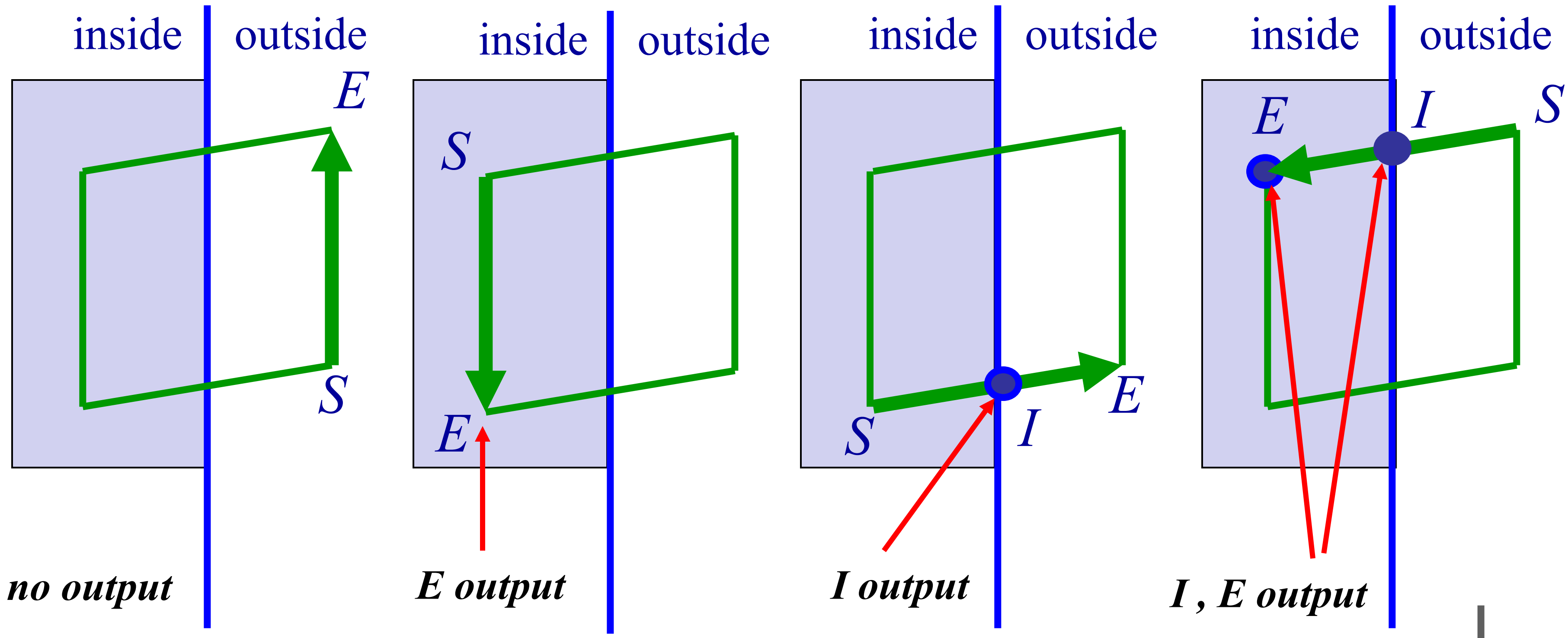
  {a, b, c, d, e, f}

- output: lista dei vertici del poligono, che consiste in alcuni di quelli iniziali e alcuni nuovi

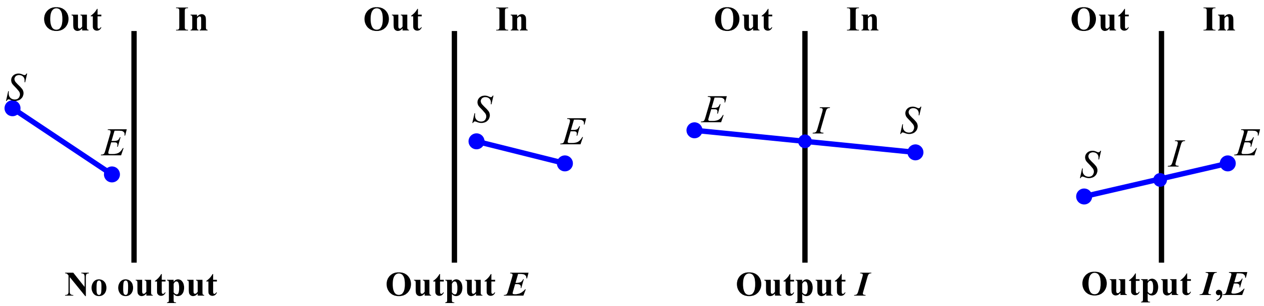


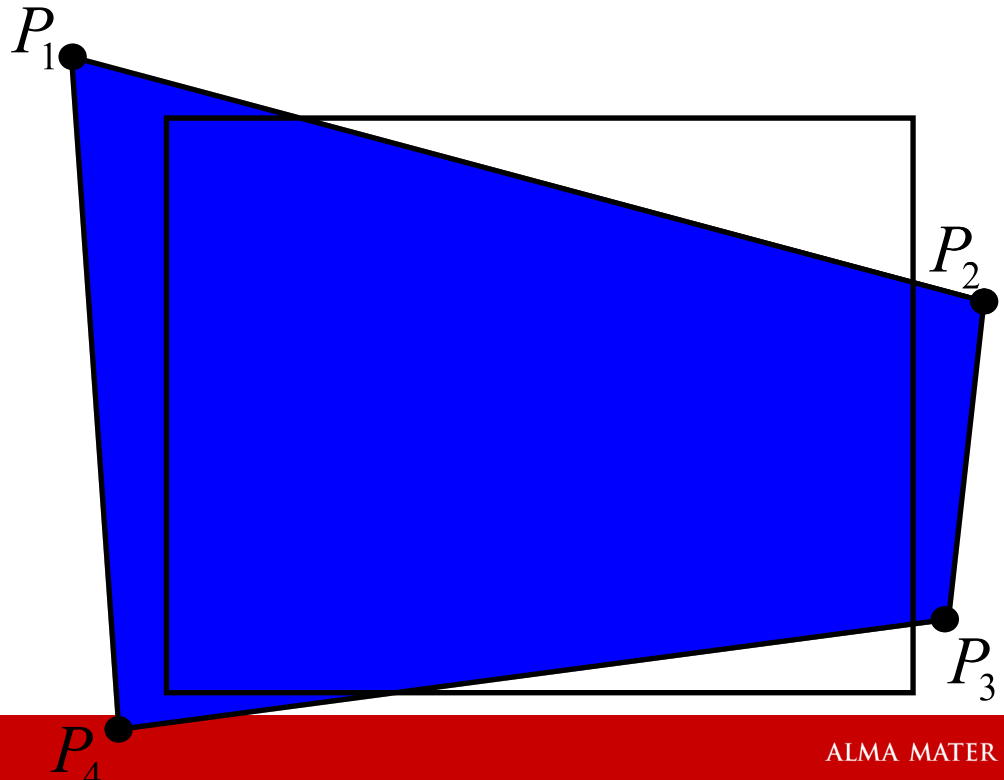
Sutherland-Hodgman Clipping

Il lato che va dal vertice S (Start) al vertice E (End) può trovarsi in uno dei seguenti quattro casi:

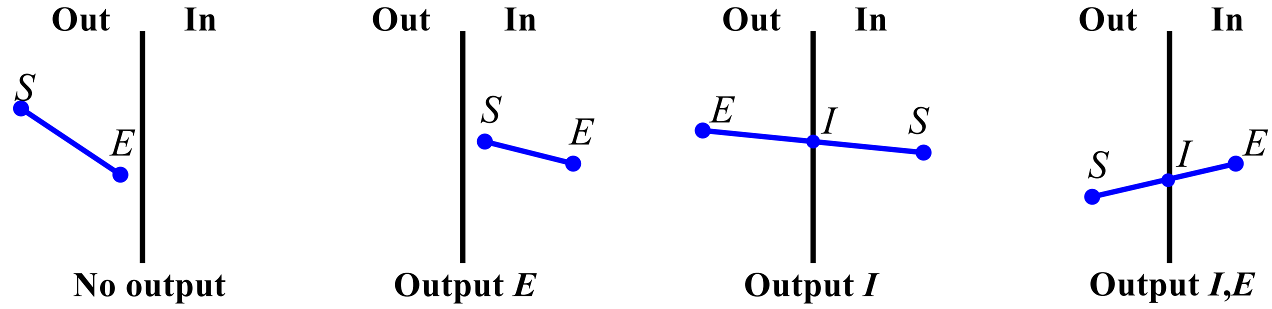


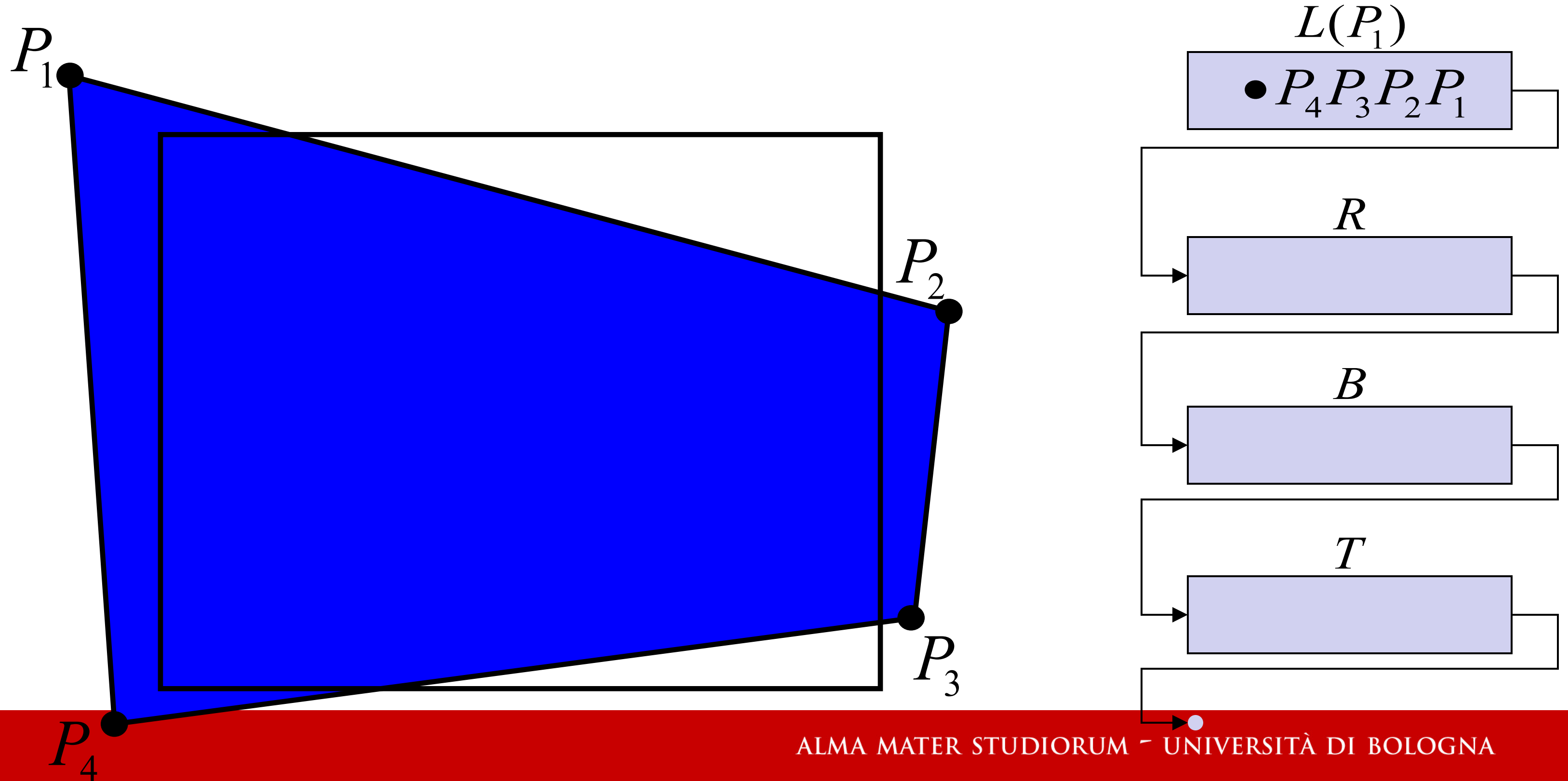
Algoritmo di Sutherland-Hodgman



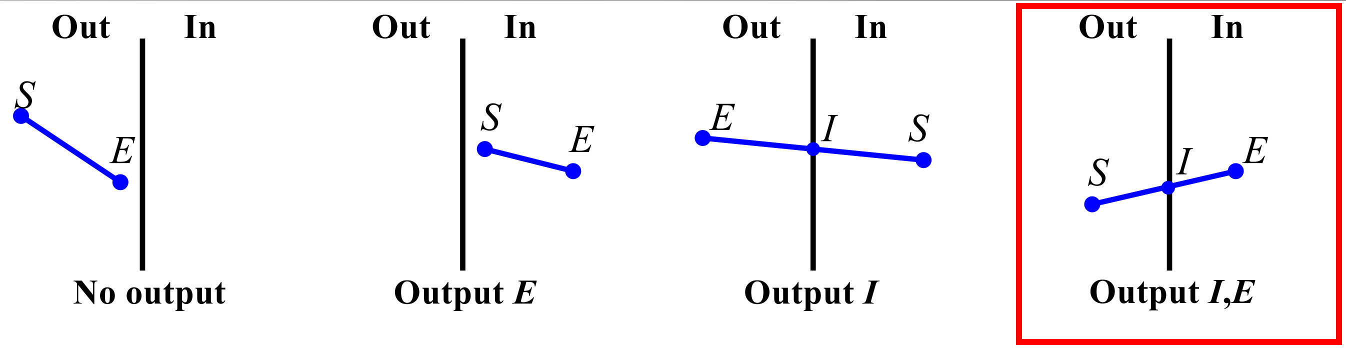


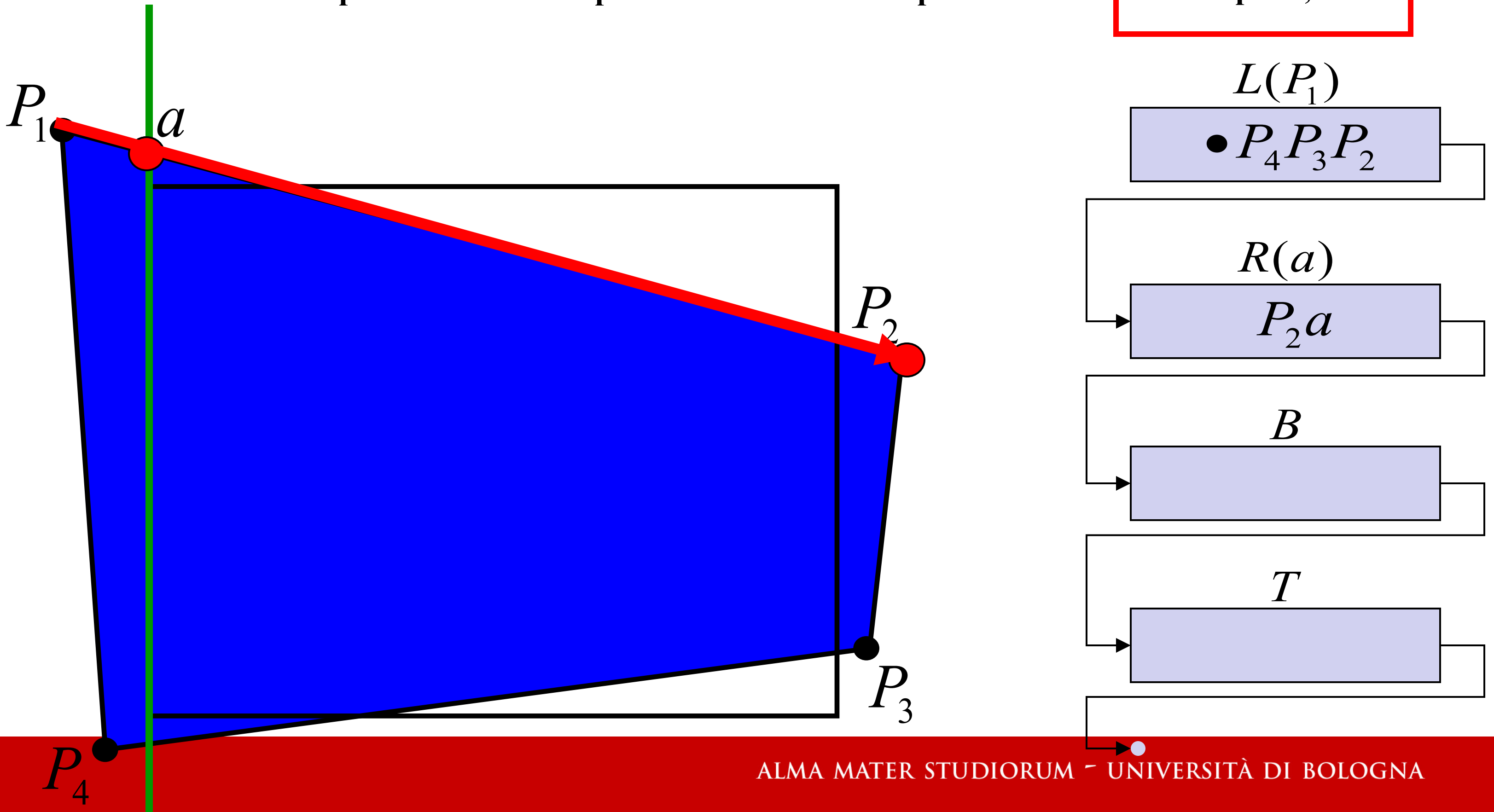
Algoritmo di Sutherland-Hodgman



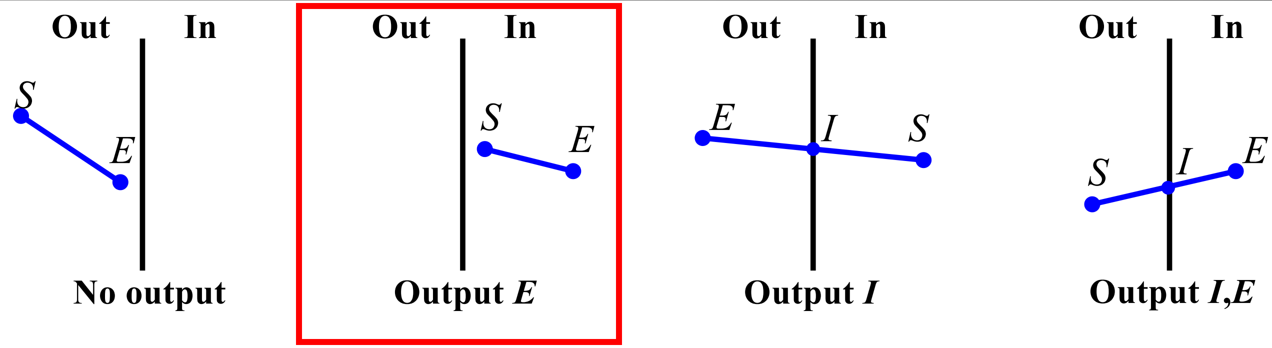


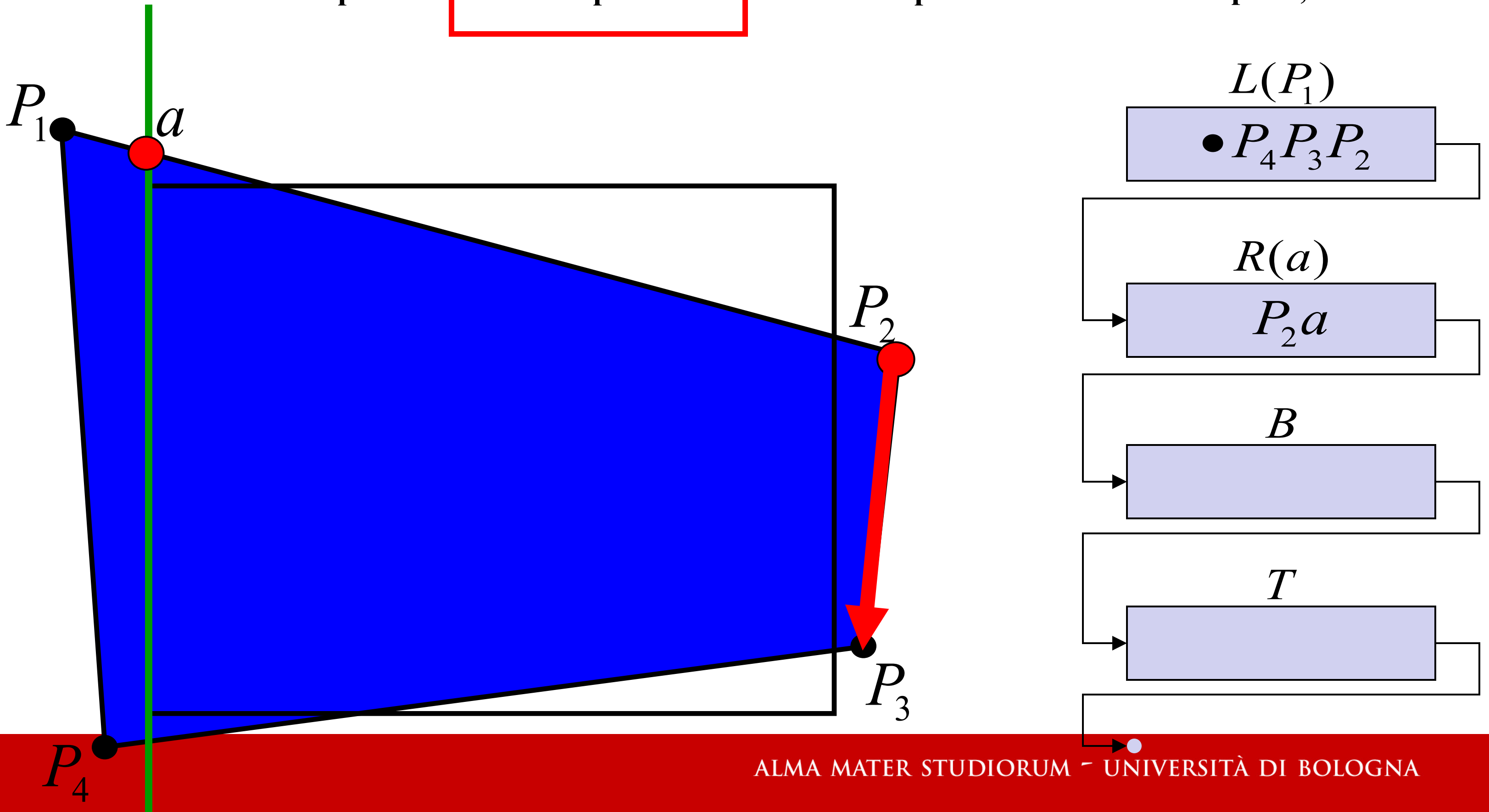
Algoritmo di Sutherland-Hodgman



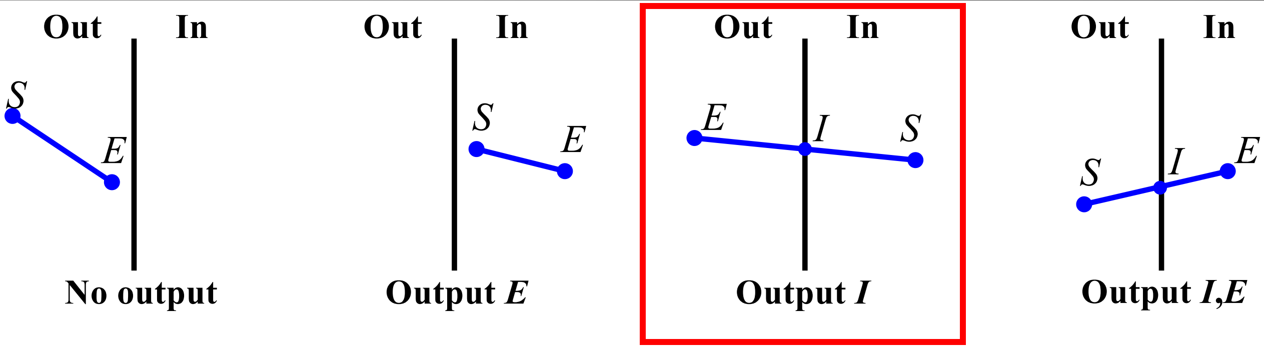


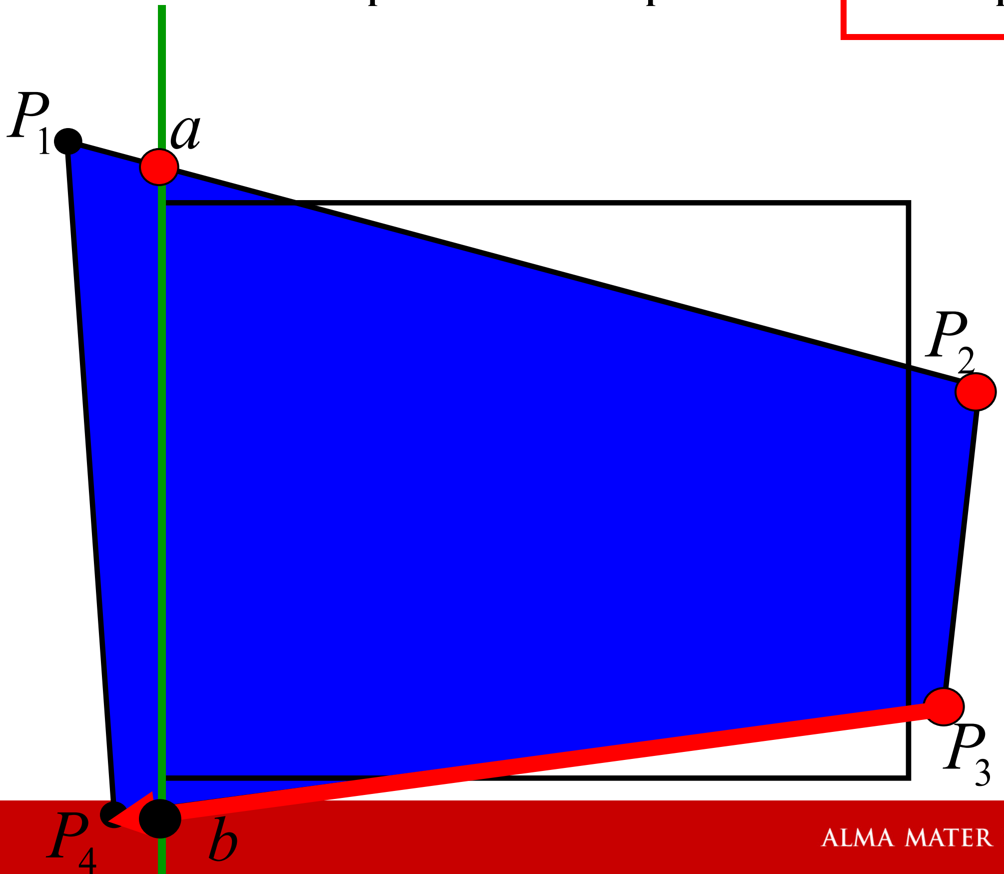
Algoritmo di Sutherland-Hodgman

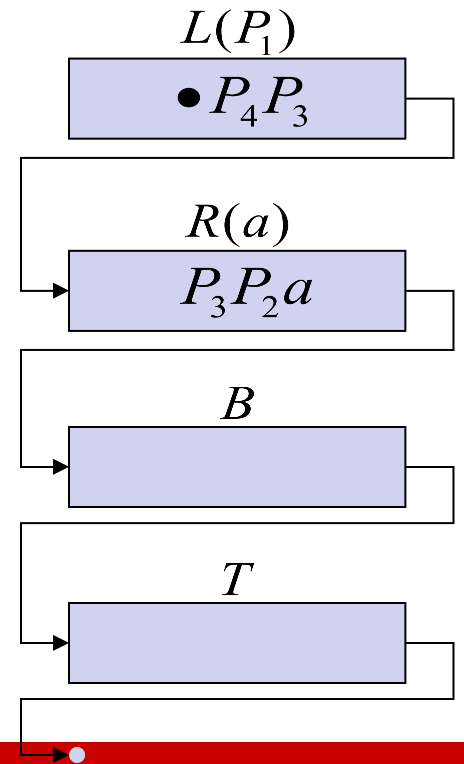




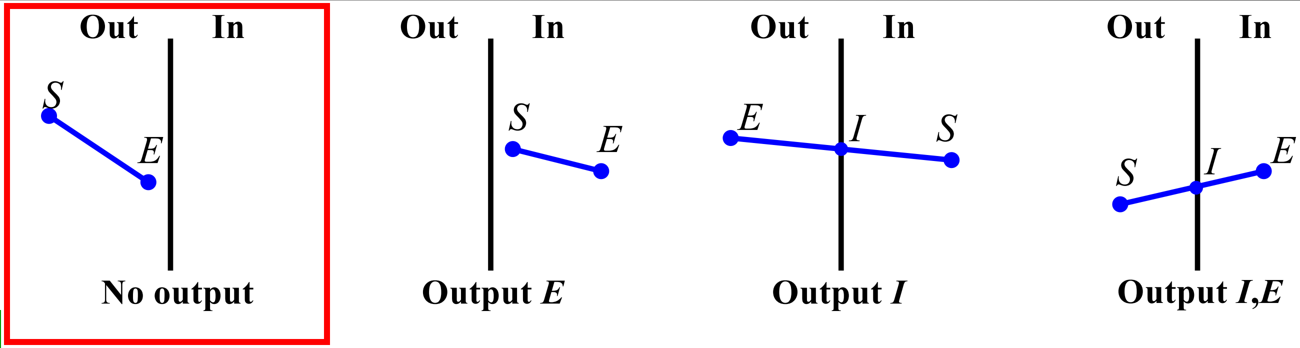
Algoritmo di Sutherland-Hodgman

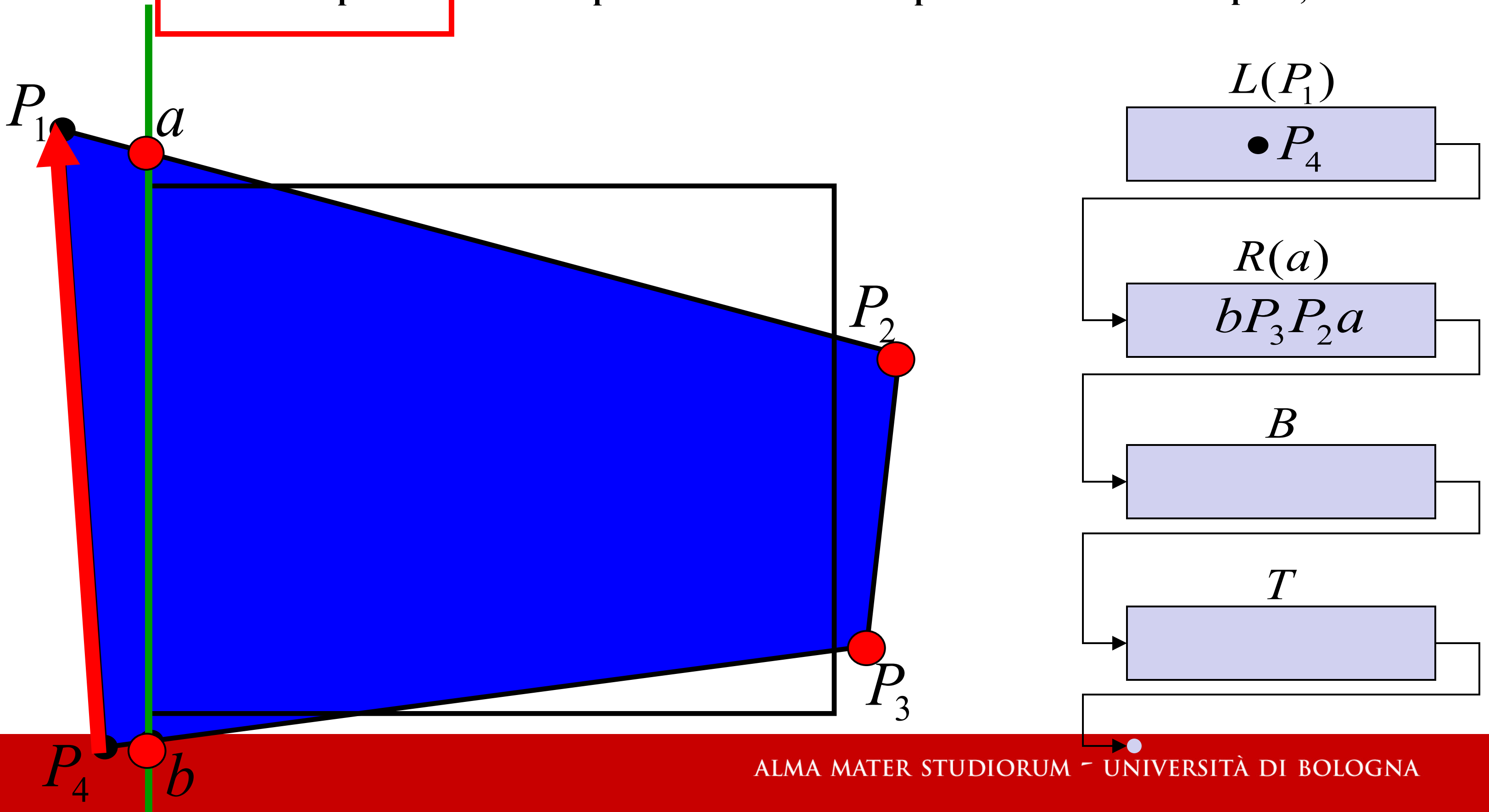




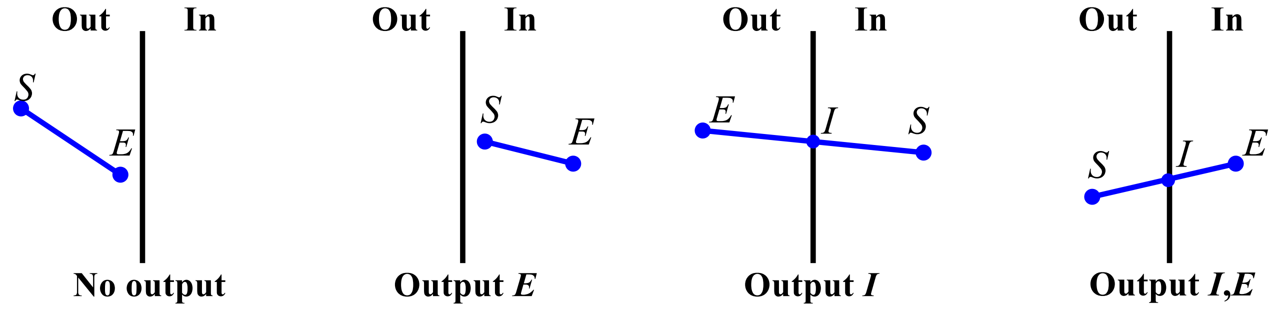


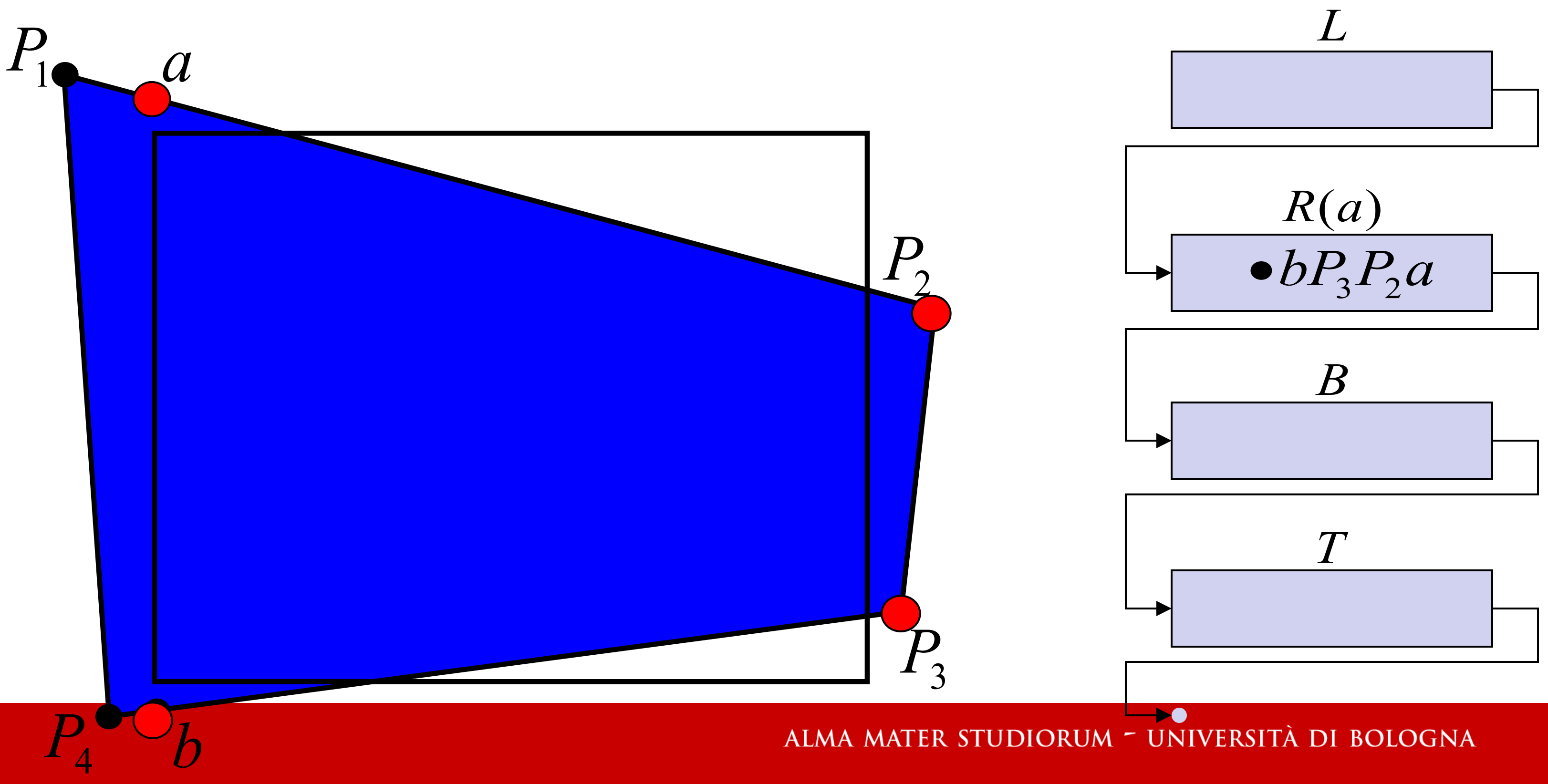
Algoritmo di Sutherland-Hodgman



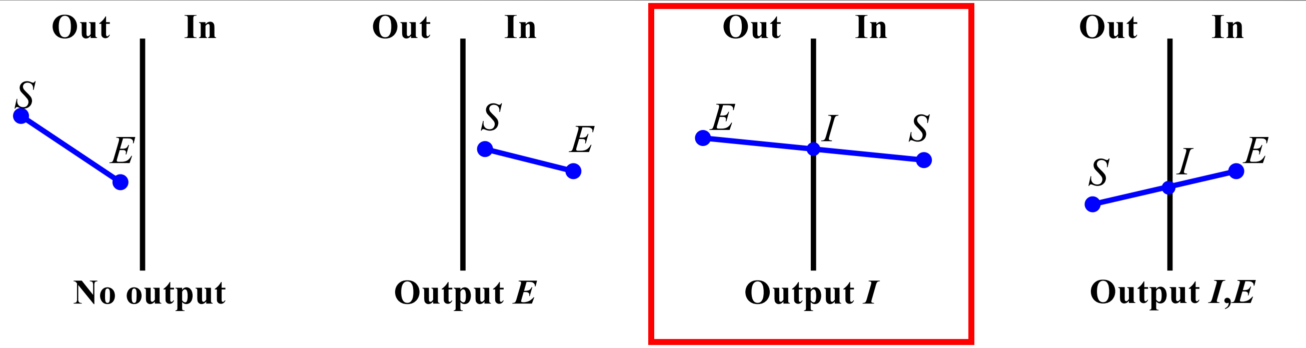


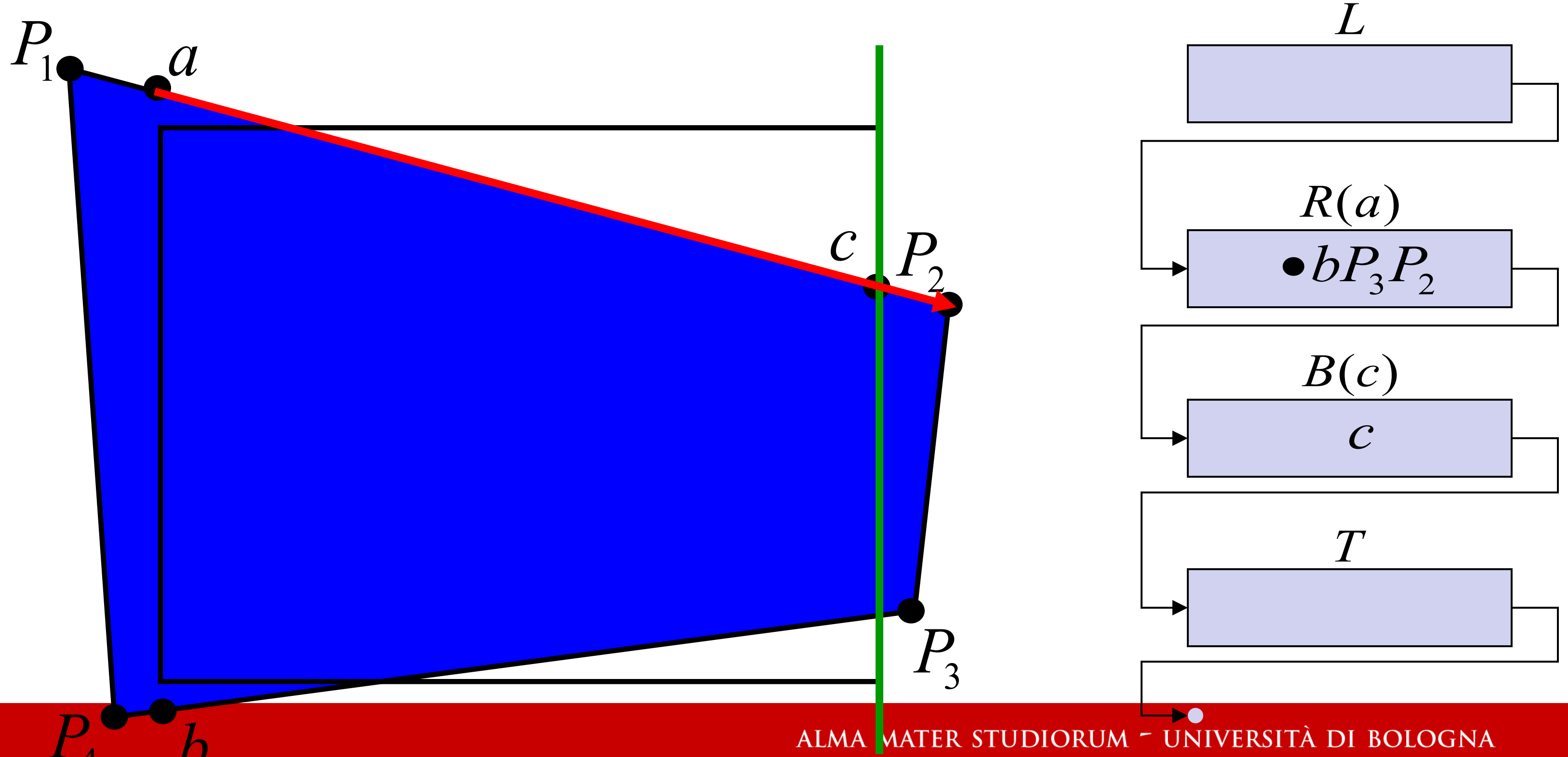
Algoritmo di Sutherland-Hodgman



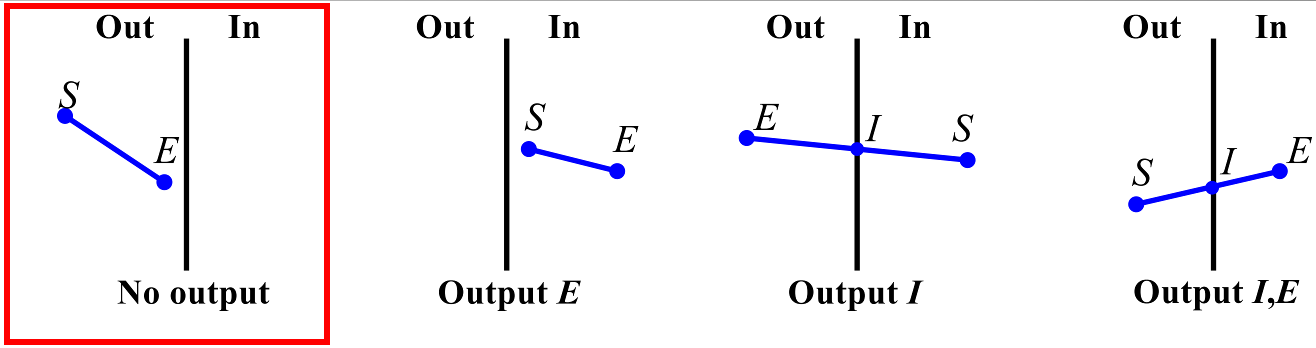


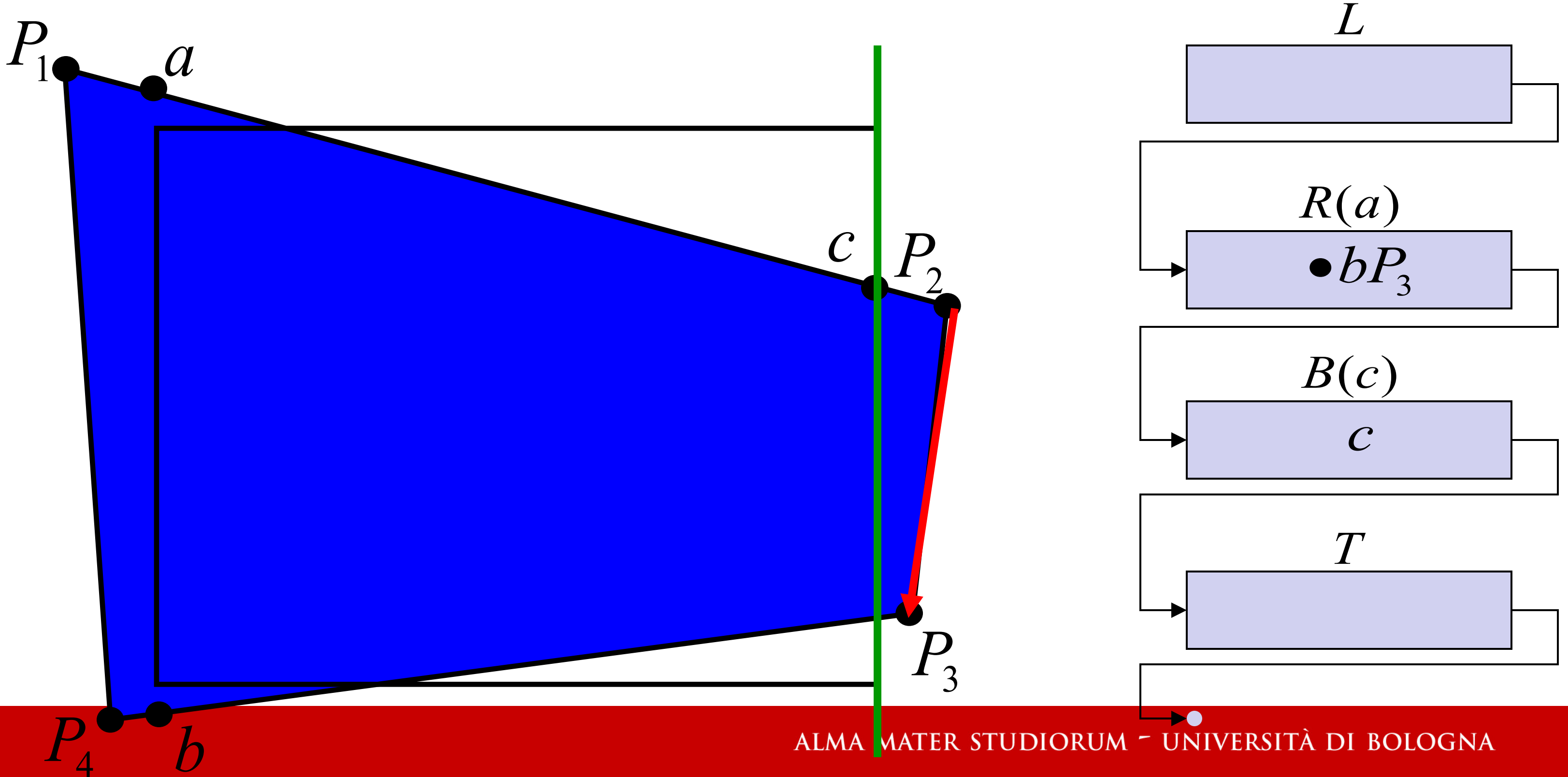
Algoritmo di Sutherland-Hodgman



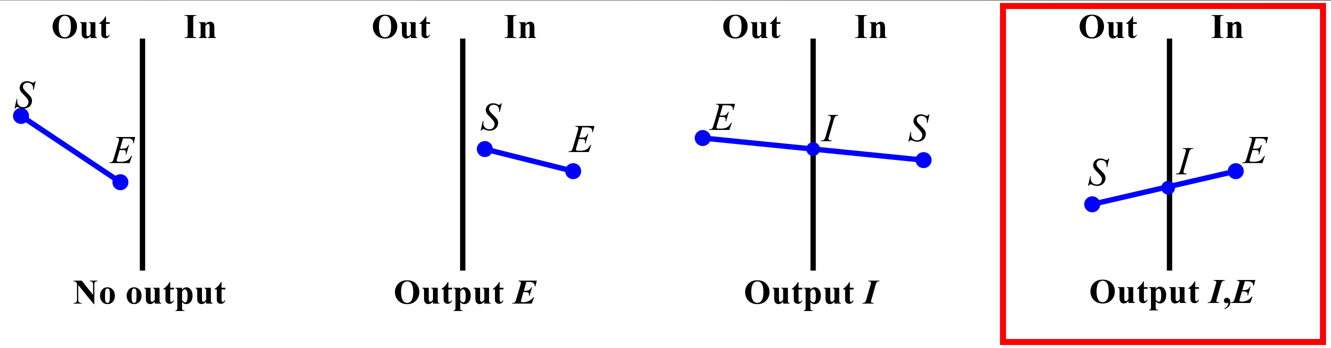


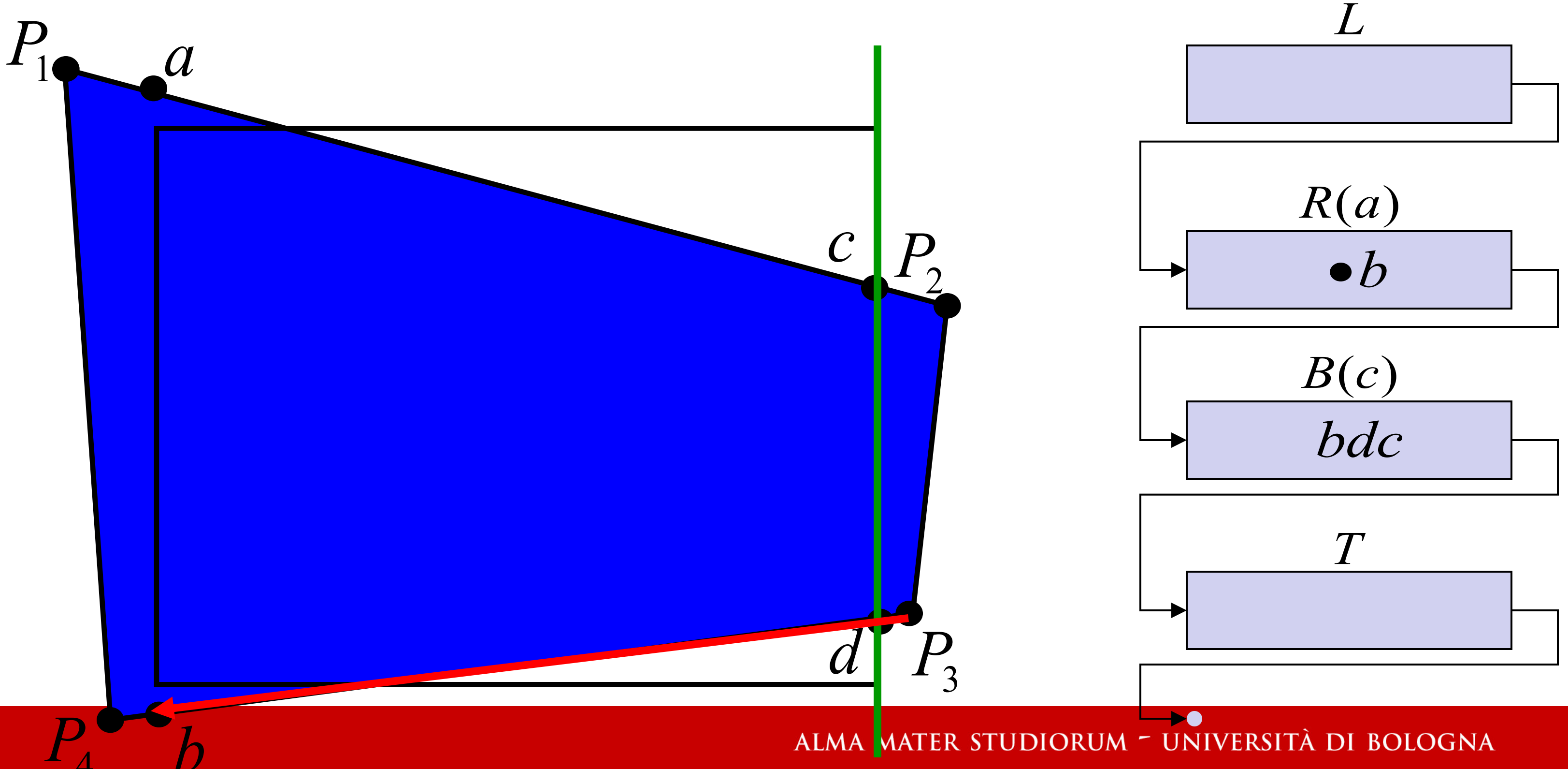
Algoritmo di Sutherland-Hodgman



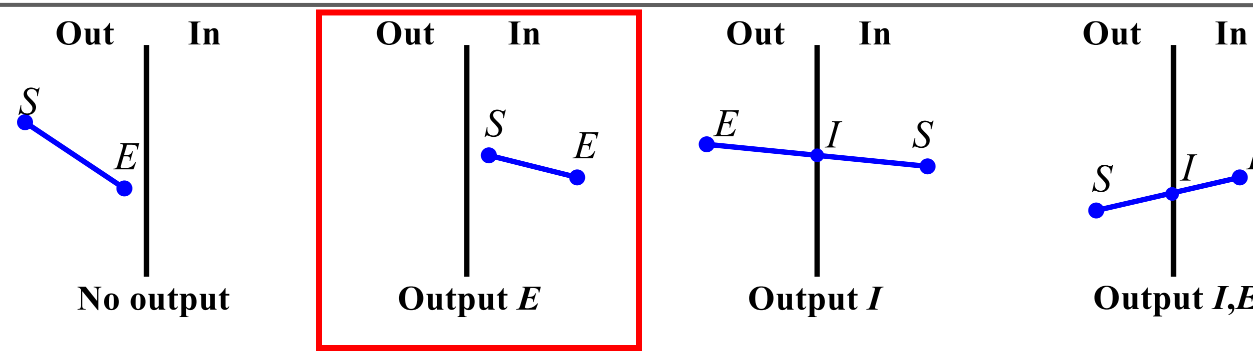


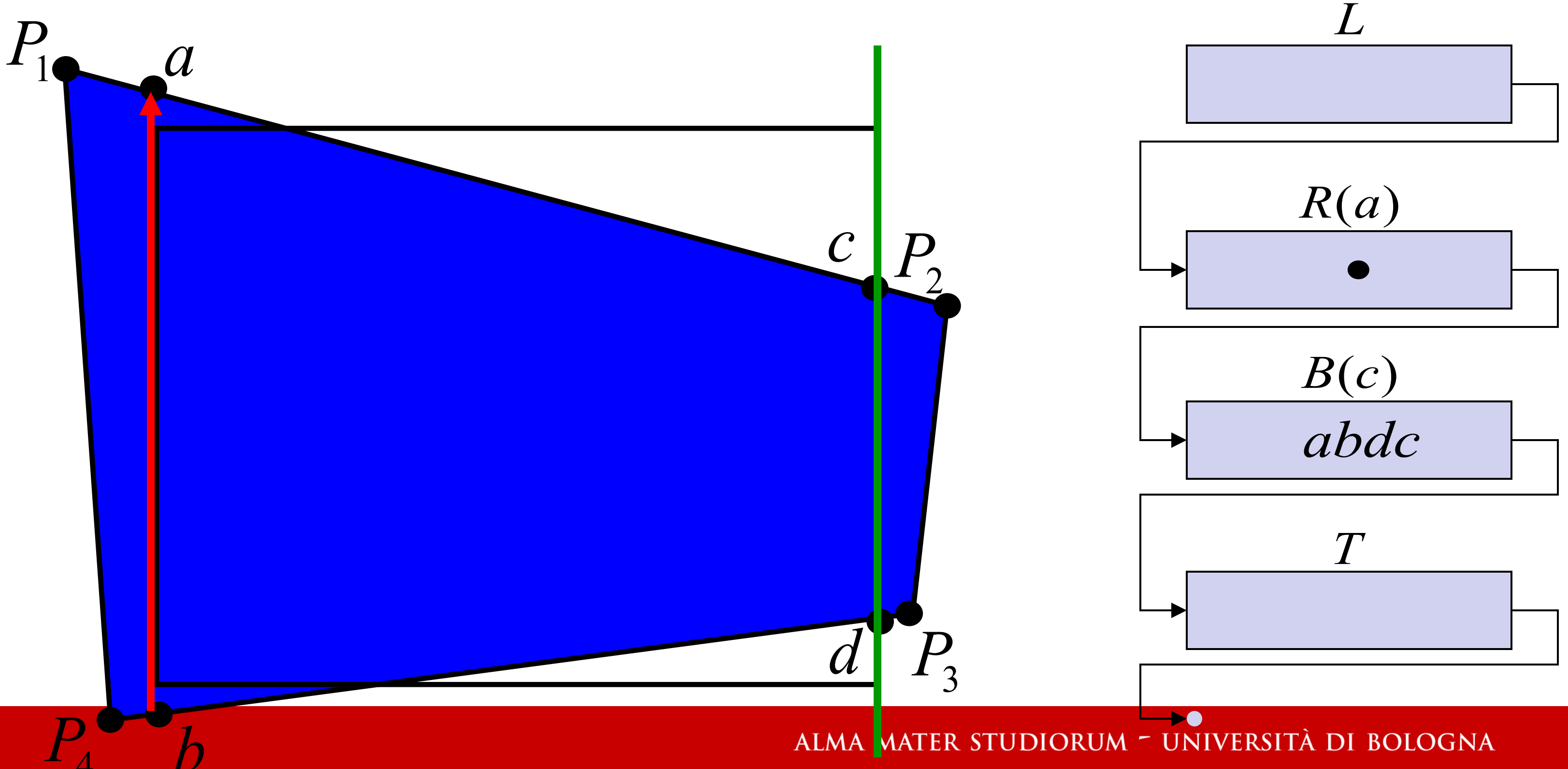
Algoritmo di Sutherland-Hodgman



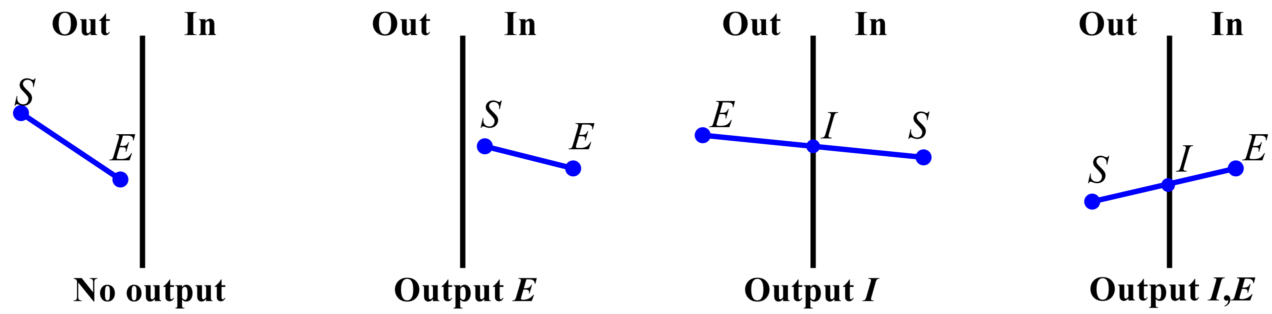


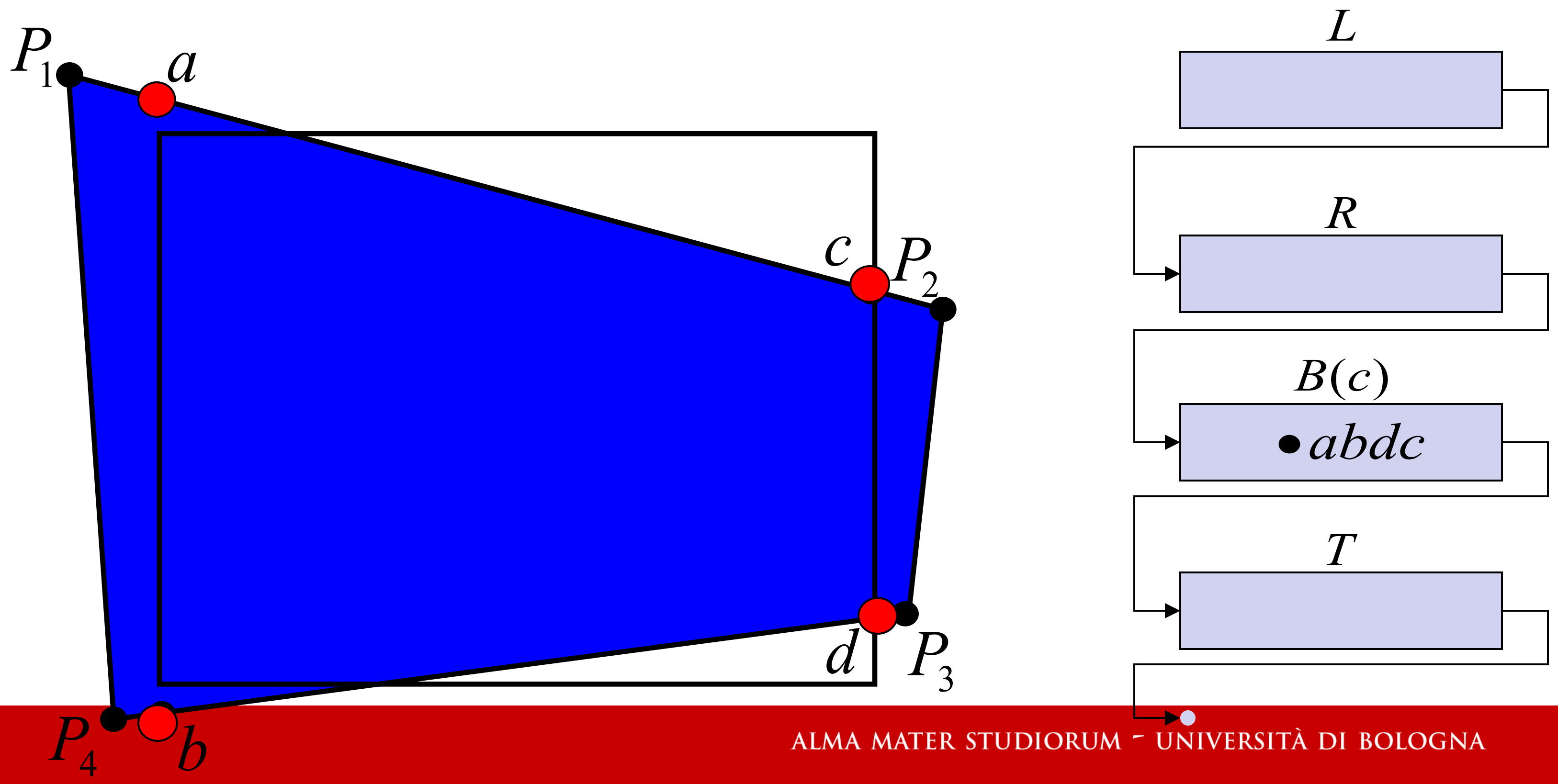
Algoritmo di Sutherland-Hodgman



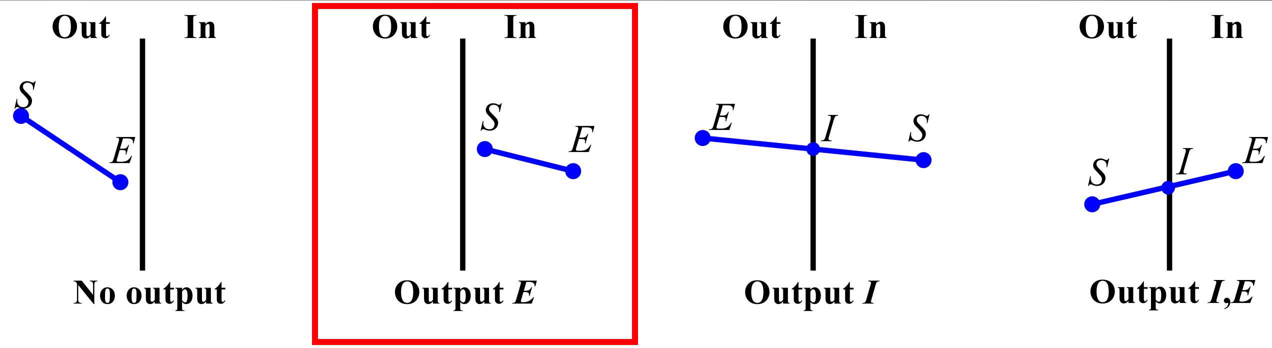


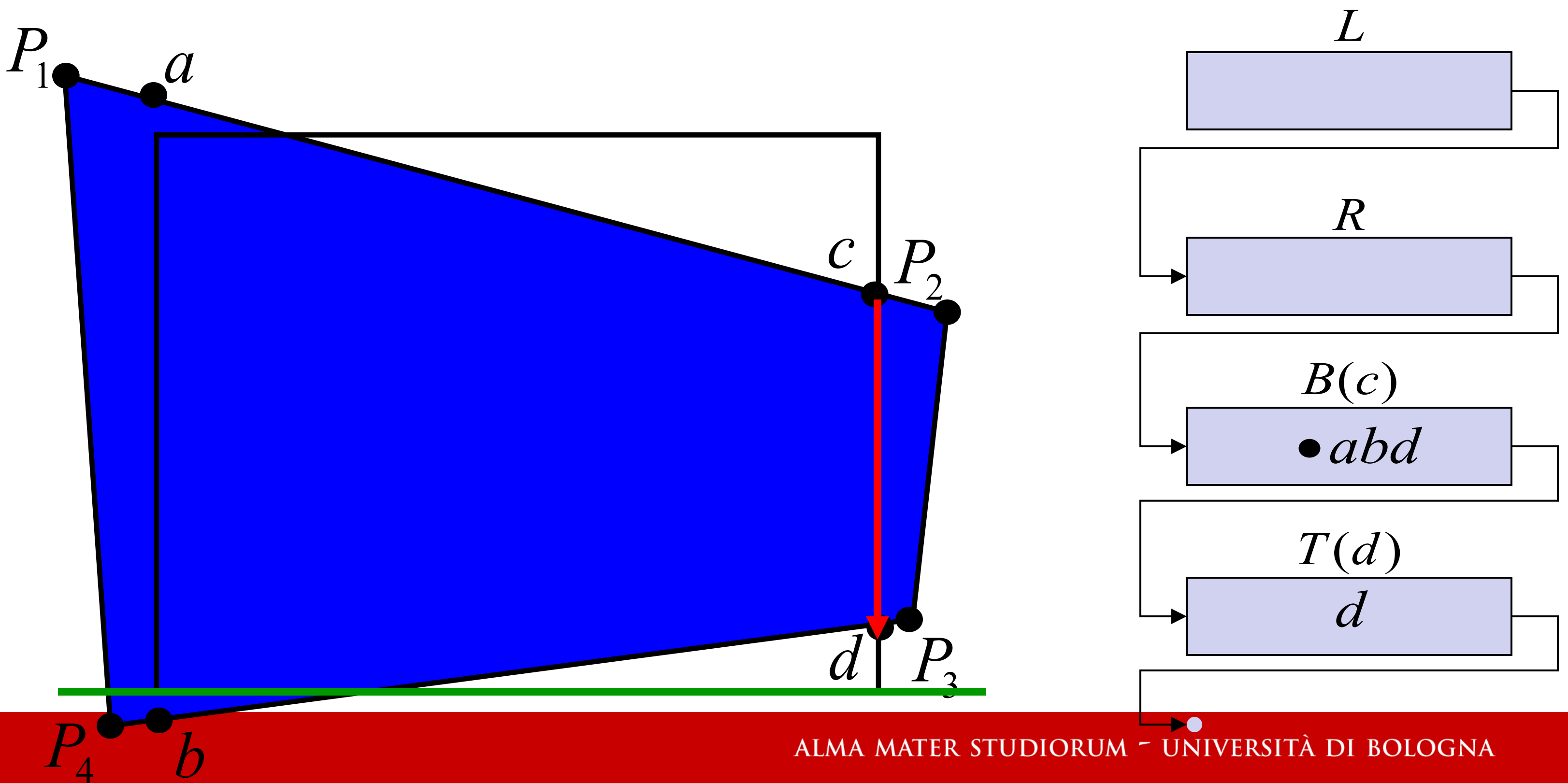
Algoritmo di Sutherland-Hodgman



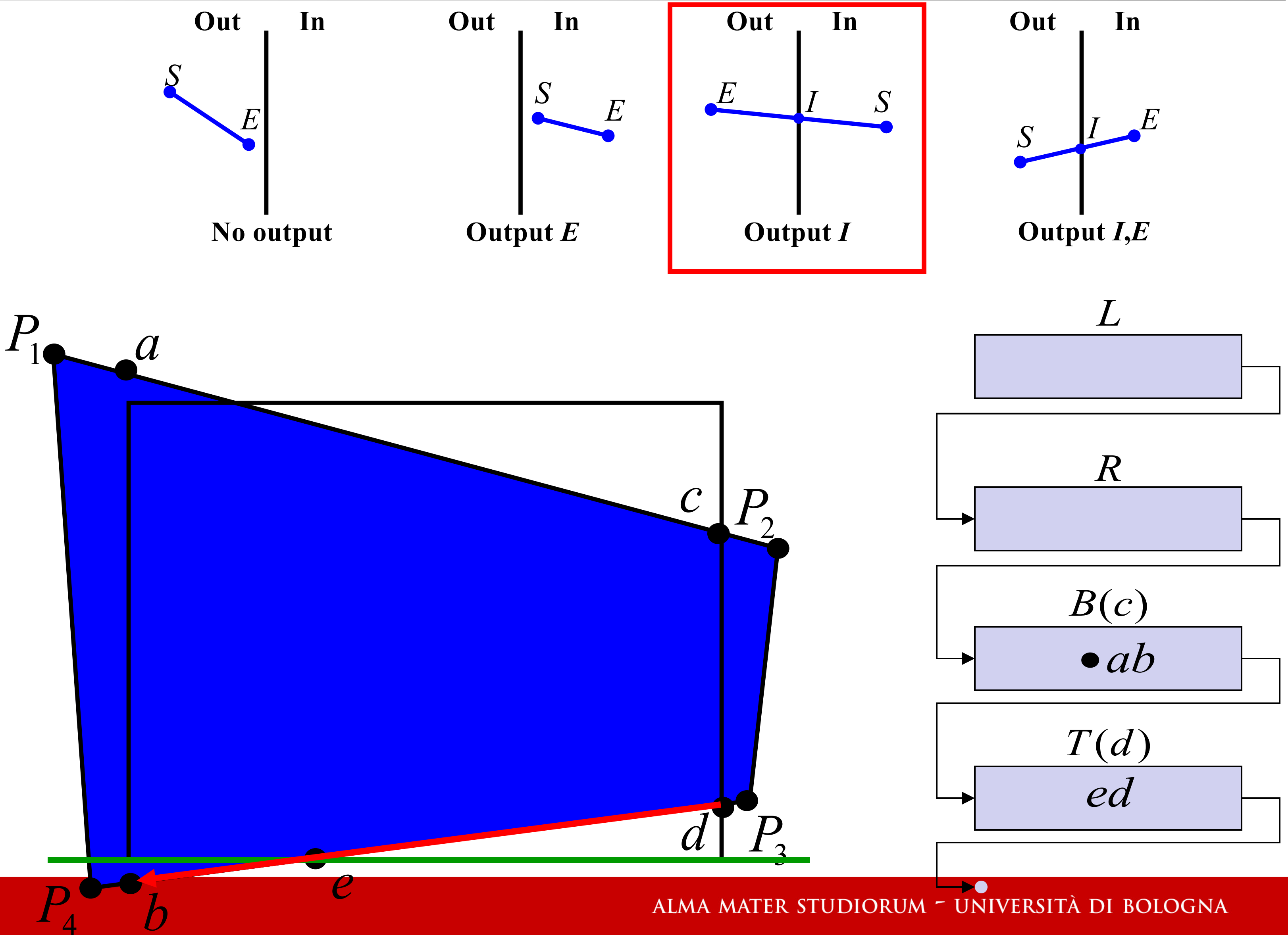


Algoritmo di Sutherland-Hodgman

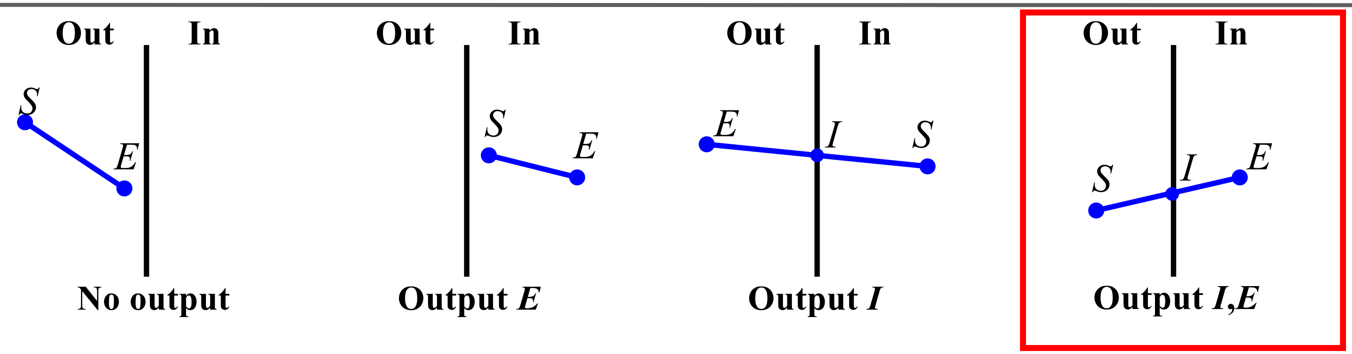


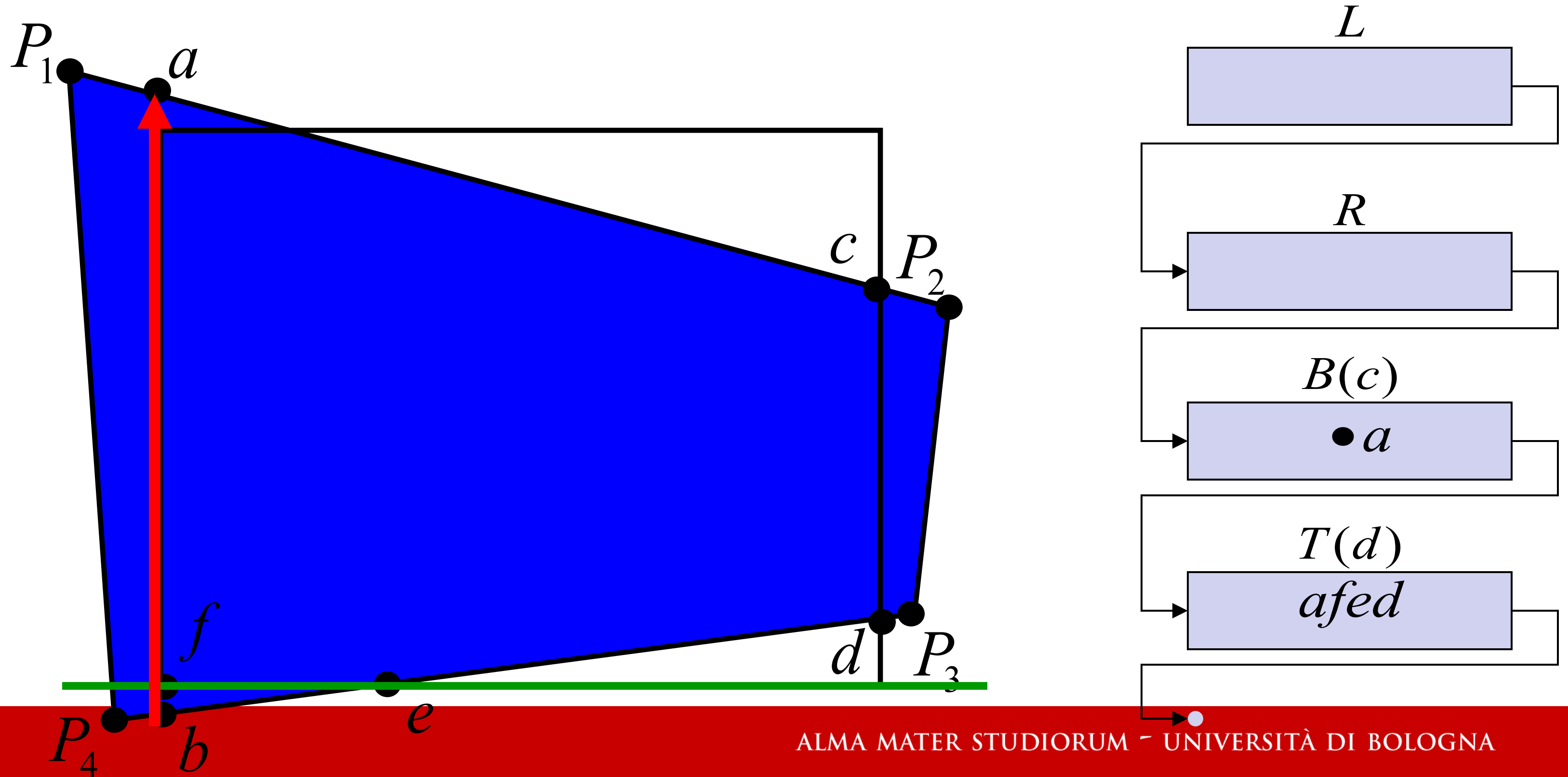


Algoritmo di Sutherland-Hodgman

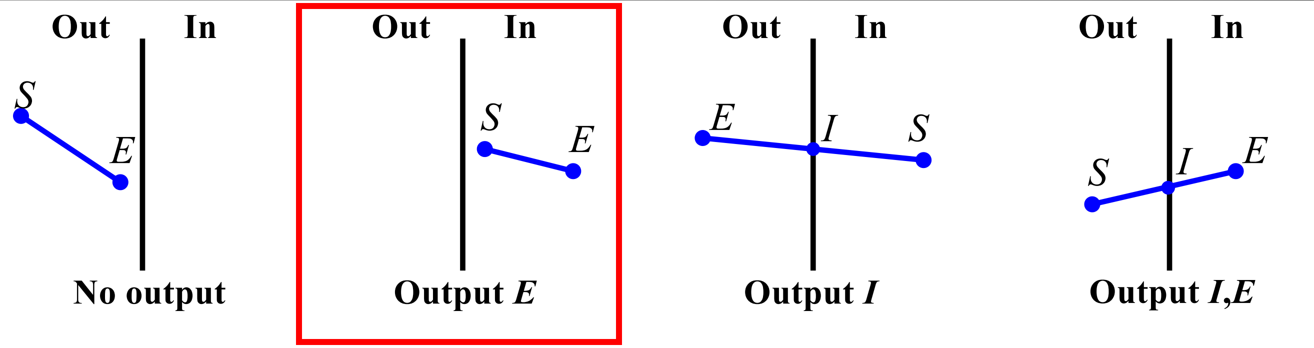


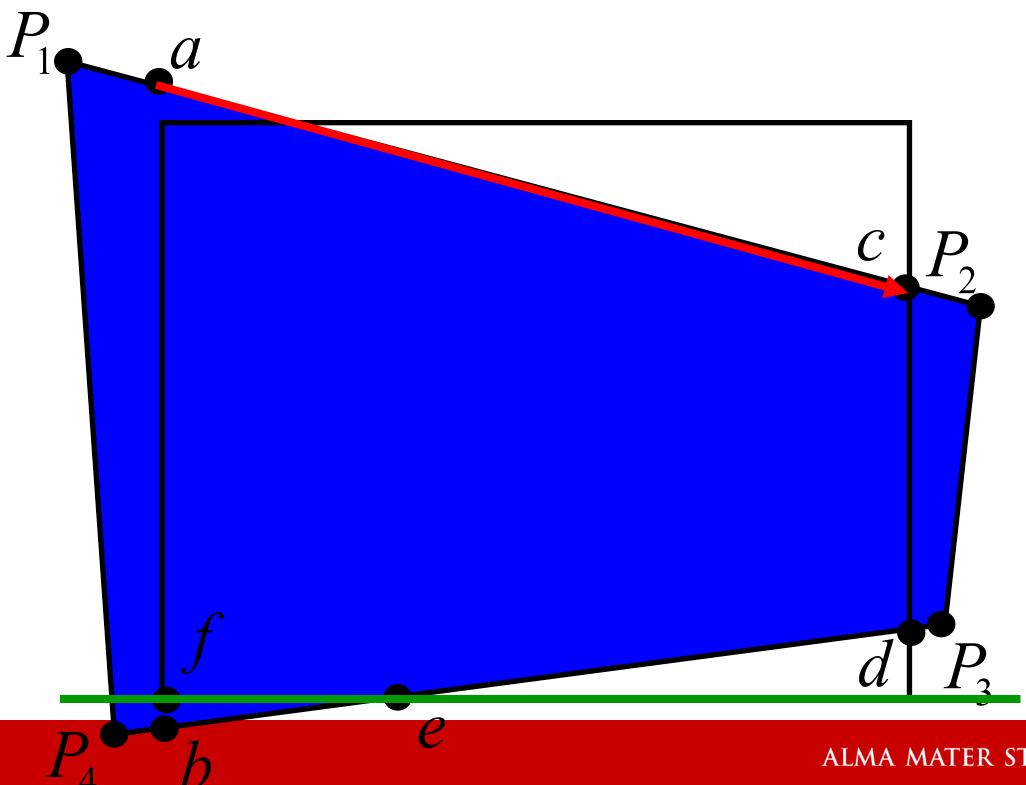
Algoritmo di Sutherland-Hodgman

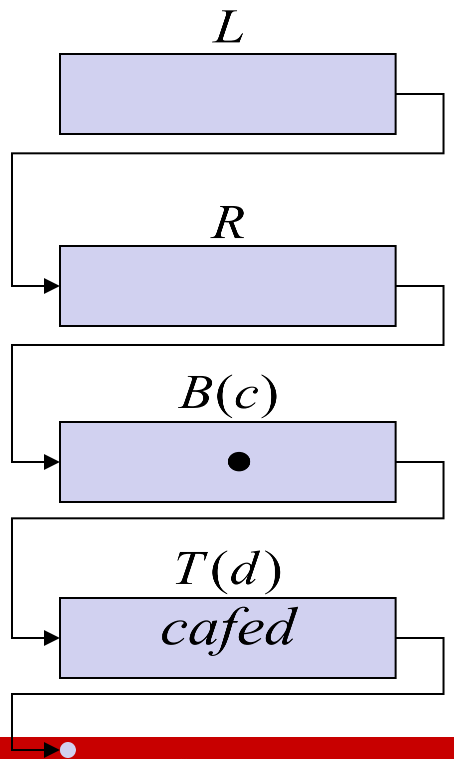




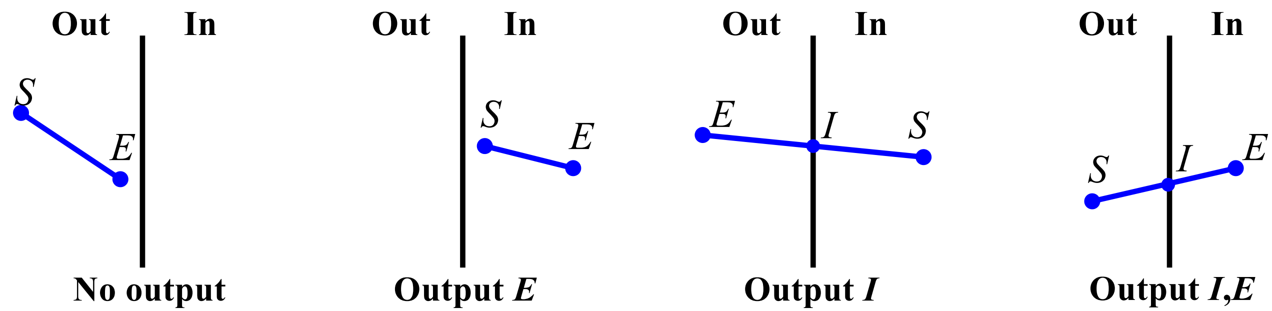
Algoritmo di Sutherland-Hodgman

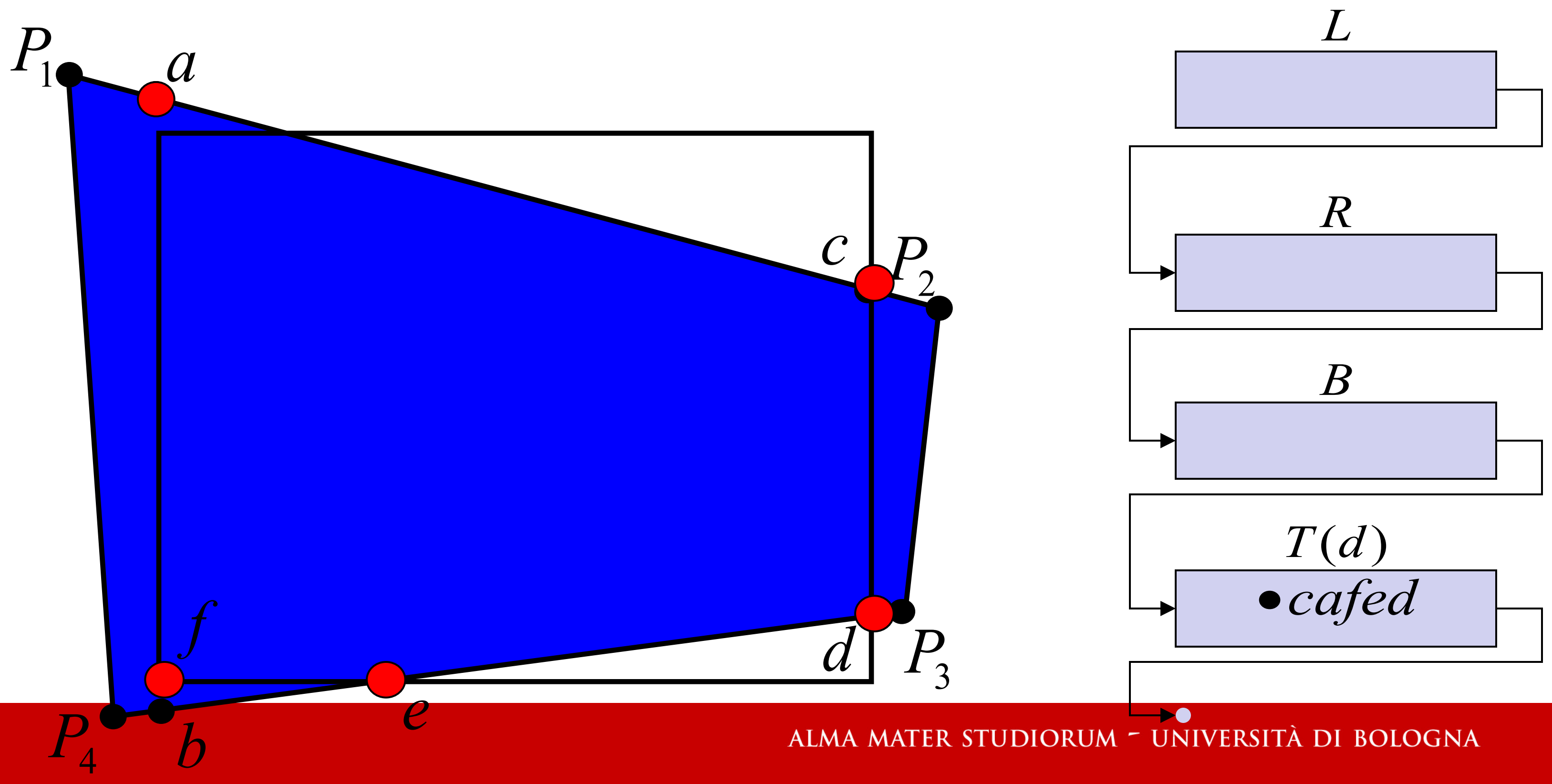




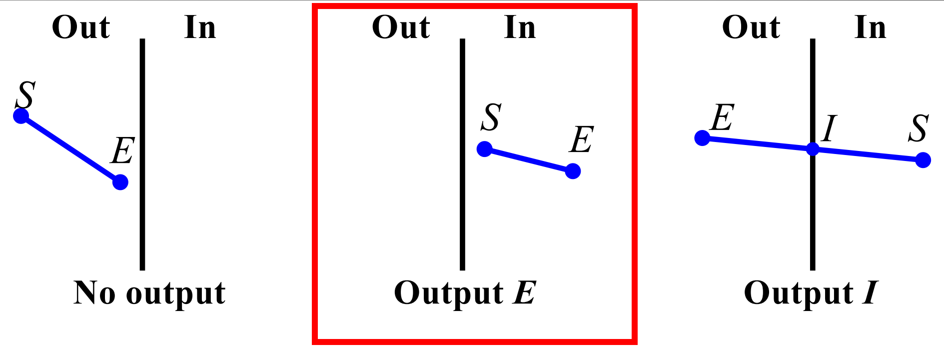


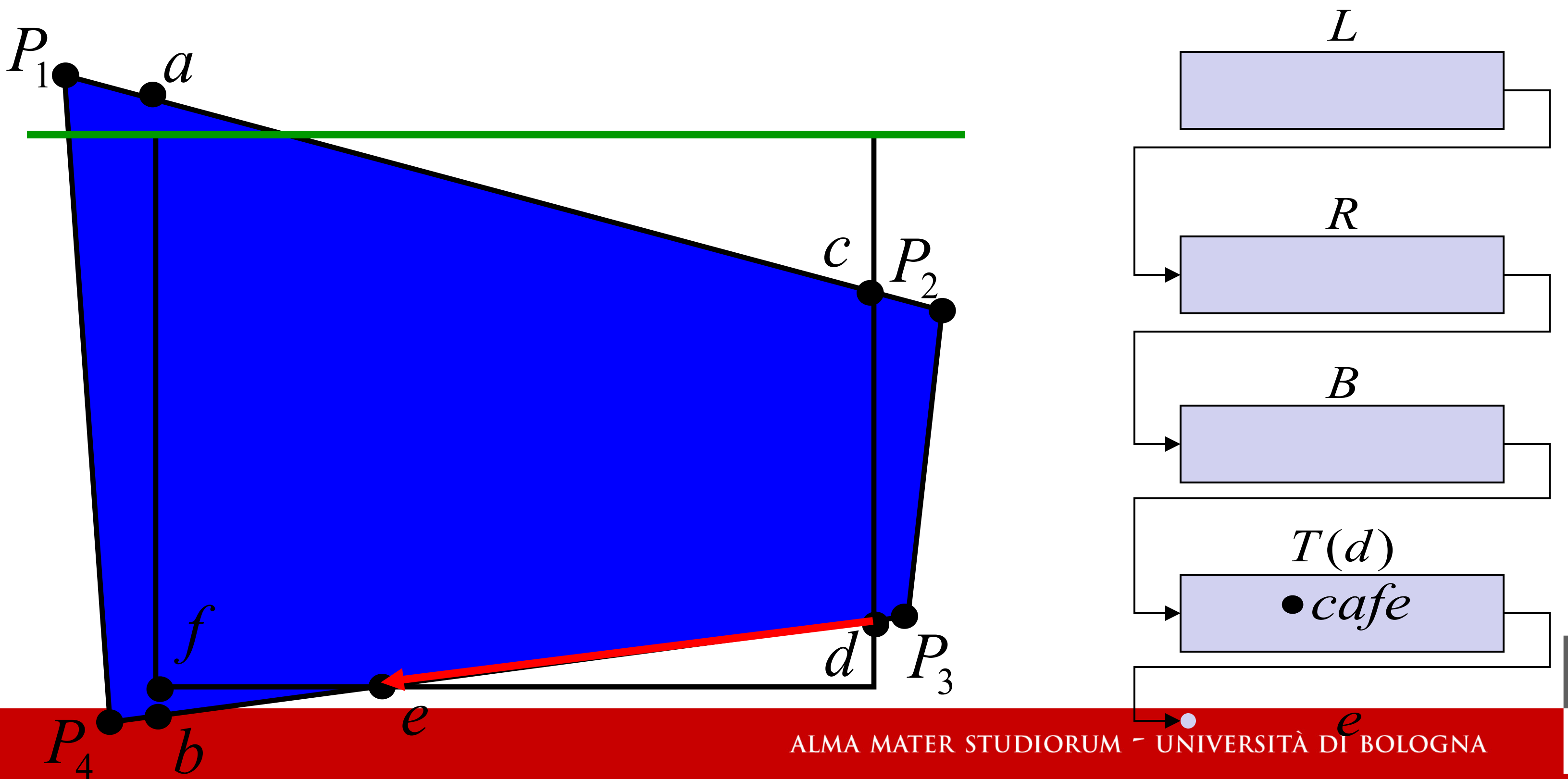
Algoritmo di Sutherland-Hodgman



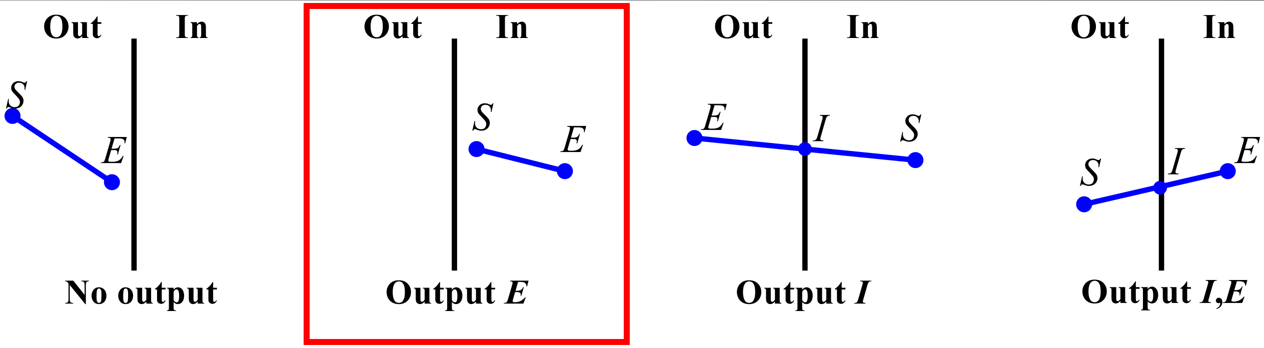


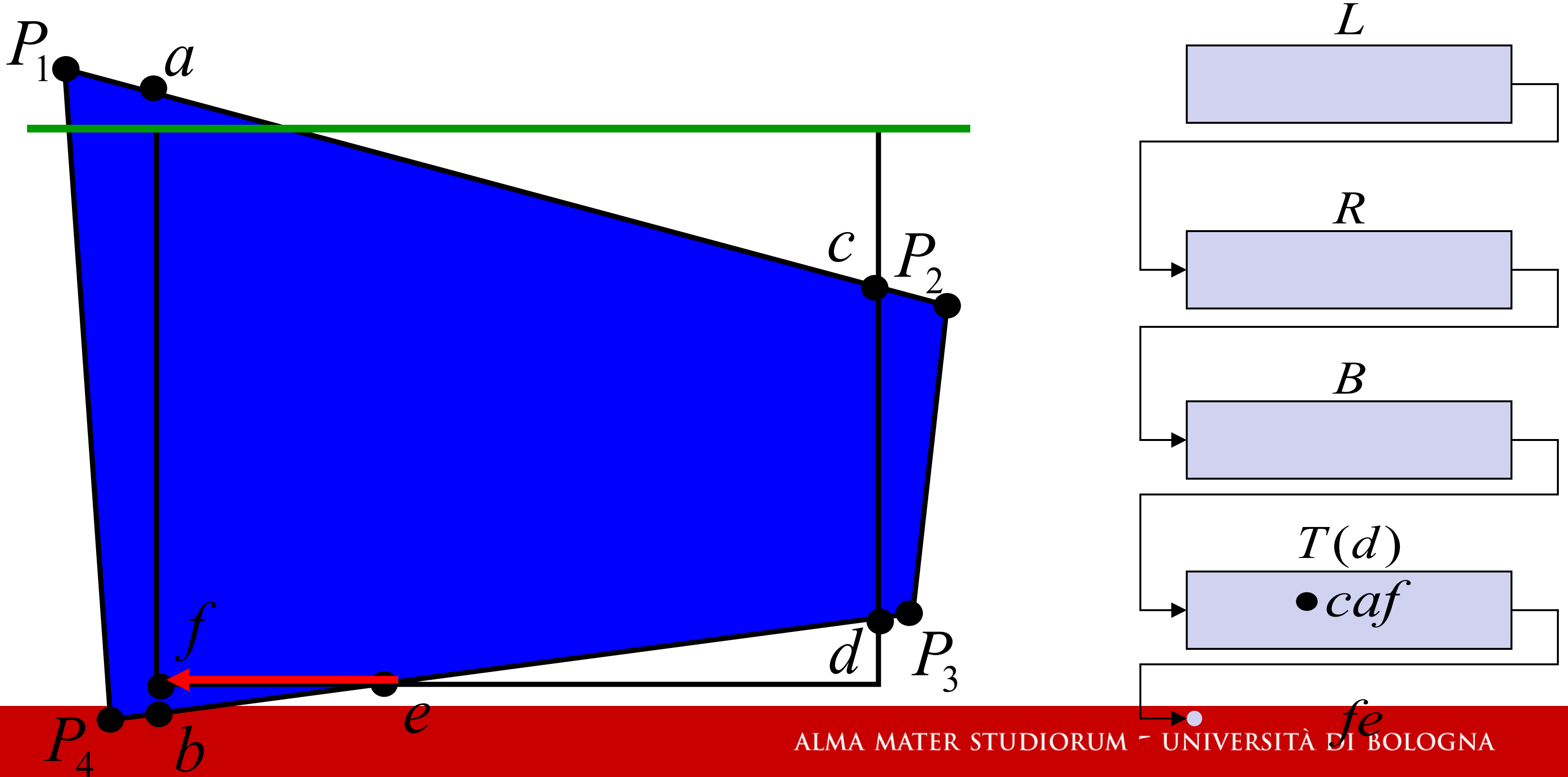
Algoritmo di Sutherland-Hodgman



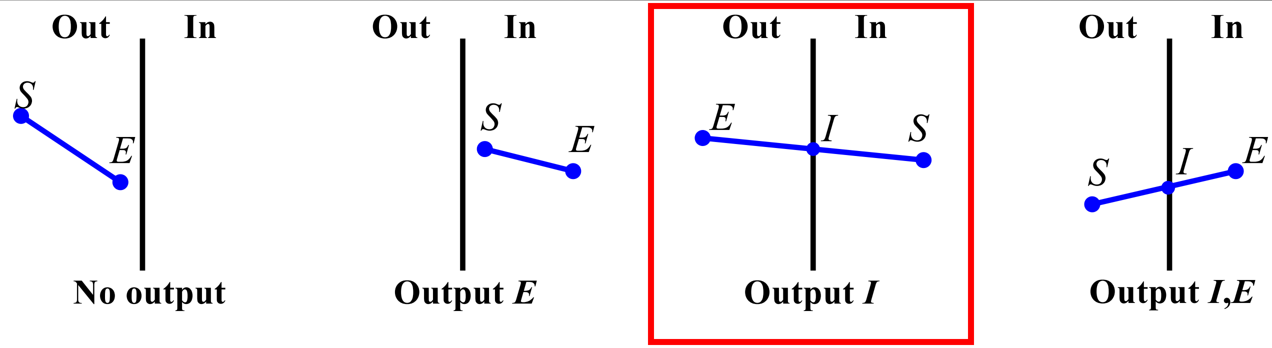


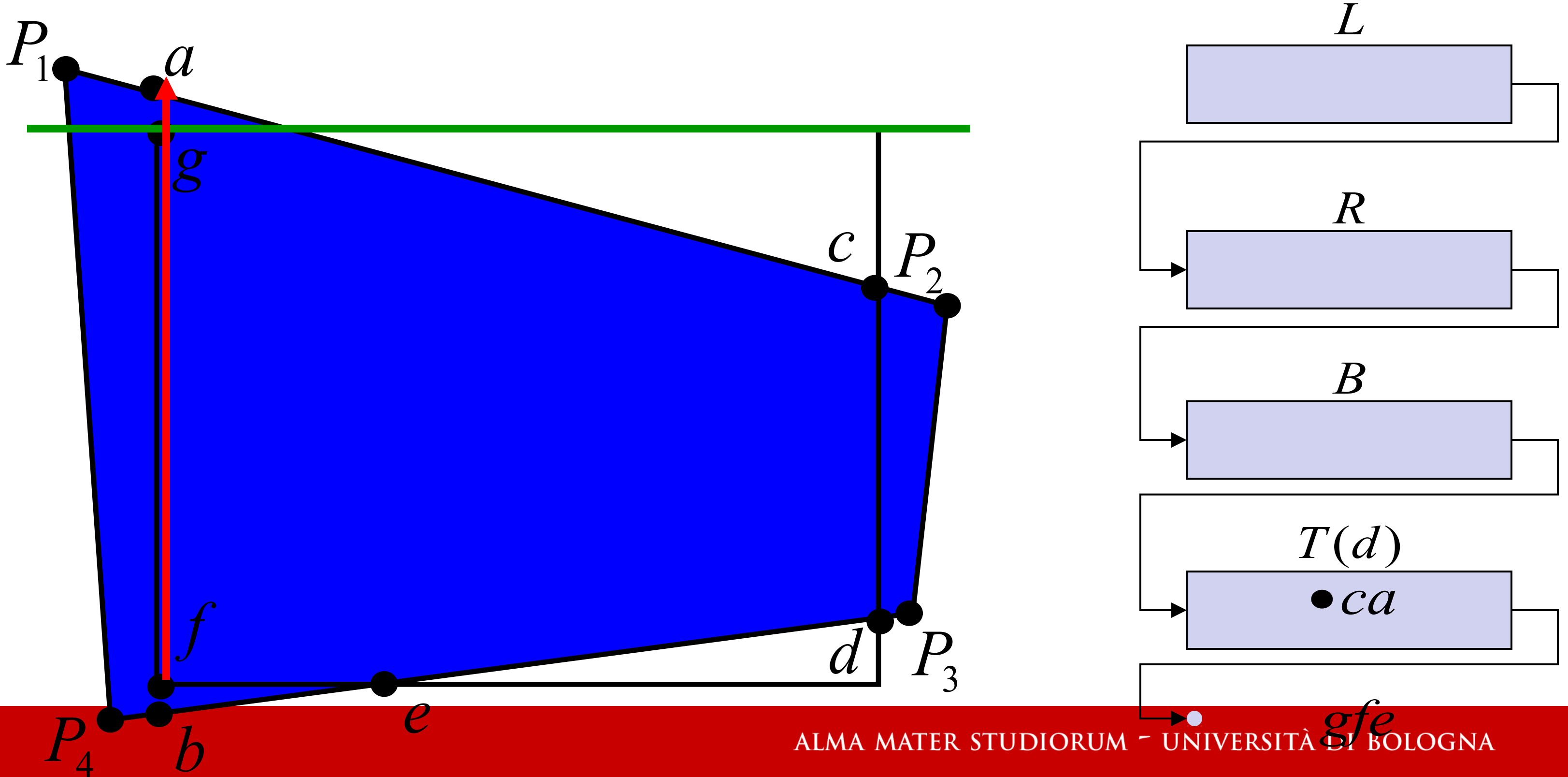
Algoritmo di Sutherland-Hodgman



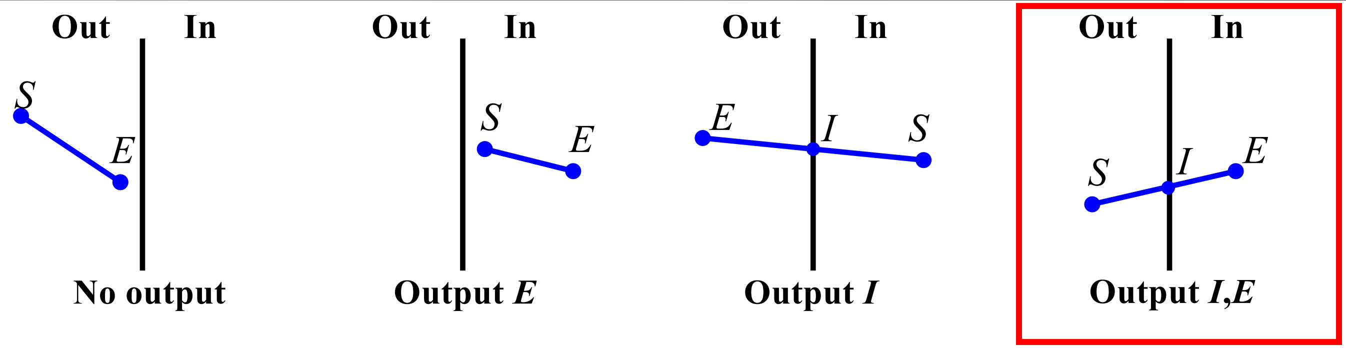


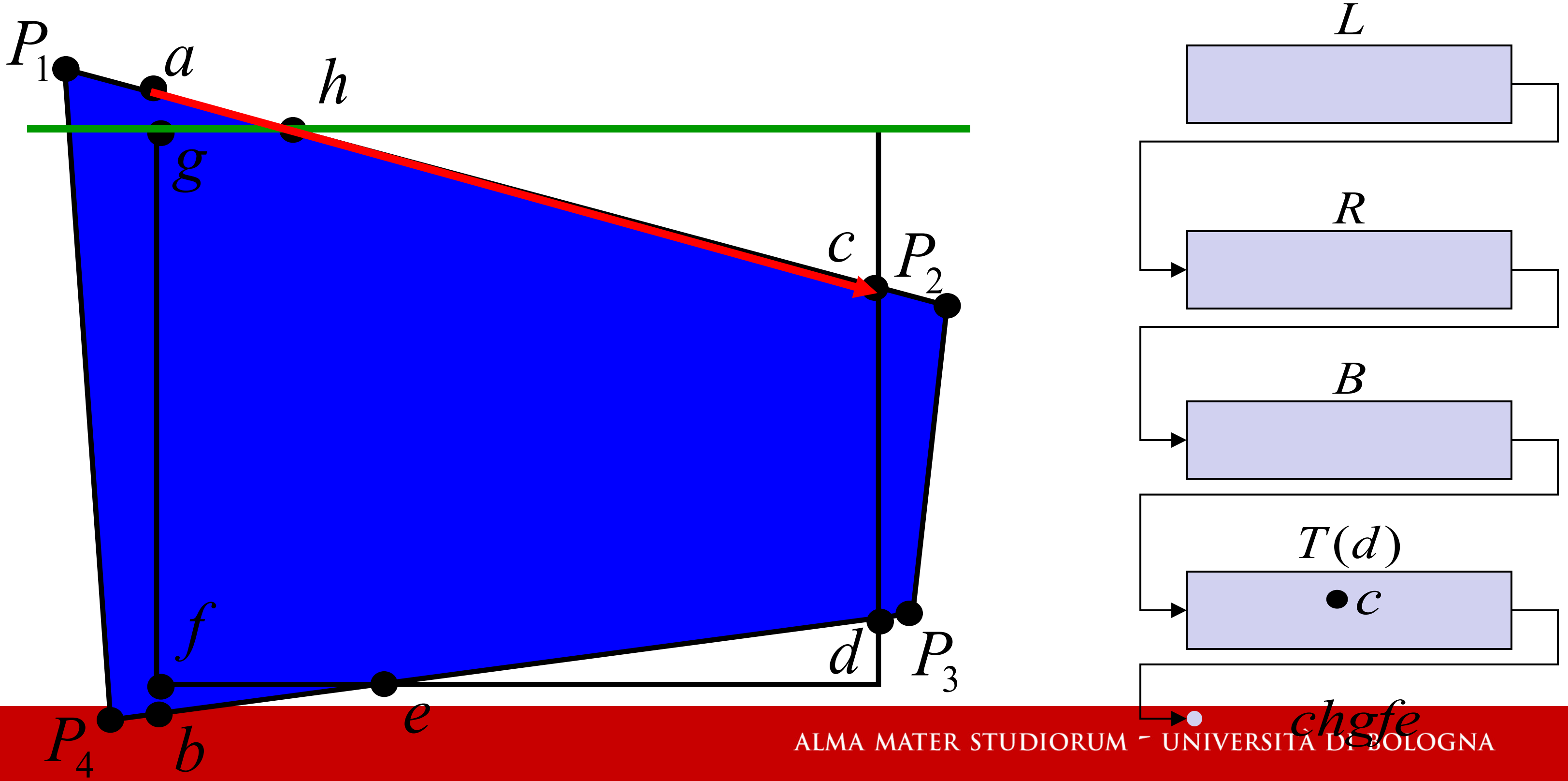
Algoritmo di Sutherland-Hodgman



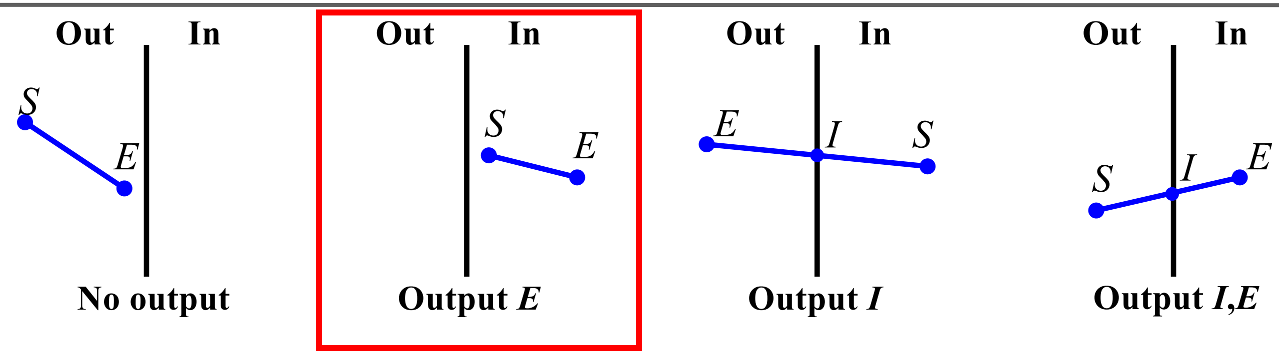


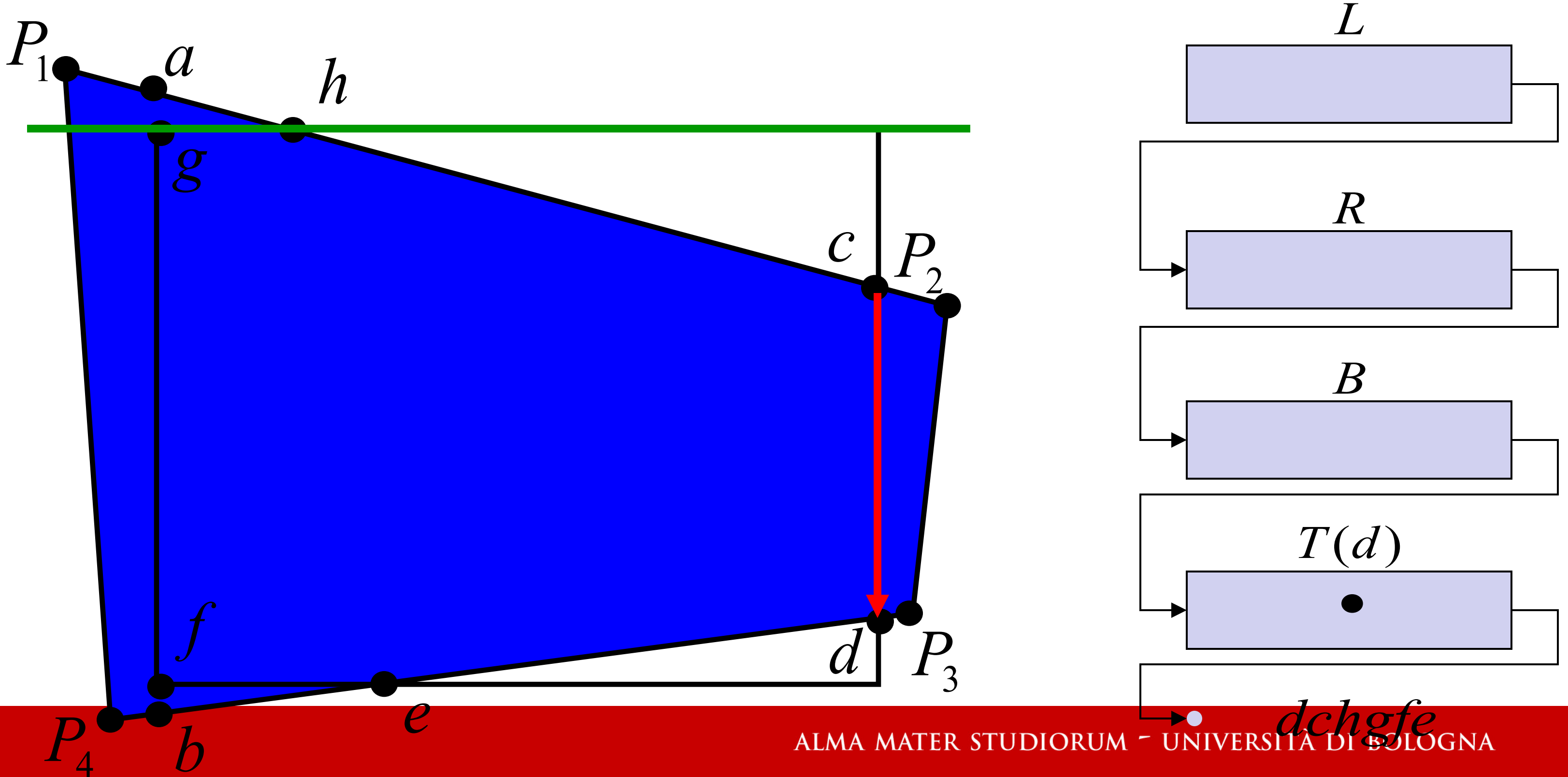
Algoritmo di Sutherland-Hodgman



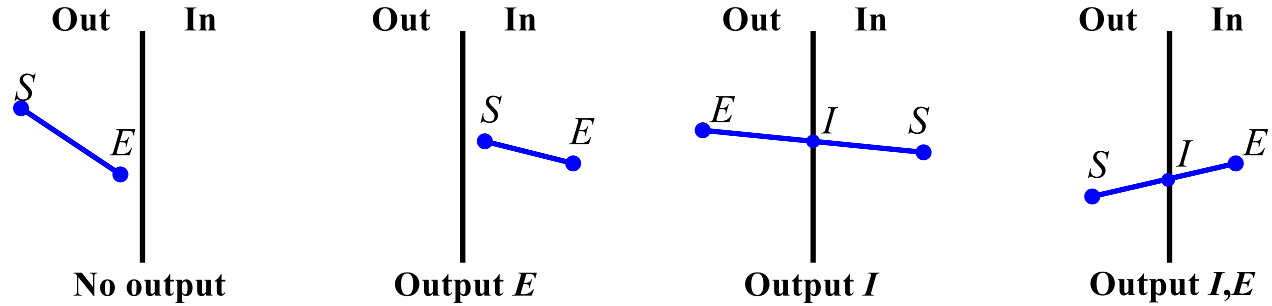


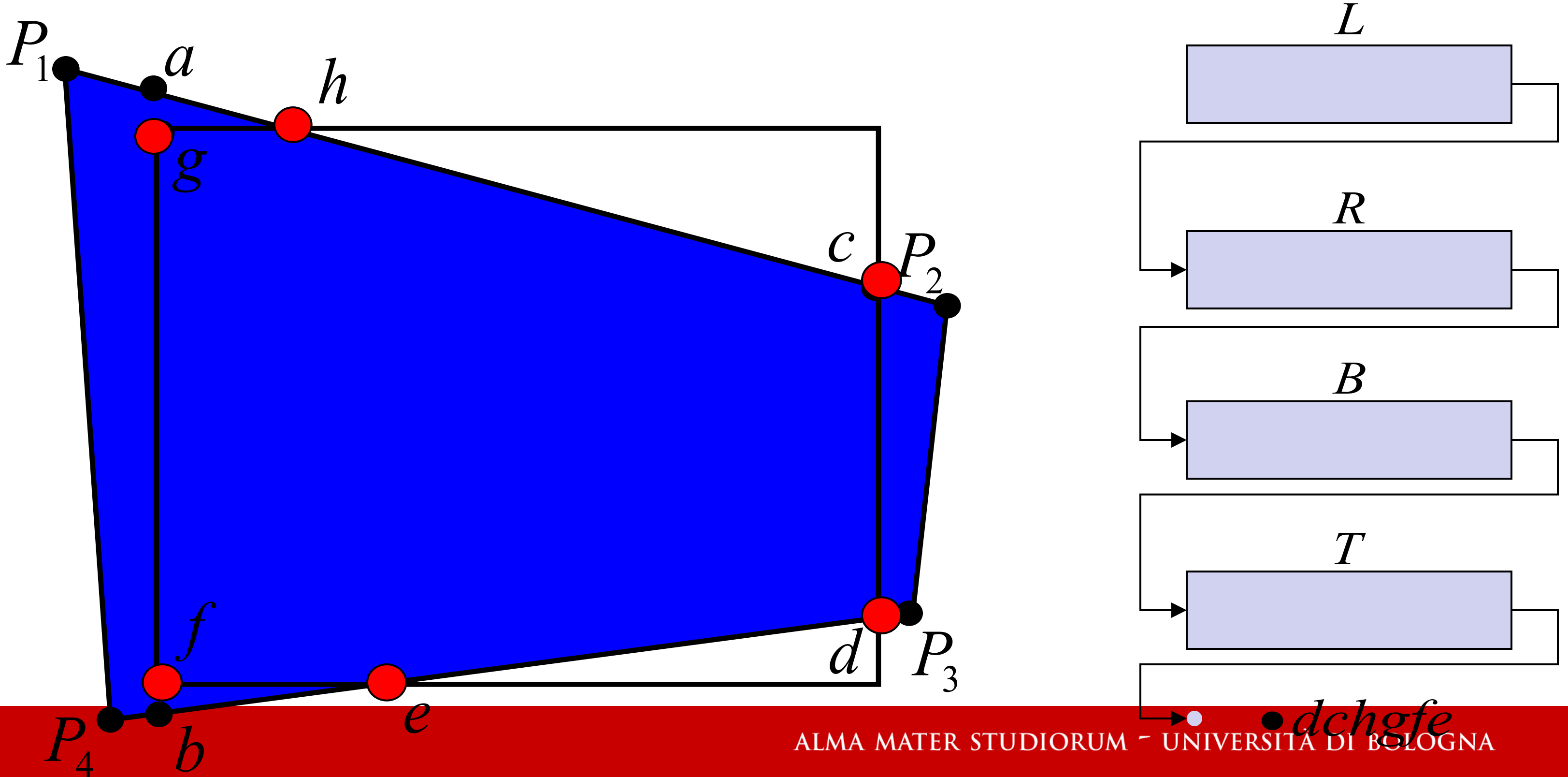
Algoritmo di Sutherland-Hodgman



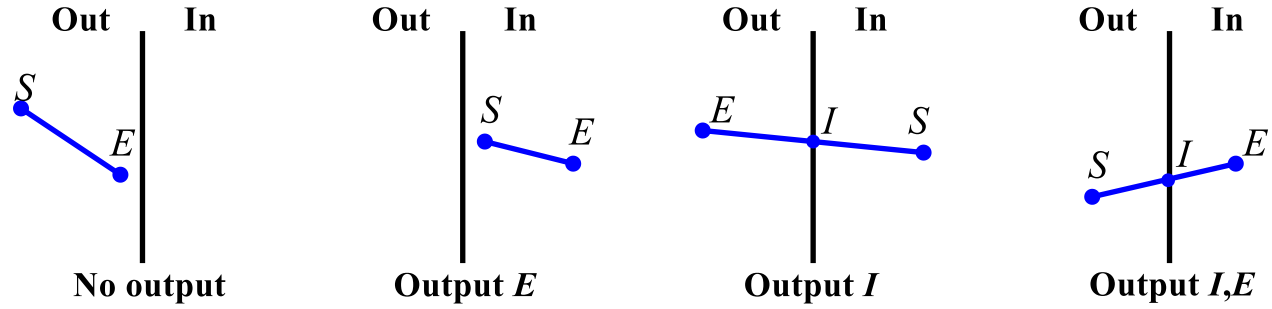


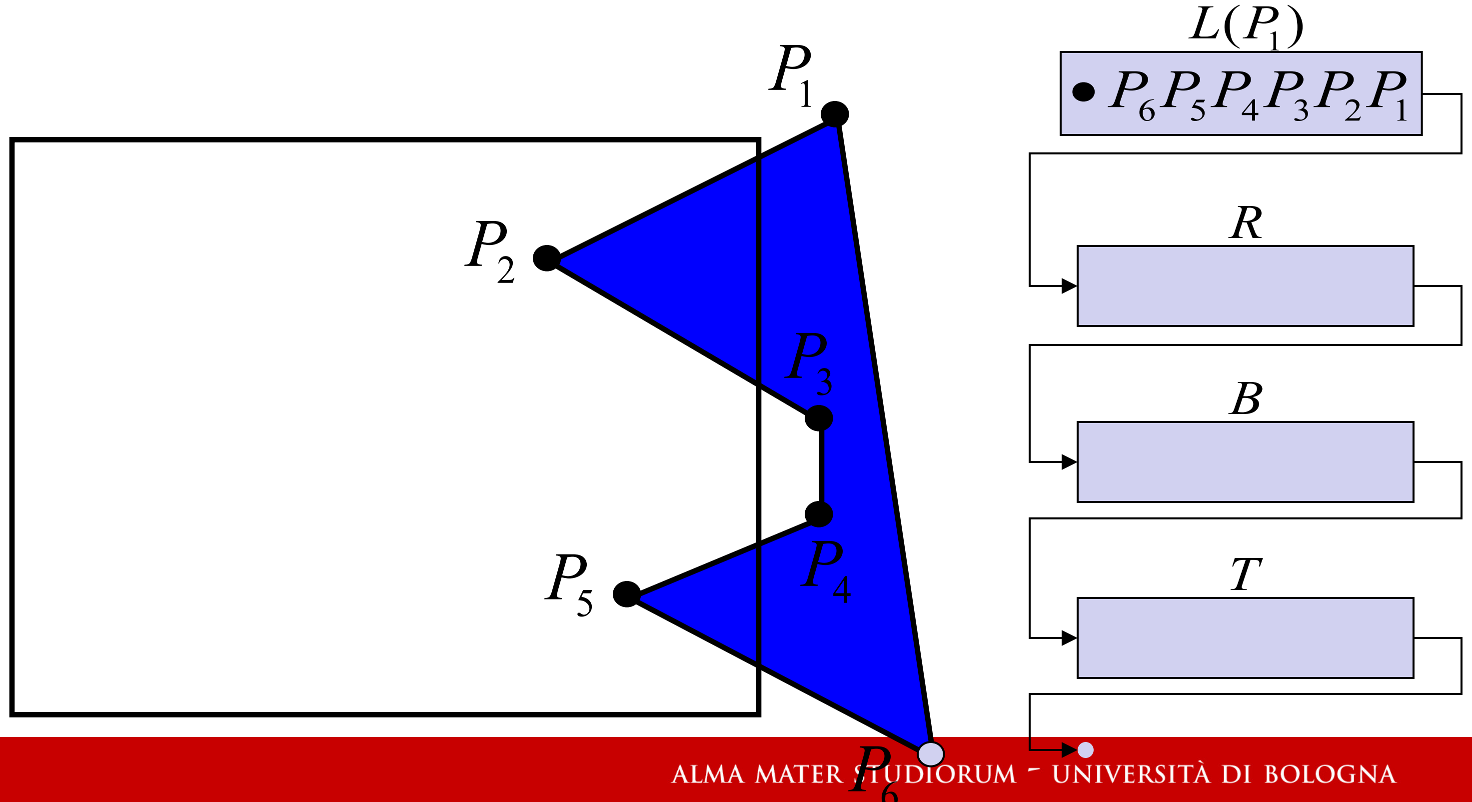
Algoritmo di Sutherland-Hodgman



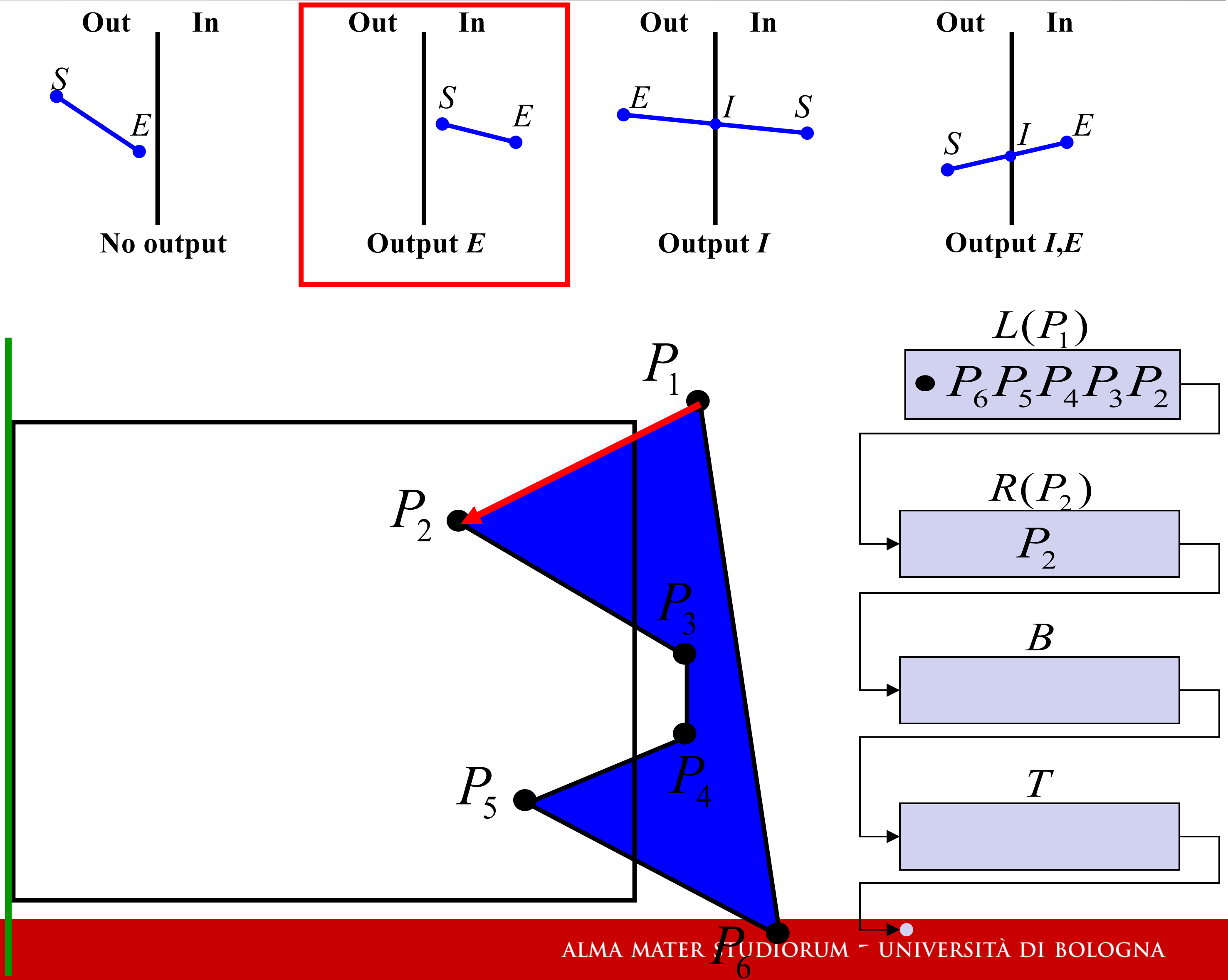


Algoritmo di Sutherland-Hodgman

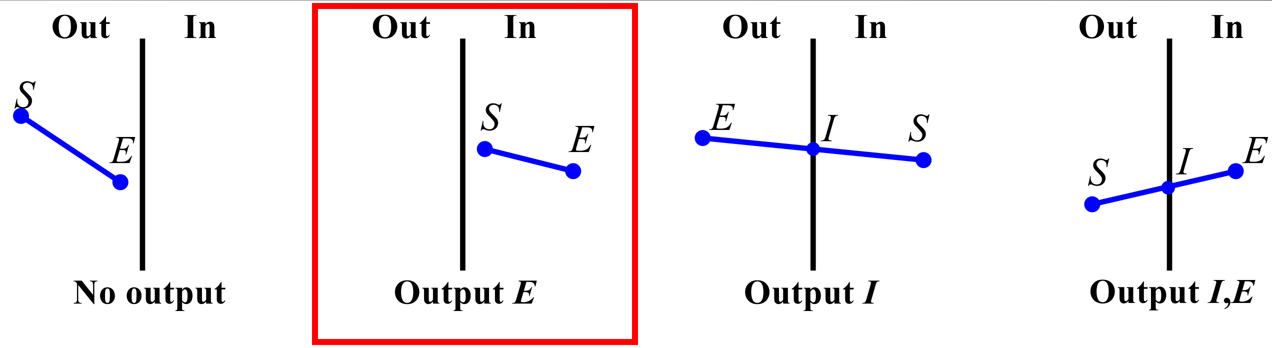


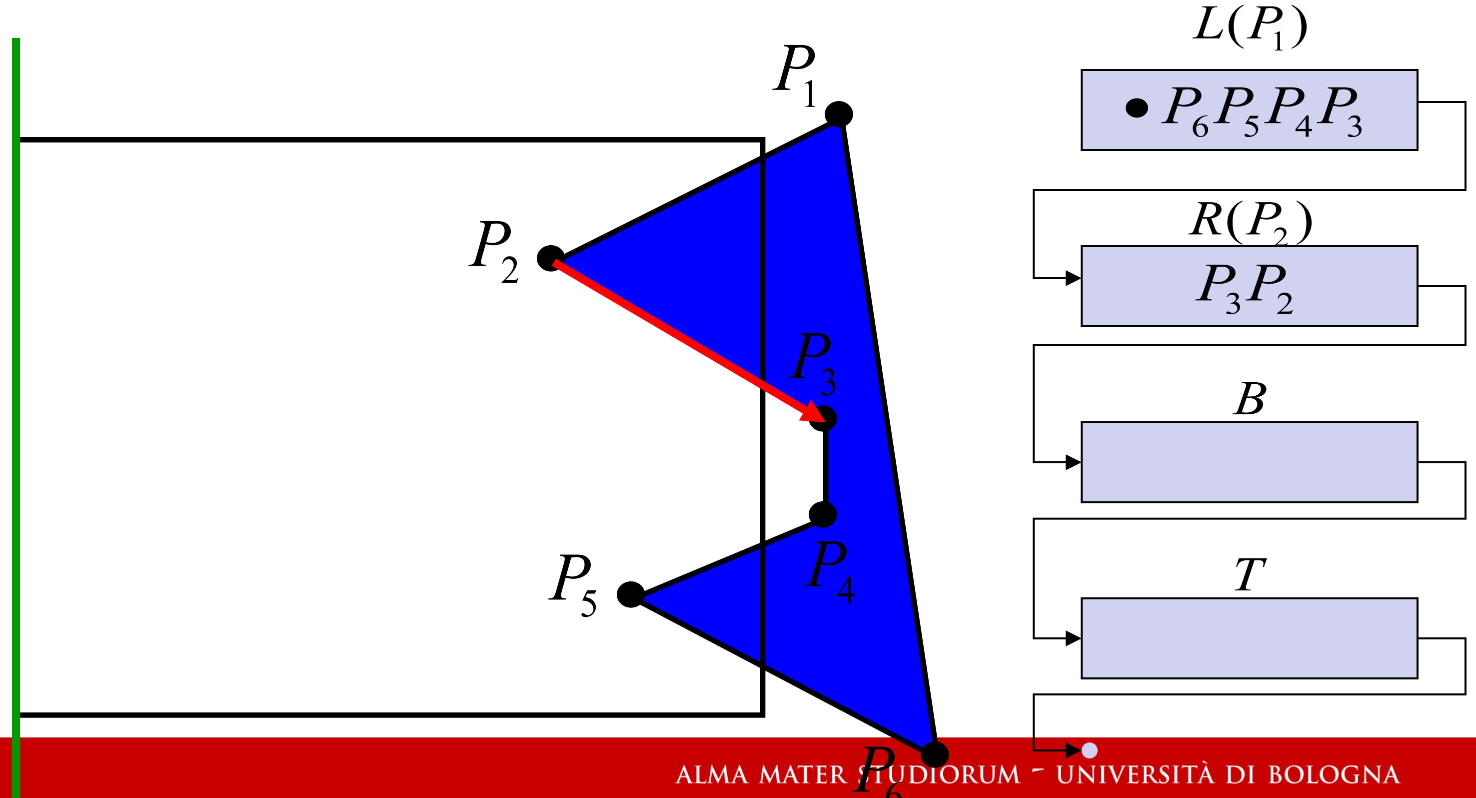


Algoritmo di Sutherland-Hodgman

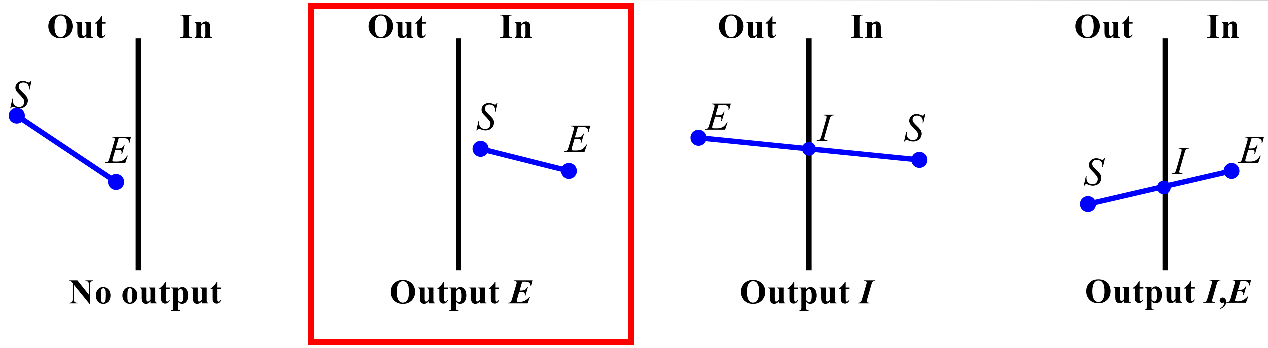


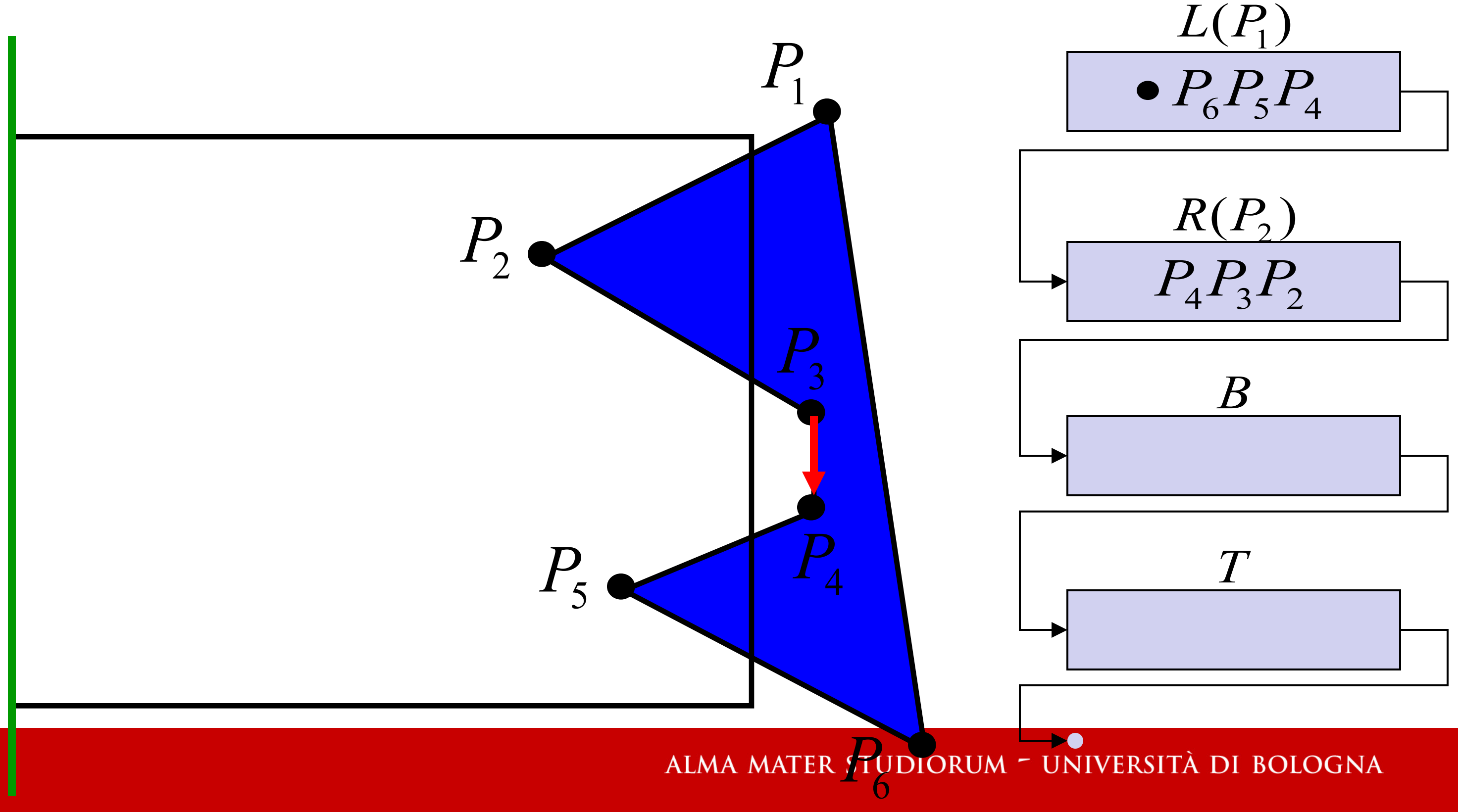
Algoritmo di Sutherland-Hodgman



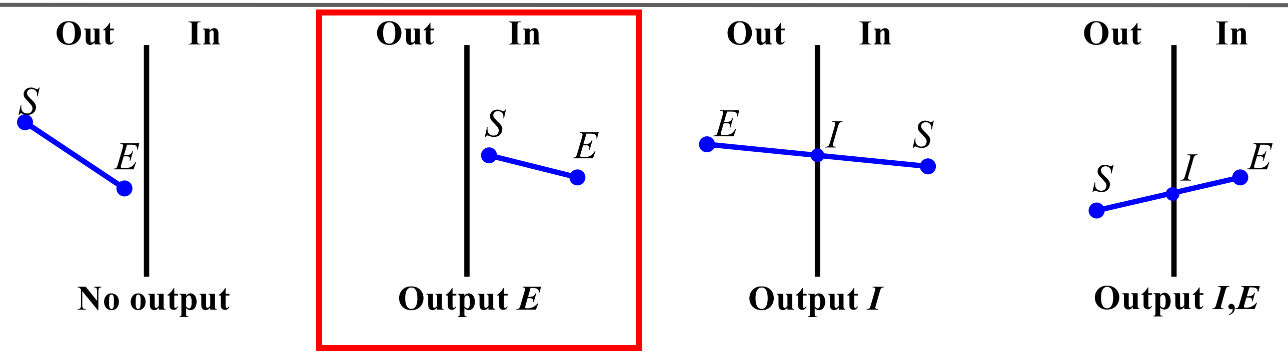


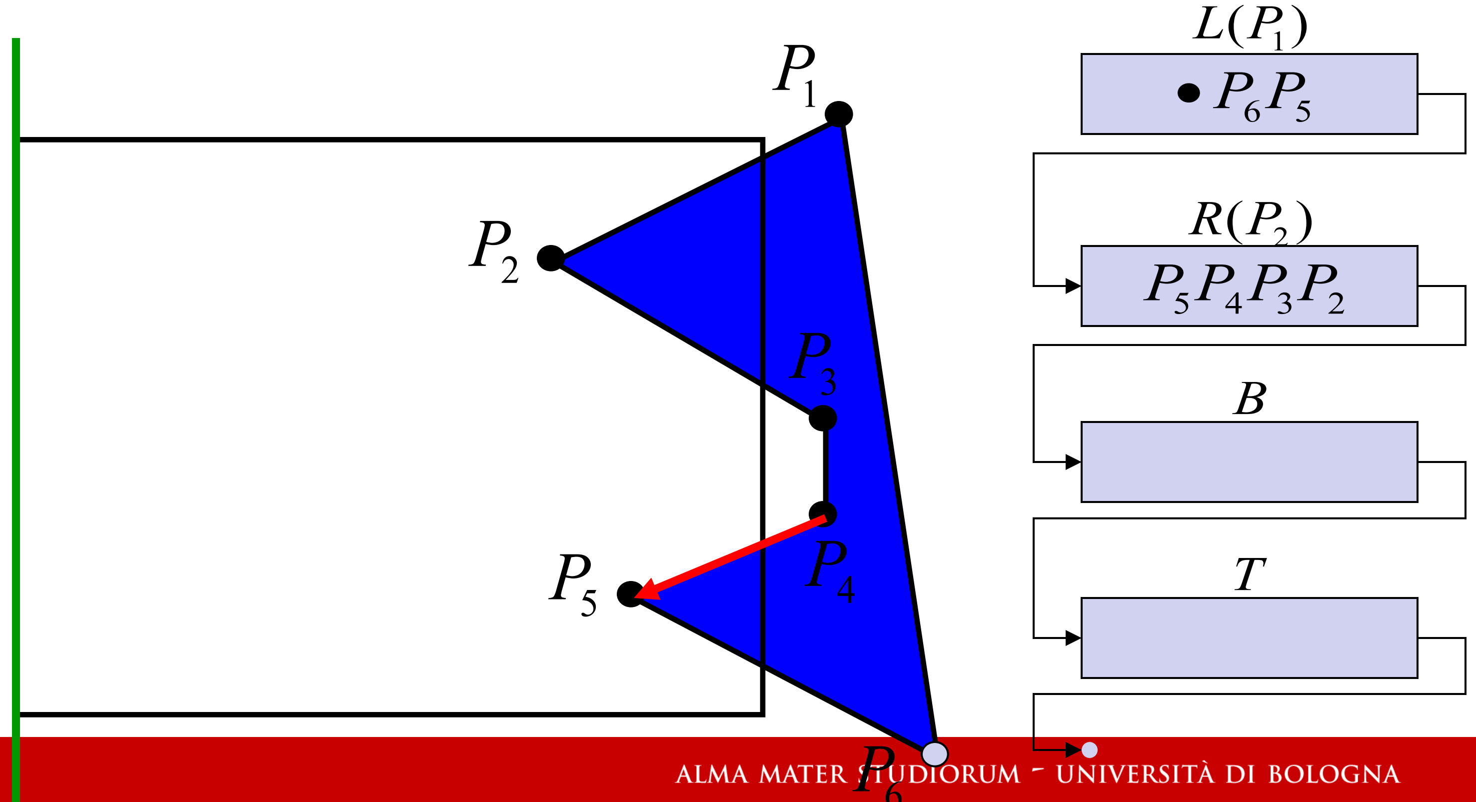
Algoritmo di Sutherland-Hodgman



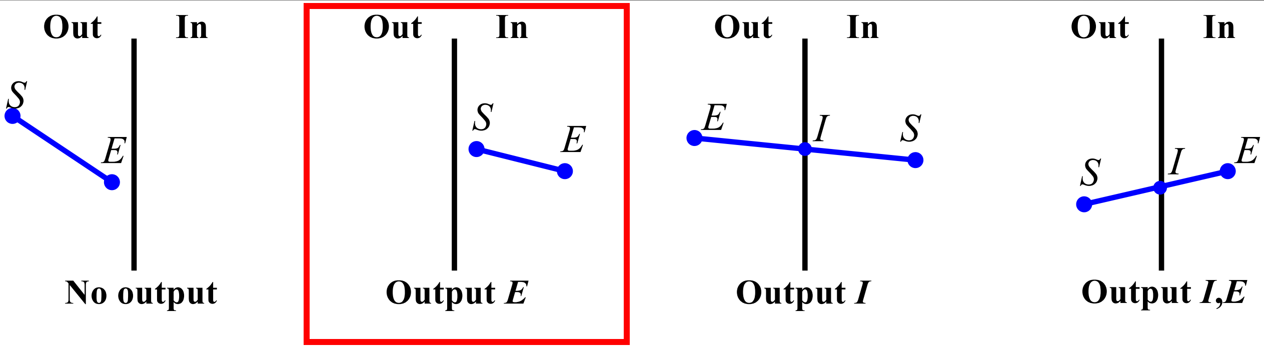


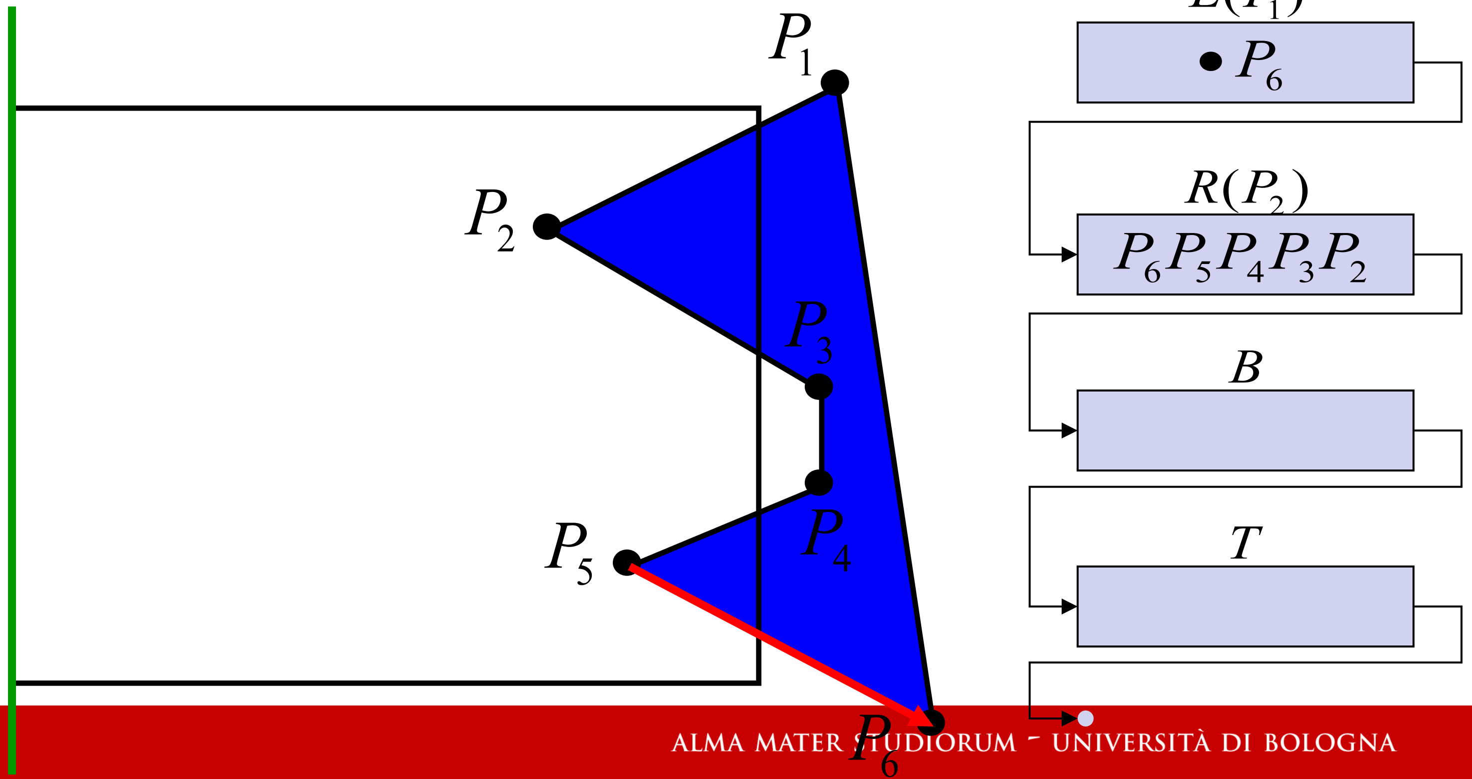
Algoritmo di Sutherland-Hodgman



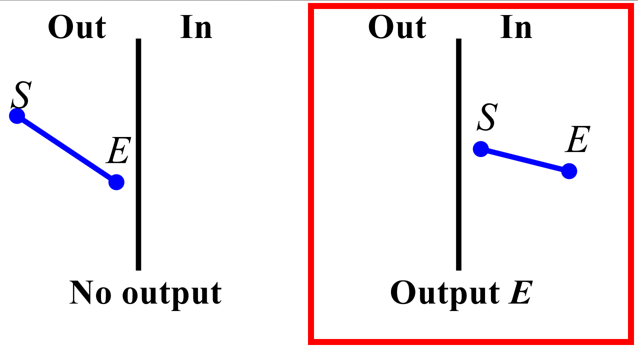


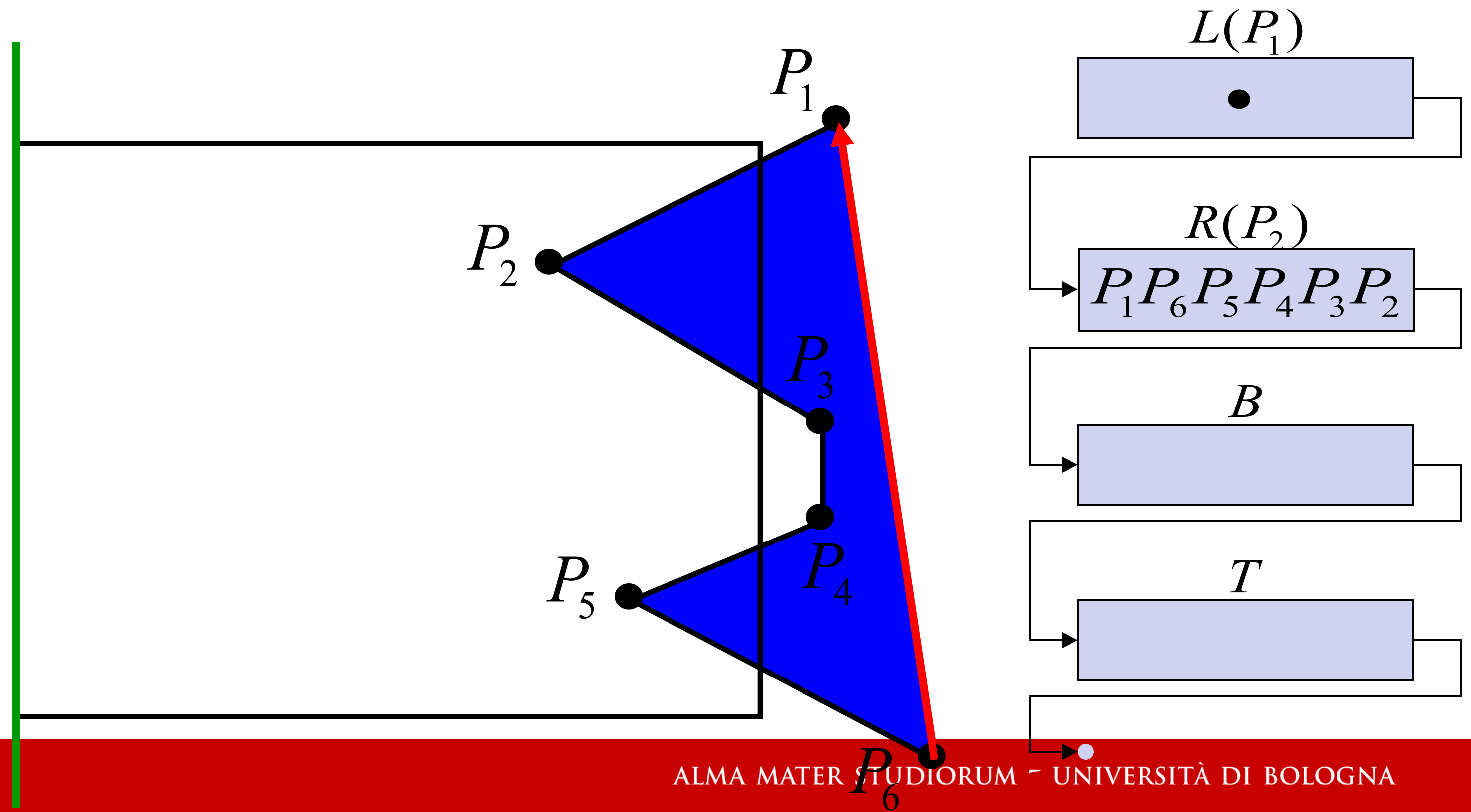
Algoritmo di Sutherland-Hodgman



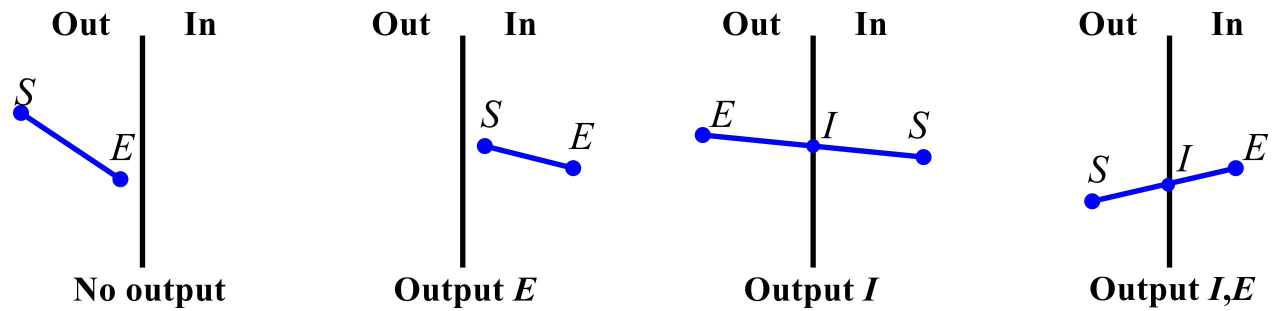


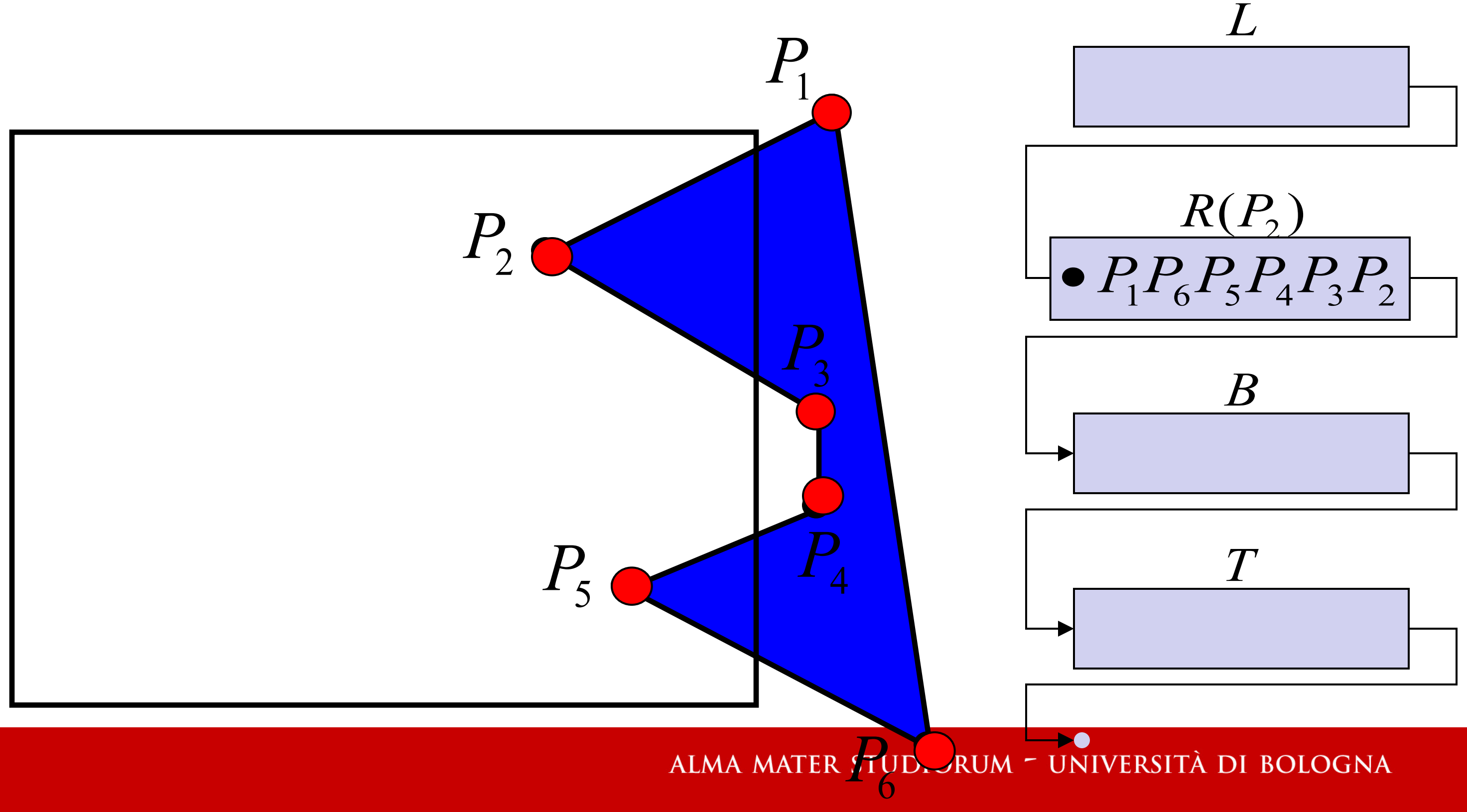
Algoritmo di Sutherland-Hodgman



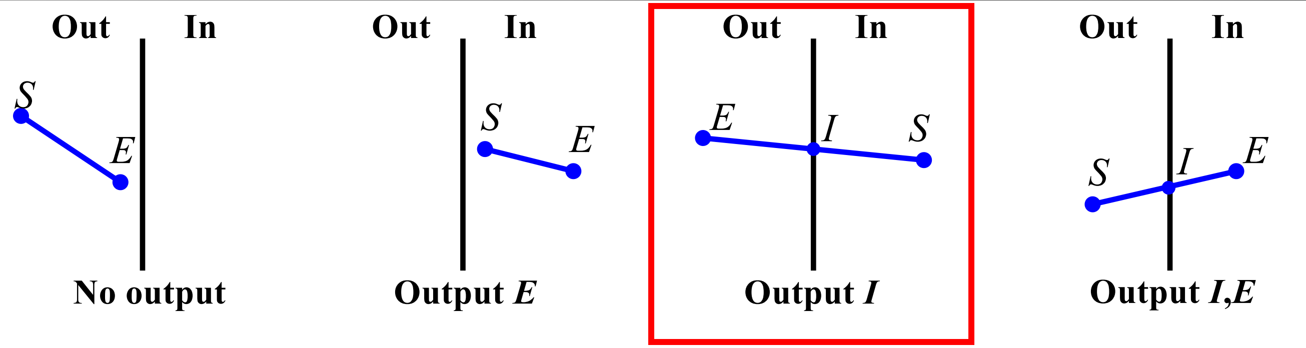


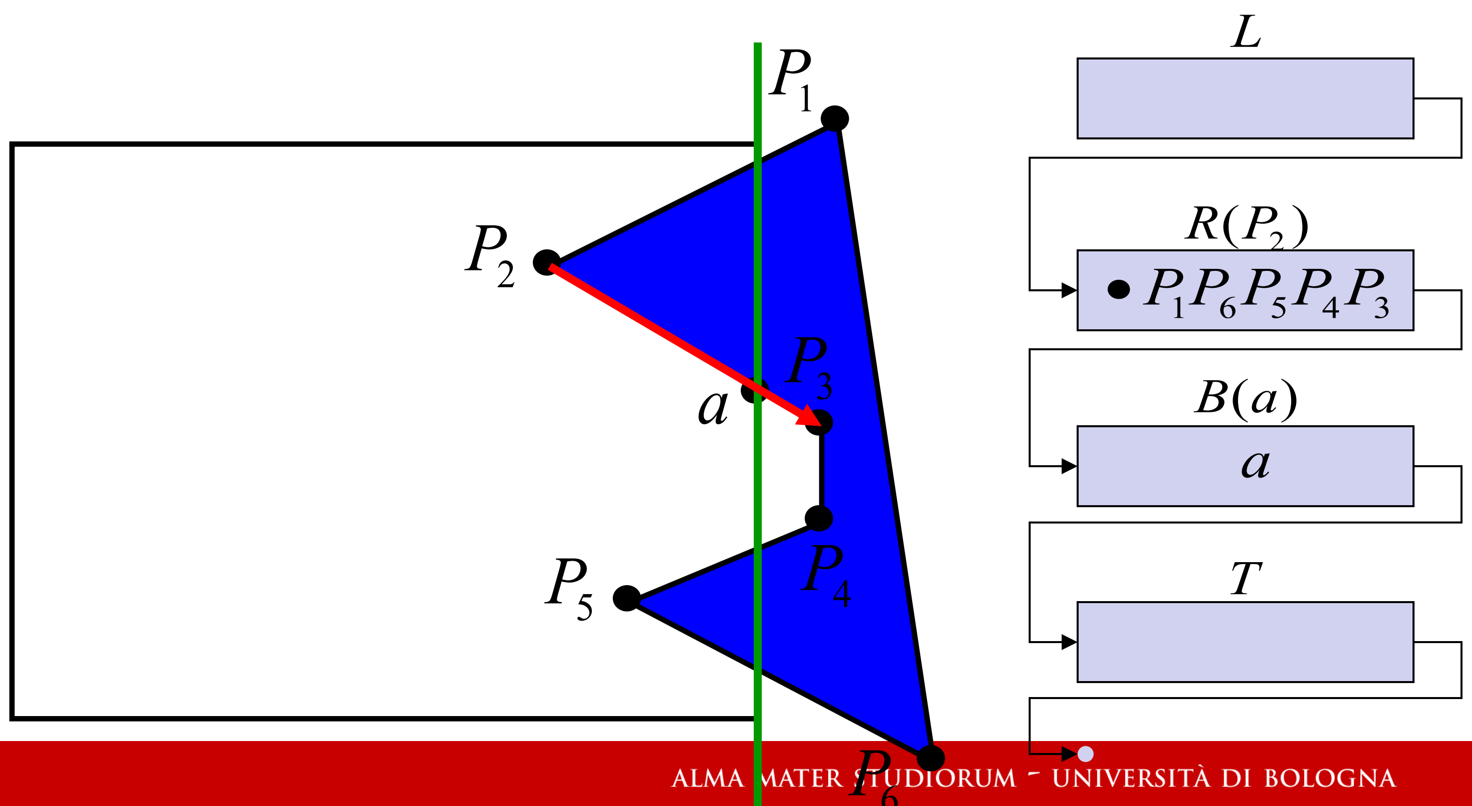
Algoritmo di Sutherland-Hodgman



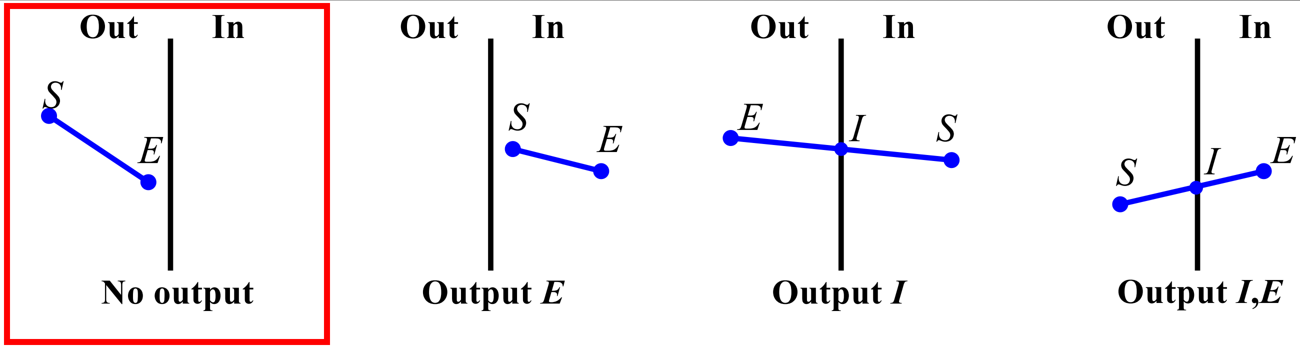


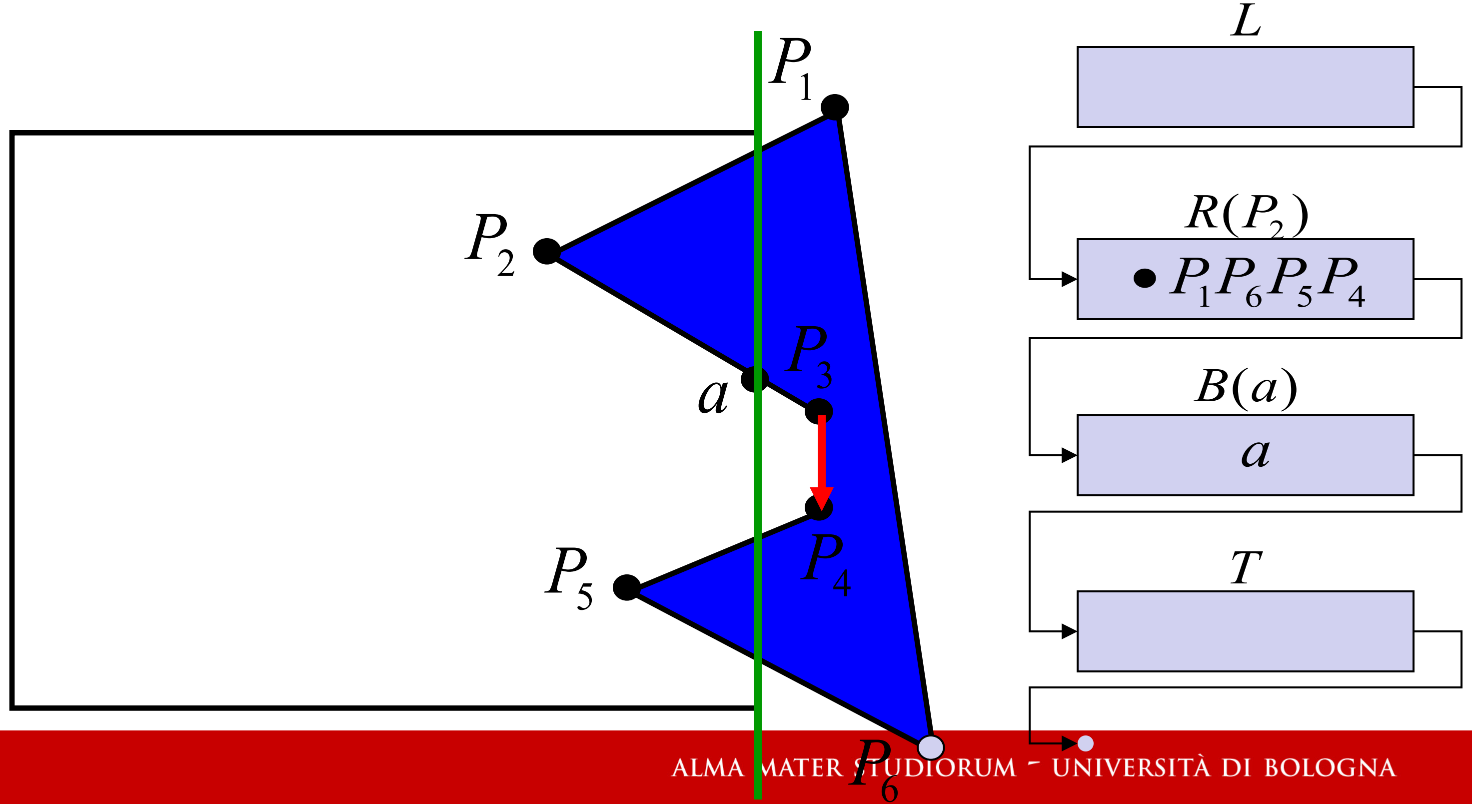
Algoritmo di Sutherland-Hodgman



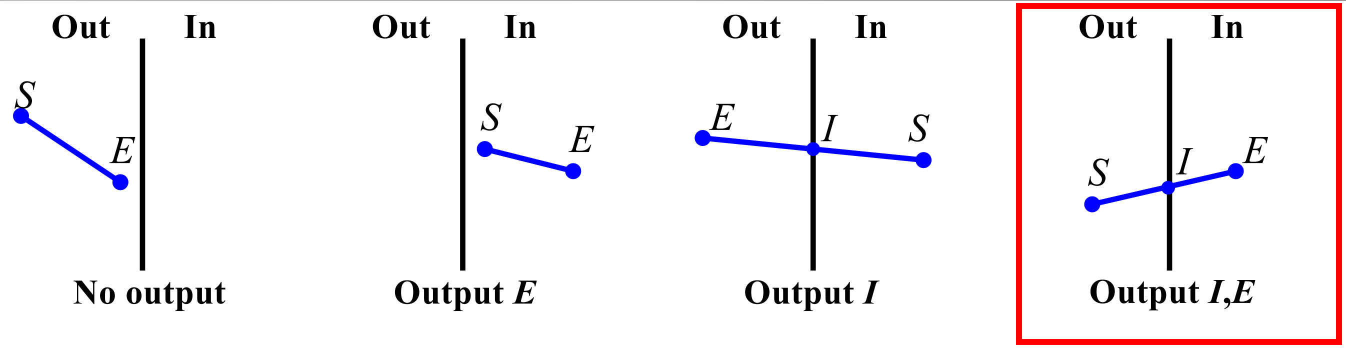


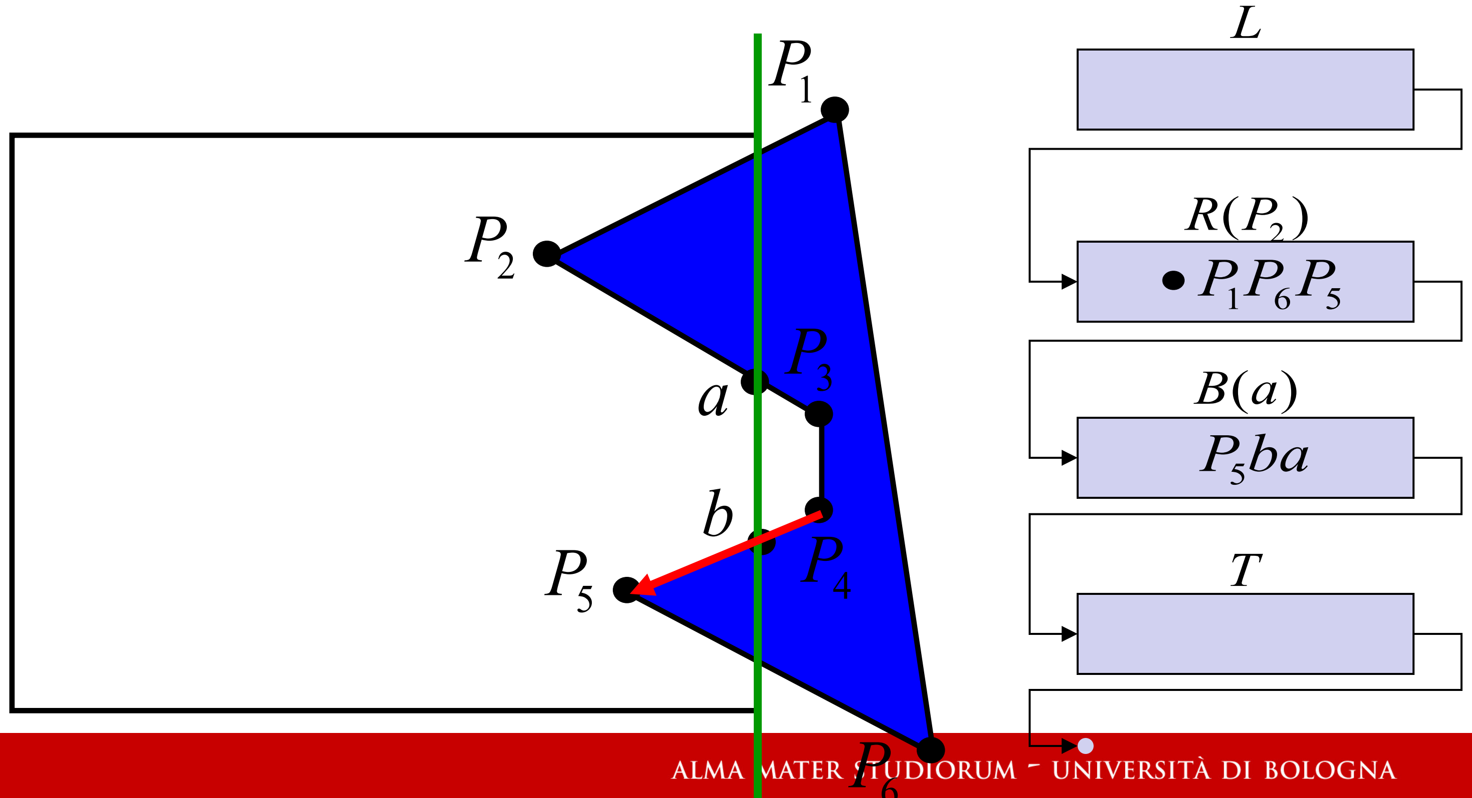
Algoritmo di Sutherland-Hodgman



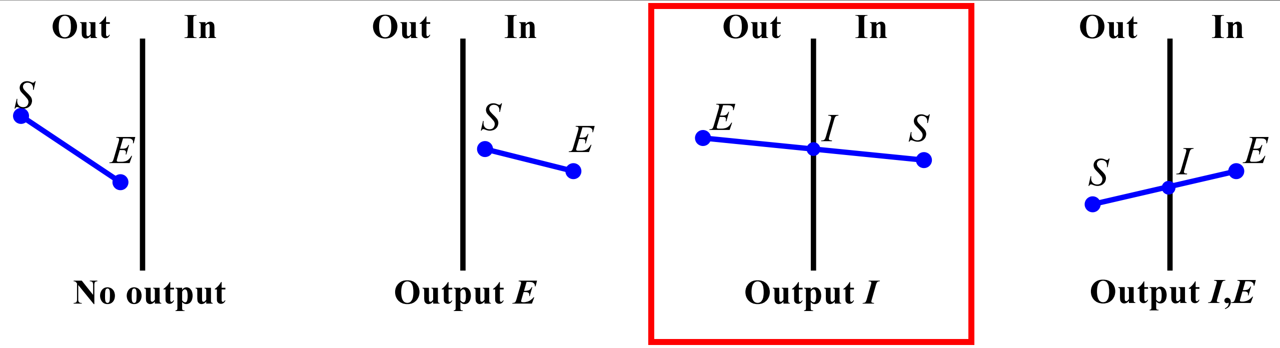


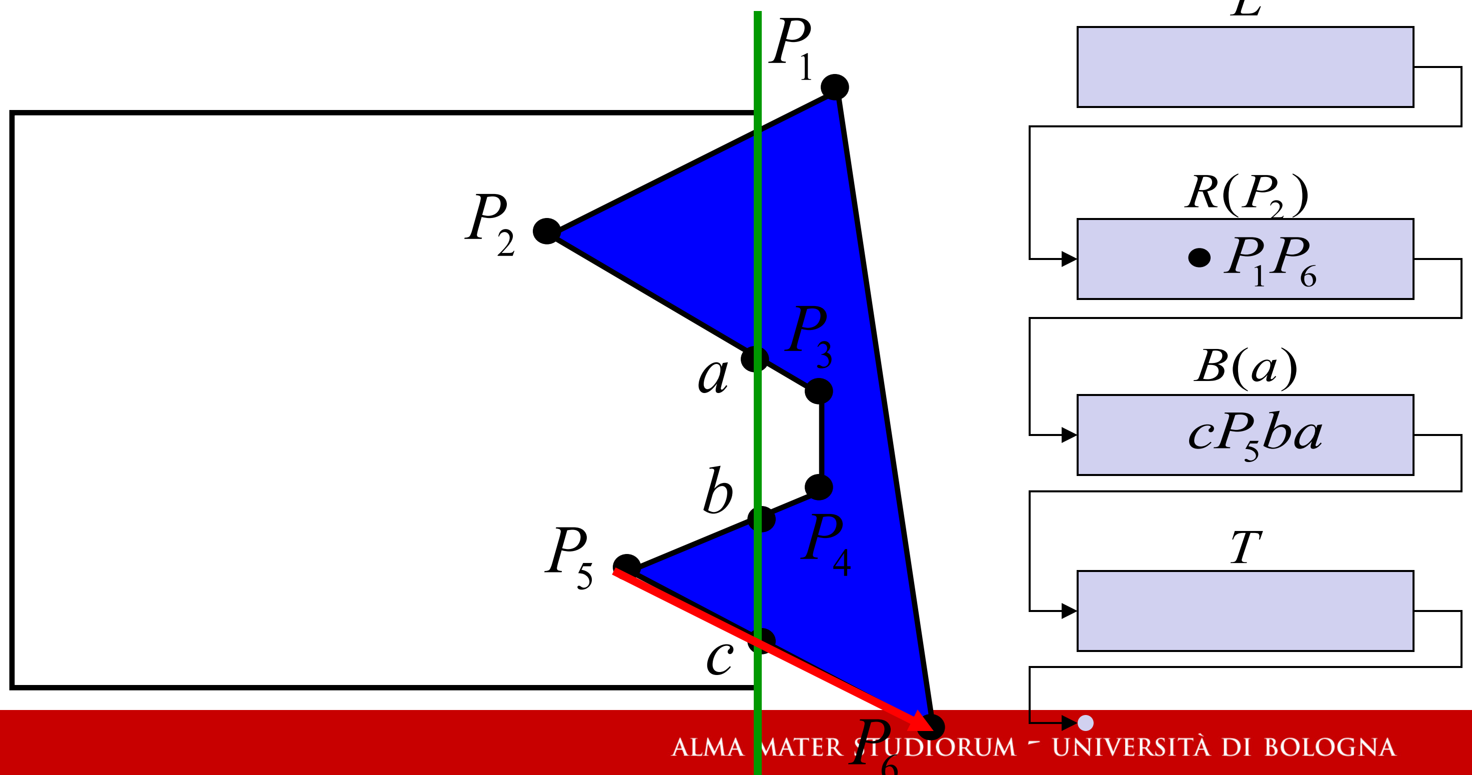
Algoritmo di Sutherland-Hodgman



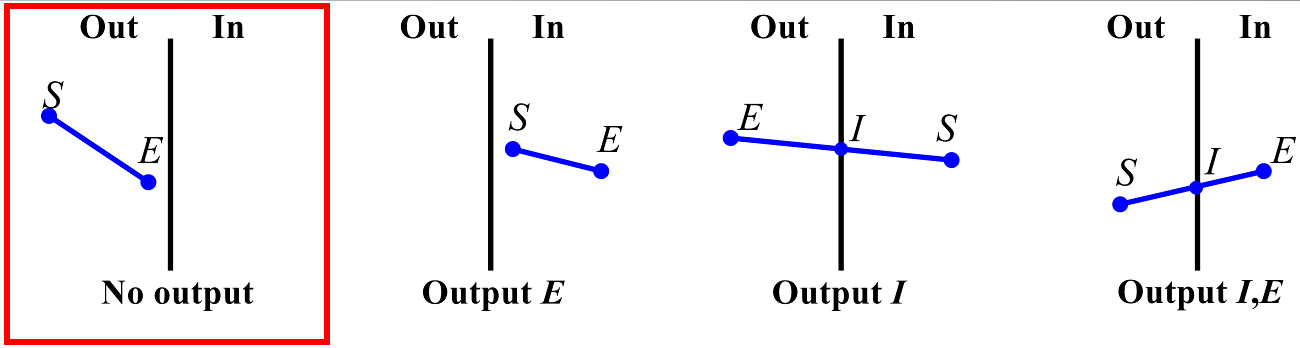


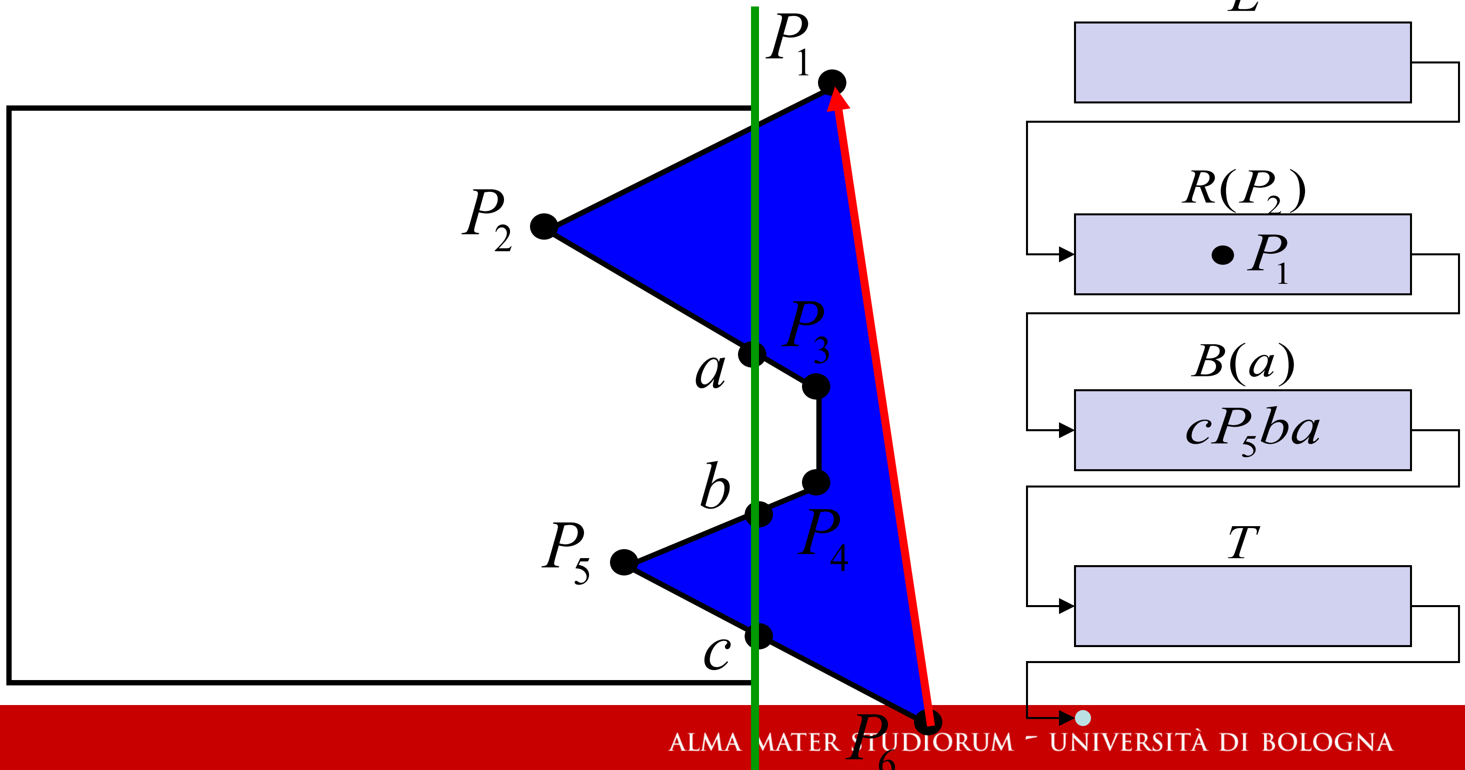
Algoritmo di Sutherland-Hodgman



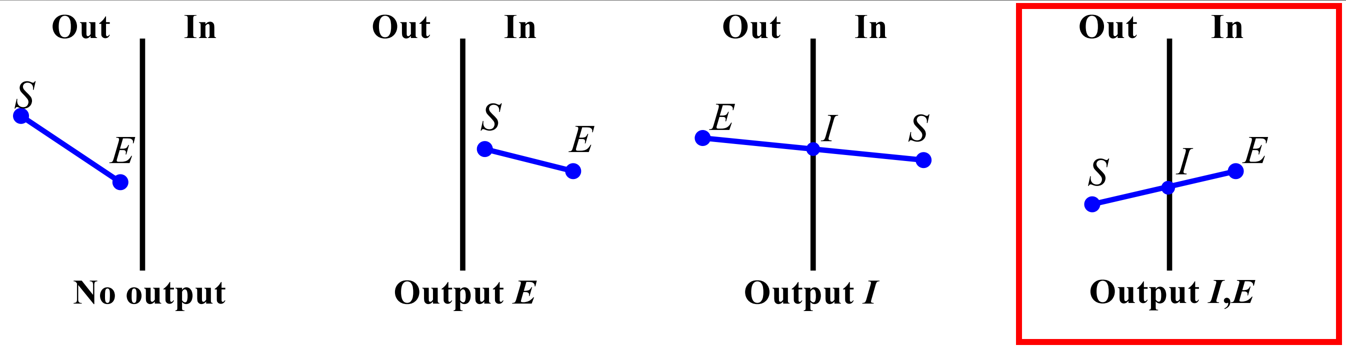


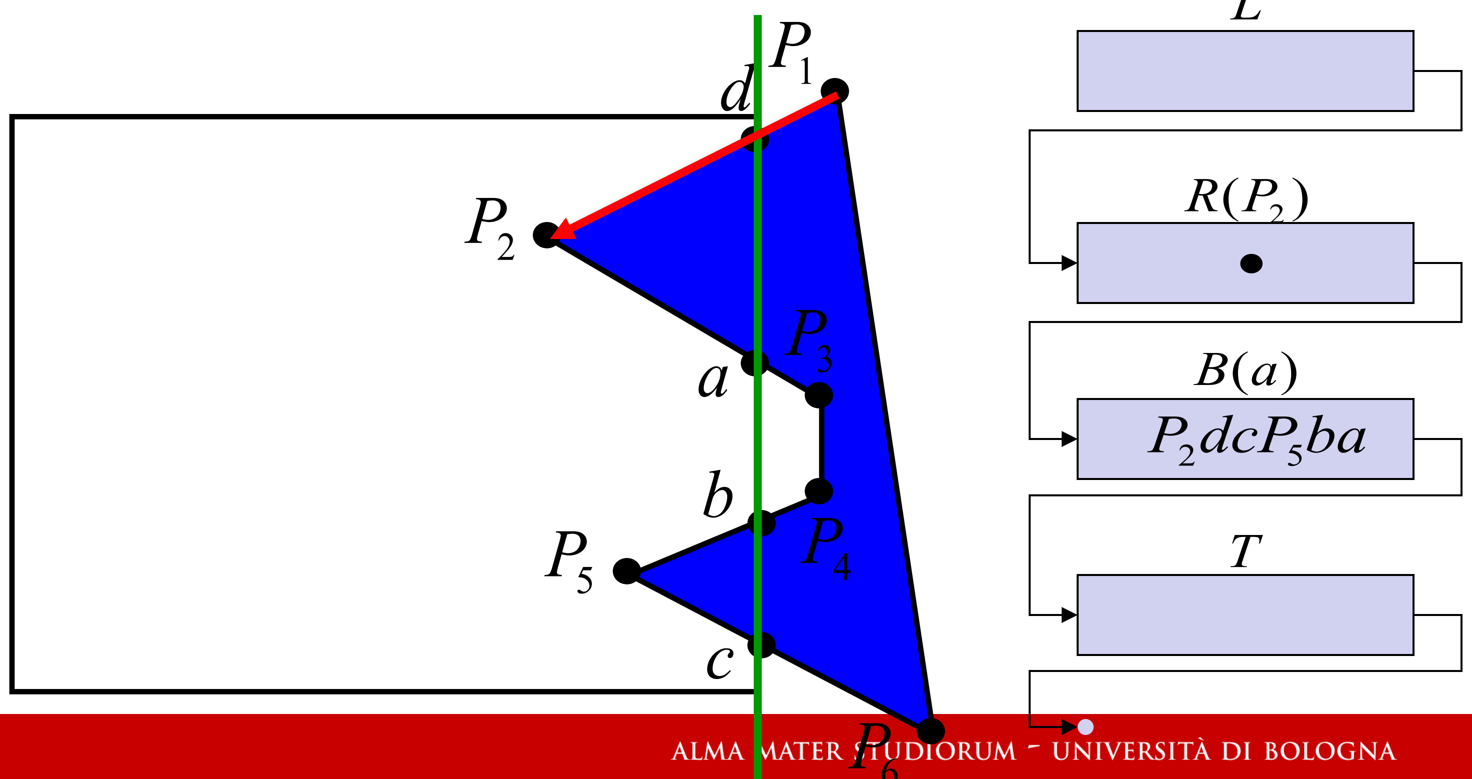
Algoritmo di Sutherland-Hodgman



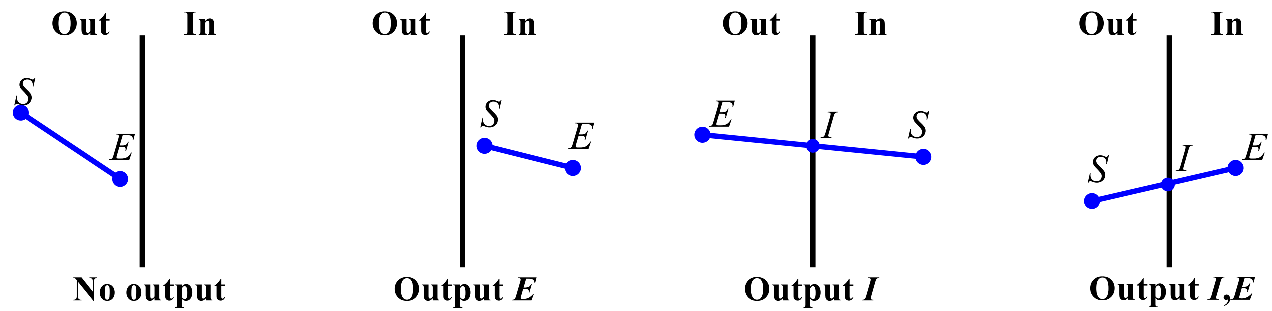


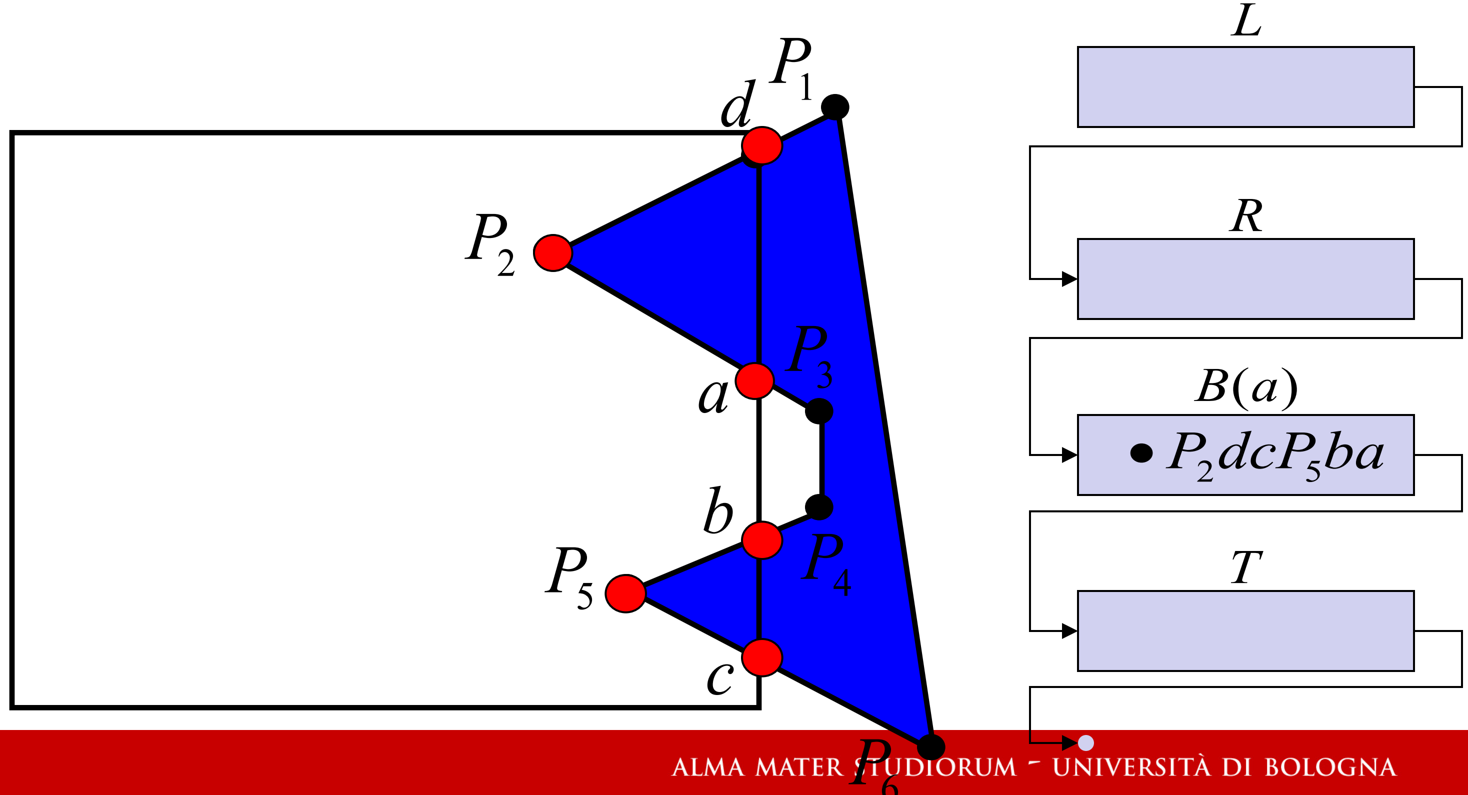
Algoritmo di Sutherland-Hodgman



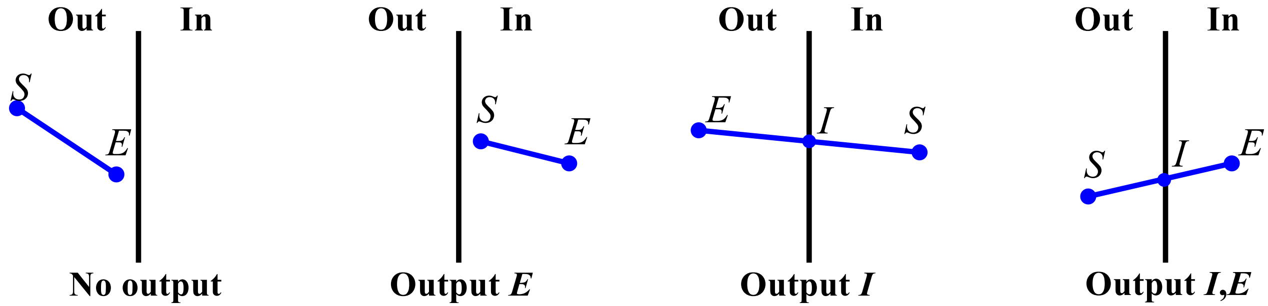


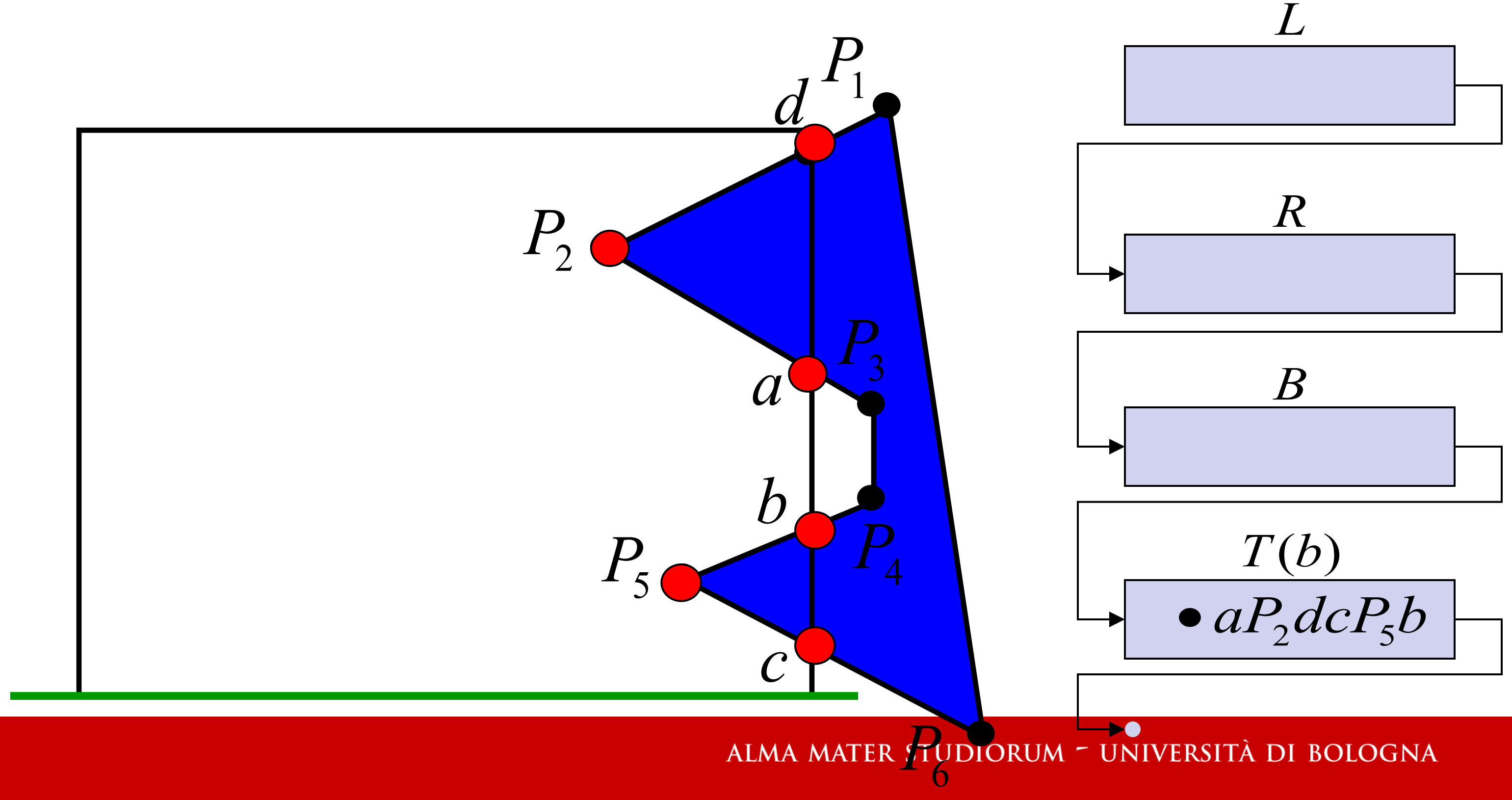
Algoritmo di Sutherland-Hodgman



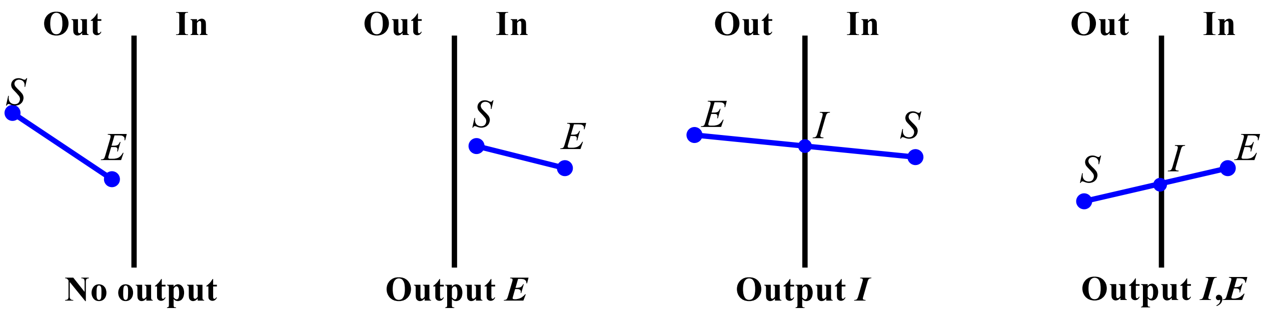


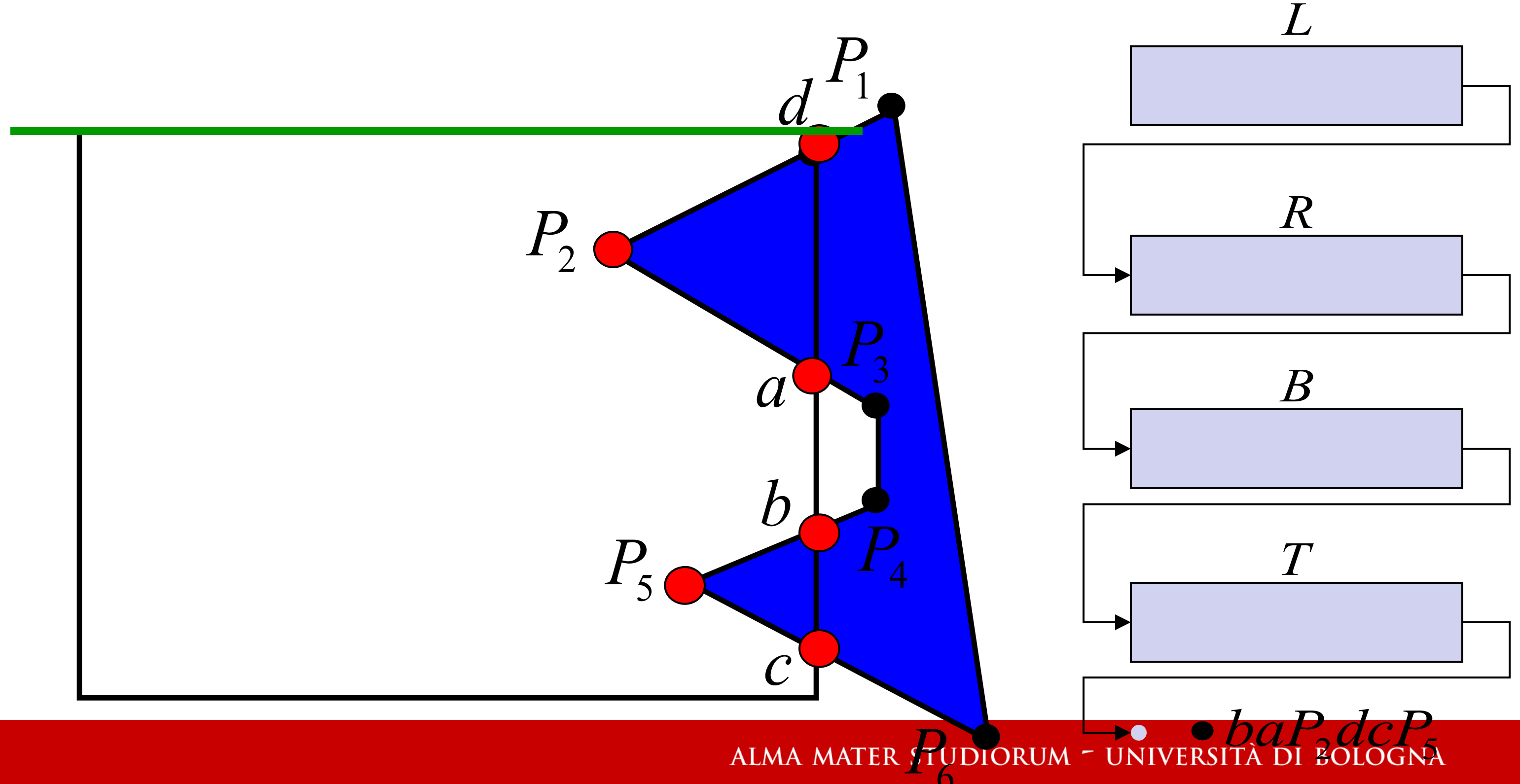
Algoritmo di Sutherland-Hodgman



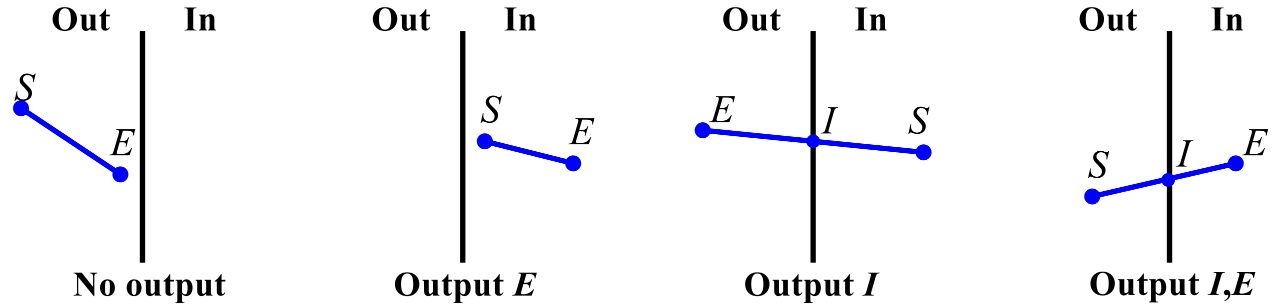


Algoritmo di Sutherland-Hodgman

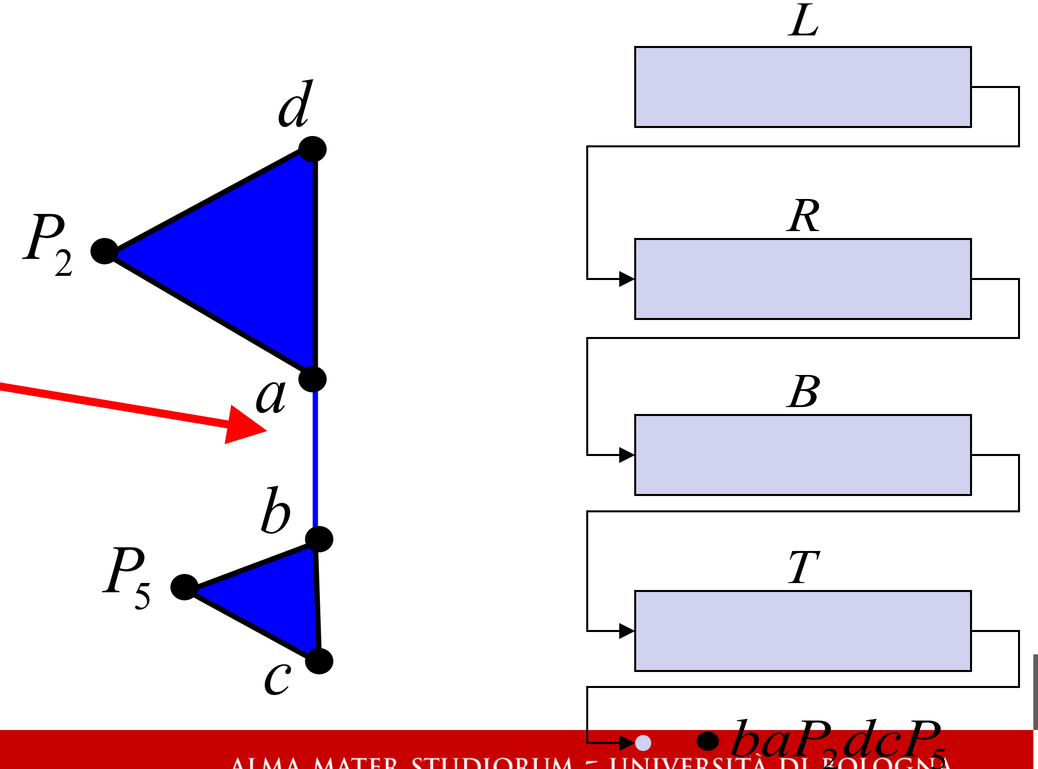




Algoritmo di Sutherland-Hodgman



Possibile errore nel caso di poligoni non convessi



Algoritmo di Sutherland-Hodgman

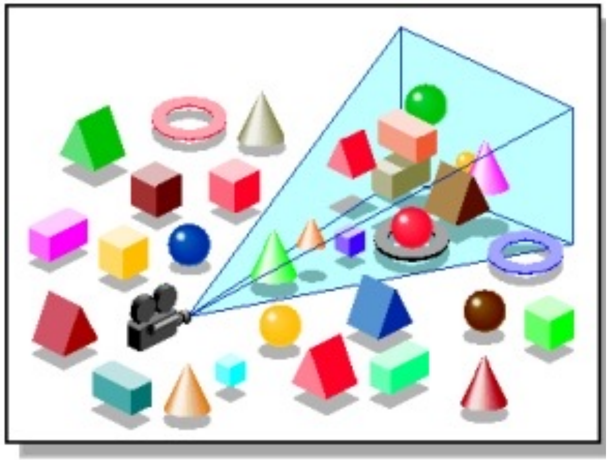
- Ideale per l'elaborazione in pipe-line.
  Il clipping di un lato del poligono non è necessario che sia terminato per cominciare con il lato successivo

- Ideale per l’elaborazione in fasi successive.
  Si può operare il clipping solo su due lati e portare a termine il procedimento in una seconda fase

Algoritmo di Sutherland-Hodgman 3D

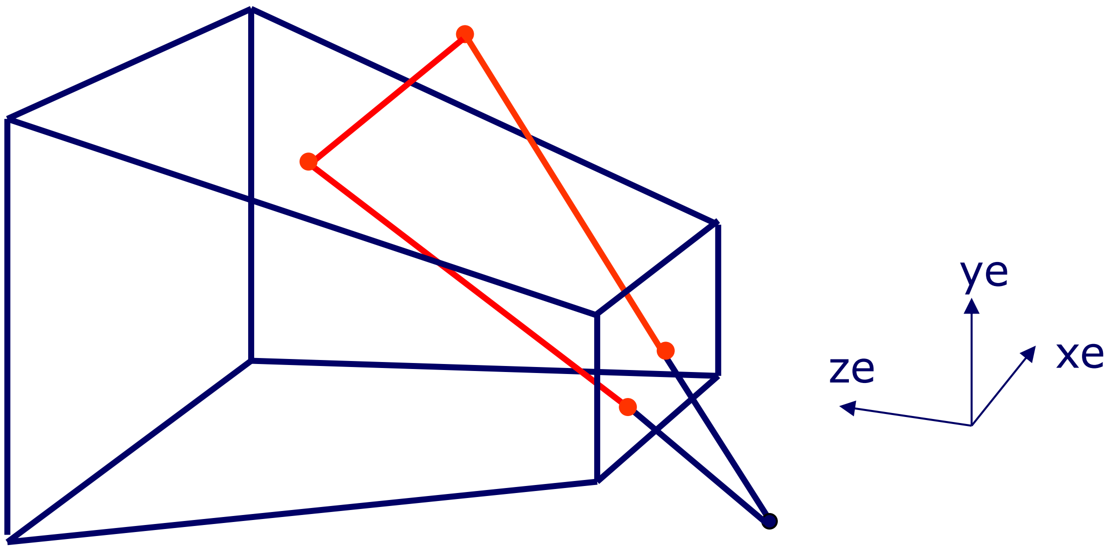
L'algoritmo di Sutherland-Hodgman può essere generalizzato rispetto ad un View Volume 3D.

Prima di applicare l'algoritmo si controlla se l'oggetto è esterno al Frustum, utilizzando un Bounding Volume, o completamente interno;



Poi si esegue il clipping dei singoli poligoni utilizzando l'algoritmo di Sutherland-Hodgman;

Algoritmo di Sutherland-Hodgman 3D



Si lavora in coordinate omogenee; siano (x1,y1,z1,w1) e (x2,y2,z2,w2) le coordinate di due punti, uno esterno ed uno interno; vogliamo trovate il punto di intersezione di questo segmento con uno dei 6 piani ,

$x=\pm w, \quad y=\pm w, \quad z=\pm w.$

Algoritmo di Sutherland-Hodgman 3D

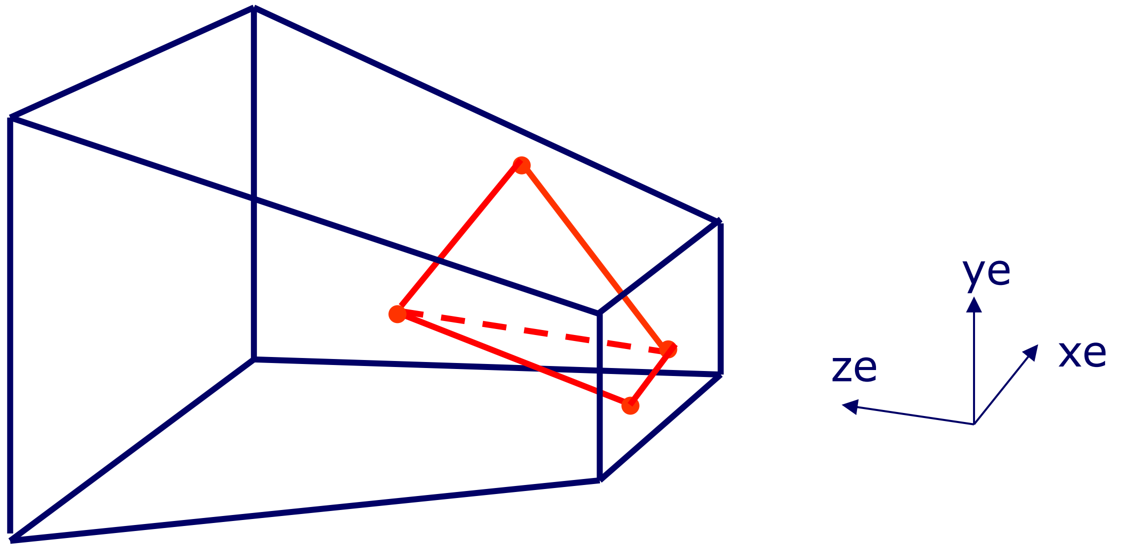
Siano $P1=(x1,y1,z1,w1)$ e $P2=(x2,y2,z2,w2)$ due vertici in clip-space; sia $P(t)=P1+t(P2-P1)=(x(t),y(t),z(t),w(t))$, con $t \in [0,1]$ il segmento in forma parametrica e lo si voglia intersecare con il piano $x=w$.

Sarà: $x(t)=x1+t(x2-x1)$ e $w(t)=w1+t(w2-w1)$

La condizione di intersezione sarà: $x(t)=w(t)$
Cioè: $x1+t(x2-x1)=w1+t(w2-w1)$

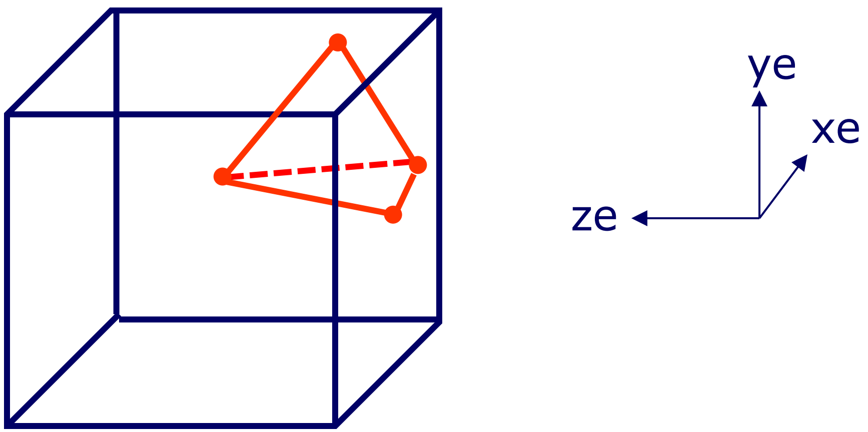
Risolvendo per t troveremo il punto in cui il segmento interseca il piano $x=w$ del frustum.

Algoritmo di Sutherland-Hodgman 3D



Una volta effettuato il clipping rispetto a tutti i sei piani, se il poligono risultante non è un triangolo, deve essere opportunamente suddiviso.

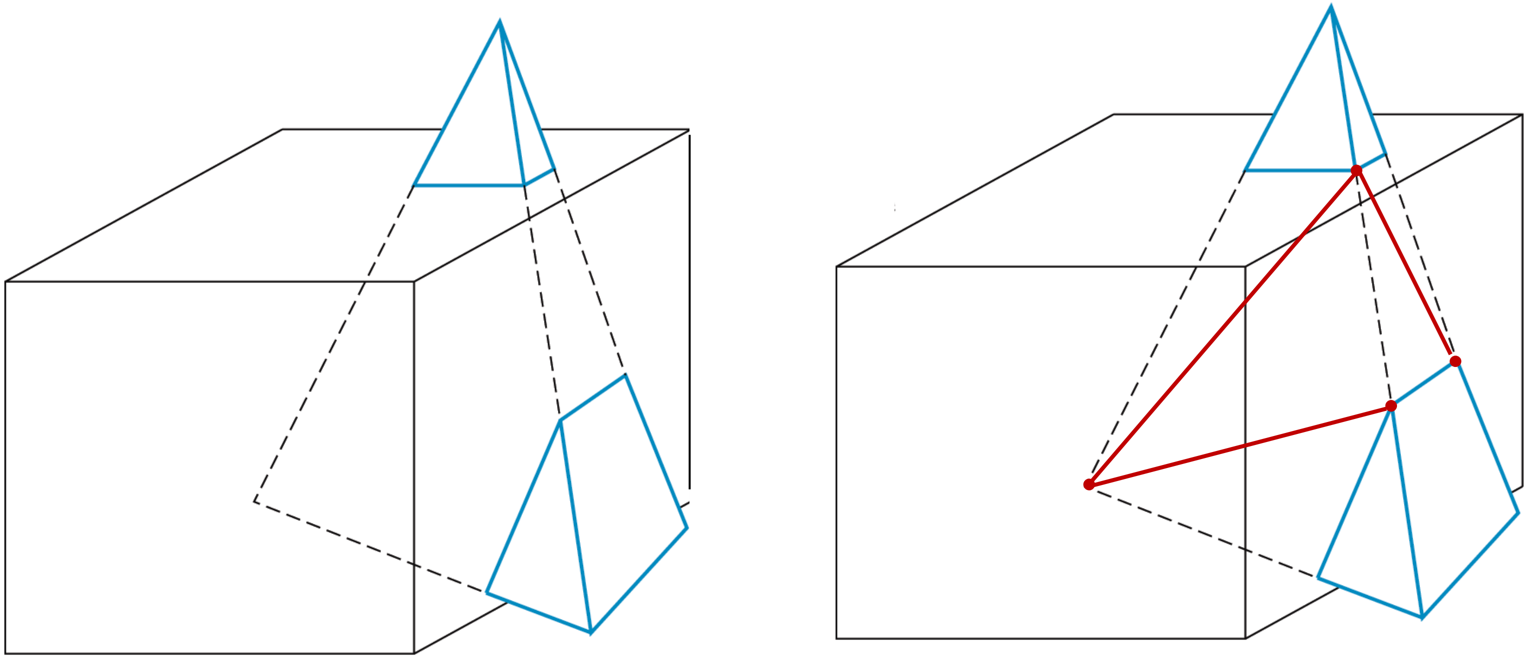
Algoritmo di Sutherland-Hodgman 3D



poi si procede con la proiezione prospettica che trasforma il tronco di piramide in un cubo mantenedo facce triangolari piane in facce triangolari piane.

Algoritmo di Sutherland-Hodgman 3D

Il caso più comune di clipping 3D è quello di oggetti mesh con facce poligonali ed in particolare triangolari.



Come già detto, poligoni risultanti, se non sono triangoli, devono essere opportunamente suddivisi.

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Giulio Casciola

Dip. di Matematica

giulio.casciola at unibo.it